International Refrigeration and Compressor Course 2026

Water-Cooled Chiller for High-Efficiency Edge Compute Data Center

Heat Exchanger Selection & Pressure-Drop Models

Project #3 — R-290 Refrigerant System — Champaign, IL

University of Illinois Urbana-Champaign / TU Dresden / KA

June 2026

Component	Status in this report
Working fluid selection	**Complete** — Chapter 2
Evaporator BPHE	**Complete** — Chapter 3
Condenser BPHE	**Complete** — Chapter 4
Compressor	**Complete** — Chapter 10
Expansion valve	**Complete** — Chapter 7
Dry cooler	**Complete** — Chapter 6
Free-cooling economizer	**Complete** — Chapter 5
Oil separator	**Complete** — Chapter 11
Piping	**Complete** — Chapter 12
Liquid receiver	**Complete** — omitted with justification, Chapter 13
System performance (COP / IPLV / PUE)	**Complete** — Chapter 14
Cycle representation (p-h chart)	**Complete** — Chapter 15
Charge analysis & safety measures	**Complete** — Chapter 16
Bill of materials	**Complete** — Chapter 17
Compressor–EEV control interaction	**Complete** — Chapter 18
System assessment (pros & cons)	**Complete** — Chapter 19
Suggestions for improvement	**Complete** — Chapter 20
Piping & instrumentation diagram	Pending
Commissioning procedure	Pending

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Chapter 1

Project Overview

1.1. System Description

This report covers the heat exchanger selection and pressure-drop modelling for **IRCC 2026, Project #3**: a 150 kW R-290 water-cooled chiller designed for a high-efficiency edge-compute data centre facility near Champaign, IL.

The system uses a vapour-compression cycle with propane (R-290) as the working fluid. Heat is absorbed from the chilled-water loop in the evaporator, compressed, and rejected to an intermediate 30 % propylene-glycol (PG) loop in the condenser. The glycol loop then carries the heat to an outdoor dry cooler with variable-speed fans. A waterside economiser (glycol-to-chilled-water plate HX) allows free cooling when outdoor temperatures are low enough to bypass the compressor entirely.

Table 1.1. Key system design parameters (Project #3).

Parameter	Value / Requirement	Reference
Maximum cooling load	150 kW (≈42 refrigeration tons)	Project spec
Minimum cooling load	30 kW (5:1 turndown)	Project spec
Chilled-water supply / return	15 / 21 °C	ASHRAE TC 9.9 W17
Refrigerant	R-290 (propane), ASHRAE A3, GWP $<$ 150	US EPA AIM Act
Dry-cooler design outdoor temp.	35 °C dry-bulb	Project spec
Full-load COP	$\geq$ 3.5 at AHRI 550/590-2023 conditions	ASHRAE 90.1-2022
Integrated Part Load Value (IPLV)	$\geq$ 5.0	AHRI 550/590-2023
Economiser activation threshold	$\leq$10 °C outdoor (full free cooling $\leq$4 °C)	ASHRAE 90.1 §6.5.1

1.2. Scope of This Report

This document covers the two refrigerant-side plate heat exchangers:

- **Chapter 3 — Evaporator BPHE**: R-290 evaporates against chilled water (21 °C → 15 °C). Thermal sizing by LMTD; selection from the Kelvion DW series; refrigerant-side pressure-drop model using the Han, Lee & Kim (2003) *evaporation* correlation as implemented in `evaporator.py`.

- **Chapter 4 — Condenser BPHE**: R-290 condensing rejects heat to the 30 % propylene-glycol loop (40 °C → 46 °C). Thermal sizing by counter-flow LMTD (three-zone approximation); same Kelvion DW model family; pressure-drop model using the Han, Lee & Kim (2003) *condensation* correlation as implemented in `condenser.py`.

Both chapters follow the same document structure: **Part A** (sizing and model selection) followed by **Part B** (refrigerant-side pressure-drop derivation and code documentation). All geometry parameters marked **ASSUMED** are placeholders pending a model-specific vendor drawing from Kelvion; parameters marked **DERIVED** follow analytically from the assumed values.

Chapter 2

Working Fluid Selection

Part A — Screening Constraints and Candidate Elimination

2.1. Project Constraints

The project brief fixes two hard constraints on refrigerant selection before any performance comparison is meaningful:

- **GWP** < 150 (US EPA AIM Act Technology Transitions, §40 CFR Part 84).
- **ASHRAE 34 safety classification A1, A2L, or A3 only.**

These two constraints alone eliminate several common candidates without needing a cycle calculation at all:

- **R-32** (GWP ≈675) and **R-454B** (GWP ≈466) — both popular low(er)-GWP HFCs, but both fail the project's GWP<150 requirement outright.
- **Ammonia (R-717)** — GWP 0, excellent thermodynamic performance, but ASHRAE safety class **B2L** (toxic). The project brief's safety-class list (§2.1) explicitly admits only A1/A2L/A3; B-class (toxic) fluids are out of scope regardless of GWP or COP.
- **CO_2 (R-744)** — GWP 1, ASHRAE class A1 (non-toxic, non-flammable), satisfies both hard constraints. It is nonetheless excluded here on a third, cycle-architecture ground: R-744's critical temperature is only 31.1 °C, well below this project's 52 °C design condensing temperature (Chapter 4, §4.2.3). A standard subcritical cycle — the architecture every other chapter of this report is built around — is not physically possible for CO_2 at this condensing condition; it would require a transcritical cycle (gas cooler instead of a condenser, a throttle/expander sized for supercritical inlet conditions, and a materially different high-side pressure regime, ∼90 bar rather than ∼18 bar). This is a legitimate refrigerant in other applications but a different design exercise from the one this report undertakes, so it is not carried into the performance comparison below.

2.2. Screened Candidates

Three candidates survive both hard constraints and the subcritical-cycle-feasibility check (all three have a critical temperature comfortably above the 52 °C condensing condition):

Table 2.1. Screened working-fluid candidates.

Refrigerant	GWP (AR5, 100-yr)	ASHRAE 34 class	Family
R-290 (propane)	3	A3	Hydrocarbon (HC)
R-1234yf	1	A2L	Hydrofluoroolefin (HFO)
R-1234ze(E)	1	A2L	Hydrofluoroolefin (HFO)

Part B — Idealized Cycle Performance Comparison

2.3. Comparison Method

The detailed compressor model used elsewhere in this report (Chapter 10) is a vendor-published AHRI-540/EN12900 performance map, which exists only for the specific refrigerant/compressor pair actually selected (R-290, Bitzer 4FEP-35Z) — no equivalent map exists yet for the other two candidates, since no compressor has been selected for them. Comparing refrigerants on that basis would really be comparing compressors, not working fluids.

Instead, all three candidates are evaluated through the **same idealized single-stage cycle**, with every non-refrigerant-dependent parameter held identical:

- Evaporating temperature 7 °C, condensing temperature 52 °C — this project's own design point (Chapter 4).
- 8 K superheat, 5 K subcooling — matching this project's design assumptions.
- A single fixed isentropic compressor efficiency, $\eta_{\text{isentropic}} = 0.70$, applied identically to all three candidates.

Holding every parameter fixed except the refrigerant itself isolates the working fluid's own thermodynamic contribution to COP from any compressor-specific efficiency curve — appropriate for a screening-level comparison made *before* a compressor is selected, which is exactly the point in the design process Chapter 10's real vendor map (developed only after R-290 was already chosen) does not yet inform. Implemented in `refrigerant_comparison.py`.

2.4. Results

Table 2.2. Working-fluid screening comparison: idealized single-stage cycle (7 °C evap. / 52 °C cond., 8 K superheat, 5 K subcooling, $\eta_{\text{isentropic}} = 0.70$ for all three candidates).

Refrigerant	GWP	Safety	p_{evap} [bar]	p_{cond} [bar]	PR [-]	T_{crit} [°C]	COP [-]
R-290 (propane)	3	A3	5.84	17.89	3.06	96.7	3.453
R-1234yf	1	A2L	3.98	13.66	3.43	94.7	3.375
R-1234ze(E)	1	A2L	2.78	10.49	3.77	109.4	3.537

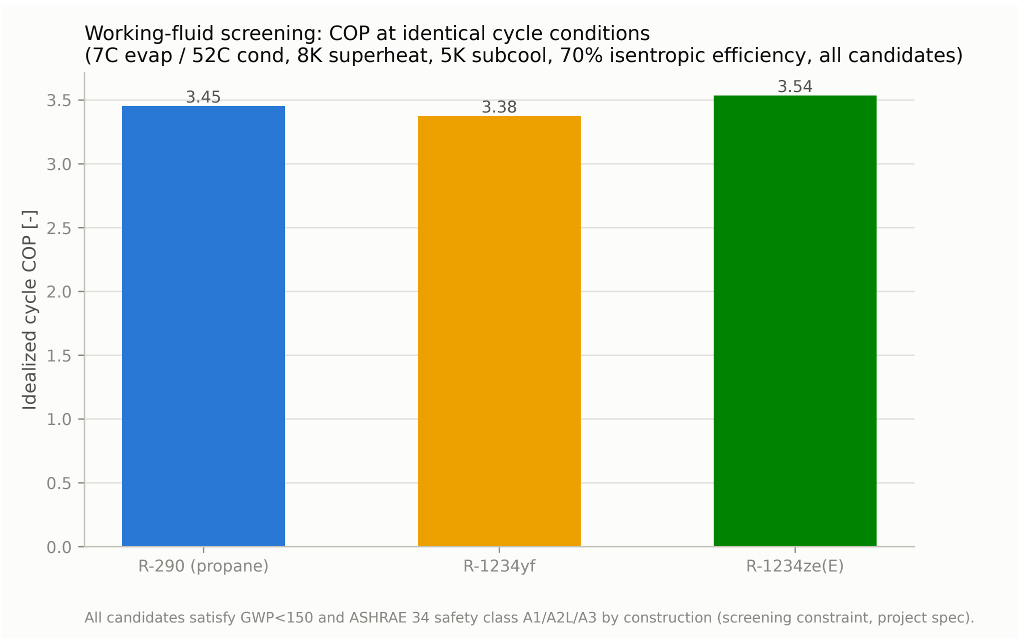


Figure 2.1. Idealized-cycle COP by candidate refrigerant, identical cycle conditions.

2.4.1 Interpreting the Result: R-1234ze(E)'s Higher COP

R-1234ze(E) shows the *highest* idealized COP of the three candidates (3.54 vs. R-290's 3.45, a ≈2.4 % margin) — yet R-290 is the refrigerant carried forward through the rest of this report. This is not an inconsistency; it reflects a second engineering axis the COP number alone does not capture: **volumetric refrigerating effect** (refrigerating capacity per unit of compressor swept volume). R-290 delivers ≈3,270 kJ/m^3 against R-1234ze(E)'s ≈1,880 kJ/m^3 — R-290 moves roughly 1.7× more cooling duty through the same size compressor. This traces directly back to R-1234ze(E)'s lower suction pressure at this evaporating temperature (2.78 bar(a) vs. R-290's 5.84 bar(a)): a lower-density suction gas needs proportionally more swept volume to move the same mass flow. R-1234yf sits between the other two on suction pressure (3.98 bar(a)) but trails both on COP (3.38), so it does not improve on R-290 on either axis.

Practically, this means an R-1234ze(E)-based system sized for the same 150 kW duty would need a meaningfully larger (and more expensive) compressor than the Bitzer 4FEP-35Z pair selected in Chapter 10, for only a marginal COP gain. Both A2L candidates would also forgo the mature hydrocarbon-refrigerant compressor, oil-separator (Chapter 11), and receiver (Chapter 13) ecosystem this report's Bitzer selections already draw on — a real switching cost that a ≈2 % idealized COP gain does not obviously outweigh.

2.5. Selection: R-290 (Propane)

R-290 is selected as the project's working fluid on the combination of:

1. Satisfies both hard constraints (GWP 3, ASHRAE class A3) with large margin against the GWP<150 threshold.

2. Competitive idealized COP (3.45, within 2.4 % of the best-performing candidate) *and* the best volumetric refrigerating effect of the three candidates, favouring a compact, cost-effective compressor selection (§2.4.1).

3. A mature, single-vendor hydrocarbon-refrigerant equipment ecosystem was available for every pressure-vessel and accessory this design needs: the Bitzer 4FEP-35Z compressor (Chapter 10), OA-series oil separator (Chapter 11), and P-series liquid receiver (Chapter 13) are all explicitly rated for A3 hydrocarbon refrigerants from a single manufacturer — simplifying procurement and spare-parts management, the same rationale already applied to the BPHE vendor selections (Chapters 3, 4).

4. R-290's A3 flammability is the trade-off accepted for the above: it drives the double-wall BPHE construction (Chapters 3, 4), the oil separator's float-valve (not solenoid) oil return (§11.4), and the liquid receiver's charge-minimisation objective (§16.4) — each already flagged and addressed in its own chapter rather than treated as a blanket risk.

2.6. Limitations and Open Items

1. **Fixed 70 % isentropic efficiency is a simplifying assumption**, applied identically across all three candidates specifically so the comparison isolates refrigerant thermodynamics rather than a compressor efficiency curve (§2.3). Real compressor efficiency varies by refrigerant, displacement, and operating point; R-290's real selected-compressor performance (Chapter 10) should be treated as the authoritative number for this project, not the idealized value here.

2. **No charge/leak-risk quantification is performed for the A2L or A3 candidates in this chapter** beyond the safety-class screen itself (§2.1); the detailed charge estimate (Chapter 16) and A3-specific mitigations are developed only for the selected fluid (R-290) in their respective downstream chapters.

3. **GWP values are approximate**, drawn from commonly cited AR5 (IPCC 2013), 100-year integrated time horizon figures per ASHRAE Standard 34 GWP tables; exact values vary slightly by source/assessment report vintage and should be confirmed against the specific regulatory basis (e.g., AIM Act Technology Transitions Table) before final compliance sign-off.

4. **R-1234yf and R-1234ze(E) are evaluated as pure single-component fluids** via CoolProp; several commercial A2L blends (e.g., R-454B, R-455A) exist but were not screened here, either because they fail the GWP<150 constraint outright (R-454B) or were judged unlikely to outperform the two pure HFOs already included on this comparison's own metrics.

2.7. References

1. U.S. EPA AIM Act Final Rule, Technology Transitions §40 CFR Part 84 (2023) — GWP limit basis for new chillers.
2. ASHRAE Standard 34, *Designation and Safety Classification of Refrigerants* — safety class (A1/A2L/A2/A3/B1/B2/B2L/B3) and GWP reference tables.
3. IPCC, *Fifth Assessment Report (AR5)*, Working Group I — 100-year global warming potential basis.

Chapter 3

Evaporator Brazed-Plate Heat Exchanger

Part A — Thermal & Mechanical Sizing and Model Selection

3.1. Design Basis

The evaporator duty and water-side mass flow rate are summarised below. The water flow rate follows directly from the specified capacity and temperature drop:

$$\dot{m} = \frac{Q}{c_p\,\Delta T} = \frac{150{,}000\ \mathrm{W}}{4{,}186\ \mathrm{J\,kg^{-1}K^{-1}} \times 6\ \mathrm{K}} = 5.97\ \mathrm{kg\,s^{-1}} \approx 21.5\ \mathrm{m^3\,h^{-1}} \approx 95\ \mathrm{US\,GPM}$$

Table 3.1. Evaporator design basis.

Parameter	Value
Application	Evaporator — chilled water generation
Cooling capacity, Q	150 kW
Refrigerant	Propane (R-290), HC, ASHRAE safety class A3
Refrigerant evaporating temperature, T_{sat}	7 °C
Water inlet / outlet temperature	21 °C / 15 °C
Water temperature drop, ΔT	6 °C
Water mass flow rate	5.97 kg/s (≈21.5 $\mathrm{m^3}$/h, ≈95 US GPM)
Assumed overall HTC, U	3,000 $\mathrm{W\,m^{-2}K^{-1}}$
Refrigerant evap. pressure, p_{in}	≈5.9 bar(a) — input to `Evaporator` class

3.2. Thermal Sizing — LMTD Method

Because the refrigerant boils at constant saturation temperature, the refrigerant side is isothermal and the LMTD formula reduces to:

$$\Delta T_{\mathrm{lm}} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1/\Delta T_2)}$$

Table 3.2. Terminal temperature differences.

Terminal	Definition	Value
ΔT_1	$T_{\text{water,in}} - T_{\text{sat}} = 21 - 7$	14 °C
ΔT_2	$T_{\text{water,out}} - T_{\text{sat}} = 15 - 7$	8 °C

$$\Delta T_{\text{lm}} = \frac{14 - 8}{\ln(14/8)} = \frac{6}{0.5596} = 10.72\ ^\circ\text{C}$$

$$A = \frac{Q}{U\,\Delta T_{\text{lm}}} = \frac{150{,}000}{3{,}000 \times 10.72} = 4.66\ \text{m}^2$$

Adding ≈20 % margin for subcooled and superheated zones gives a practical target of approximately 5.5–6.0 m^2.

3.2.1 Reasonableness of $U = 3{,}000\ \text{W}\,\text{m}^{-2}\text{K}^{-1}$

For a BPHE evaporator (water one side, propane boiling on the other), published values of U typically fall in the 2,500–4,500 $\text{W}\,\text{m}^{-2}\text{K}^{-1}$ range. Propane's favourable transport properties (low viscosity, good wetting) push achievable U toward the upper-middle of that range, so 3,000 $\text{W}\,\text{m}^{-2}\text{K}^{-1}$ is a reasonable, slightly conservative preliminary assumption.

3.3. Design Pressure Determination

Although the evaporator operates at ≈5 bar(g), the design pressure is governed by the *standstill condition*: with no flow, refrigerant equalises to the highest ambient temperature the system may reach. Using 55 °C (ASHRAE 15 / EN 378 / IEC 60335-2-40), propane saturation pressure is ≈19.1 bar(a) / 18.1 bar(g). The BPHE design pressure must exceed 18 bar(g); standard ratings of 30–31 bar or 45 bar comfortably satisfy this requirement.

3.4. Heat Exchanger Technology and Safety Considerations

3.4.1 Flammable Refrigerant (A3) and Occupied-Space Considerations

Propane is classified as an A3 (flammable) refrigerant. Where a BPHE separates an A3 refrigerant from chilled water circulating into a data centre, codes such as EN 378, ASHRAE 15, and IEC 60335-2-89 typically require double-wall / leak-safe construction or strict charge limits. A double-wall ("DW") construction is adopted for this design pending final confirmation of the applicable code requirement.

3.5. Candidate Screening — Kelvion Brazed Plate Heat Exchanger Range

The Kelvion **GB-DW series** (45 bar, double-wall) satisfies both the standstill pressure requirement and the A3 safety requirement. The **GBH-DW 500H** is the leading candidate.

3.6. Detailed Comparison — Double-Wall (DW) Series

3.6.1 Estimating Area per Plate

Table 3.3. Estimated plate count using the $A \times B$ method.

Method	Area basis	Plates needed (500H)	Feasible?
$A \times B$	Outer port width × full height, no factor	84	Yes

The 500H requires 70–84 plates for the 4.66–5.6 m^2 area range, within its 100-plate limit.

3.7. Final Selection

Selected model: Kelvion GBH-DW 500H

- 45 bar rating clears the ≈19–20 bar minimum design pressure.
- Double-wall construction appropriate for A3 refrigerant feeding a chilled-water loop into an occupied data centre (Section 2.4.1).
- 84 plates needed for the 5.6 m^2 margined target — within the 100-plate limit.

Part B — Refrigerant-Side Pressure-Drop Model

3.8. Role of the Pressure-Drop Model

The model is implemented in `evaporator.py` and feeds directly into the refrigeration cycle simulation. Accurate refrigerant-side pressure drop is required to correctly determine the compressor suction pressure and hence the overall system COP.

3.9. Source Correlation: Han, Lee & Kim (2003) — Evaporation

3.9.1 Experimental Basis

The correlation covers R410A and R22 evaporating inside BPHEs with chevron angles 20°, 35°, and 45°. Test conditions:

- Mass flux: 13–34 $\mathrm{kg\,m^{-2}s^{-1}}$; evaporation temperature: 5–15 °C.
- Heat flux: 2.5–8.5 $\mathrm{kW\,m^{-2}}$; vapour quality: $x \approx 0.15$–0.95.

Test plate geometry: $L_w = 115\,\mathrm{mm}$, $L_v = 476\,\mathrm{mm}$, $D_p = 19.5\,\mathrm{mm}$; 2 refrigerant channels against 3 water channels. Friction-factor correlation accuracy: ±15% (90 % of data).

3.9.2 Geometric Nomenclature

Table 3.4. Geometric symbols used in the evaporation correlation.

Symbol	Code var.	Meaning	Typical range
D_h	`D_h`	Hydraulic diameter of a refrigerant channel	3–5 mm
b	(derived)	Mean channel spacing (plate gap)	2–3 mm
ϕ	(constant)	Developed-to-projected corrugation length ratio	1.17
L_w	(in A_{flow})	Horizontal plate width	0.1–0.3 m
L_v	`L`	Vertical flow length between ports	0.3–0.6 m
p_{co}	`Lambda`	Corrugation pitch	5–7 mm
β	`beta`	Chevron angle (from horizontal axis)	20–45°
N_{cp}	`N_cp`	Number of refrigerant channels per pass	design-dep.

3.9.3 Hydraulic Diameter

$$D_h = \frac{4 \times (\text{channel flow area})}{(\text{wetted surface})} = \frac{2b}{\phi}, \qquad \phi = 1.17 \tag{Eq. 4}$$

3.9.4 Quality and Equivalent Flow Parameters

$$\Delta x = x_{\text{out}} - x_{\text{in}} = \frac{Q_w}{\dot{m}_r\, i_{fg}} \tag{Eq. 7}$$

$$G_{\text{Eq}} = G_c\left[(1-x) + x\left(\frac{\rho_f}{\rho_g}\right)^{1/2}\right], \qquad G_c = \frac{\dot{m}}{N_{cp}\, b\, L_w} \tag{Eq. 11}$$

$$Re_{\text{Eq}} = \frac{G_{\text{Eq}}\, D_h}{\mu_f} \tag{Eq. 20}$$

3.9.5 Two-Phase Friction Factor Correlation

$$f = Ge_3 \cdot Re_{\text{Eq}}^{Ge_4} \tag{Eq. 17}$$

$$Ge_3 = 64{,}710 \times \left(\frac{p_{co}}{D_h}\right)^{-5.27} \times \left(\frac{\pi}{2} - \beta\right)^{-3.03} \tag{Eq. 18}$$

$$Ge_4 = -1.314 \times \left(\frac{p_{co}}{D_h}\right)^{-0.62} \times \left(\frac{\pi}{2} - \beta\right)^{-0.47} \tag{Eq. 19}$$

3.9.6 Frictional Pressure Drop

$$\Delta P_{\mathrm{fr}} = f \times \frac{L_v\, N_{cp}}{D_h} \times \frac{G_{\mathrm{Eq}}^2}{\rho_f} \tag{Eq. 12}$$

Acceleration, static-head, and port losses account for less than 5 % of the total; only the frictional term is modelled.

3.9.7 Validity Ranges

- β: 20°–45°; G: 13–34 kg m^{-2}s^{-1}; x: 0.15–0.95; evaporation temperature: 5–15 °C.
- Fluids tested: R410A and R22 — not R290. Transfer to propane is treated as reasonable by practitioners but has not been independently validated (Open Item 14.2.5).

3.10. Implementation in `evaporator.py`

3.10.1 Two-Zone Structure

The model splits the refrigerant path into a **boiling zone** ($x_{\mathrm{in}} \to 1$, Han two-phase correlation) and a **superheat zone** (saturated vapour → outlet, Blasius single-phase). Zone lengths are apportioned by enthalpy:

$$L_{\mathrm{boiling}} = L \times \frac{h_g - h_{\mathrm{in}}}{h_{\mathrm{out}} - h_{\mathrm{in}}}$$

3.10.2 Boiling-Zone Integration

The local pressure gradient is integrated across 200 quality steps from x_{in} to 1 using `np.trapezoid`, capturing the continuous variation of G_{Eq}, Re_{Eq}, and f with quality.

3.10.3 Superheat-Zone Treatment

A smooth-channel Blasius friction factor is used ($16/Re$ for laminar, $0.316\, Re^{-0.25}$ for turbulent). No reference to N_{cp} is needed in this zone — the per-channel flux G already accounts for channel count via A_{flow}.

3.11. Geometry Parameter Assignment for the GBH-DW Family

3.11.1 Available Reference Material

The Kelvion DW-series product flyer is a generic catalog page; it does not publish β, p_{co}, D_h, b, or L_w for any specific model. All values below are estimated from the Han et al. (2003) paper geometry and typical commercial BPHE ranges.

3.11.2 Chevron Angle (β) and Corrugation Pitch (p_{co})

- $\beta = 30°$ — mid-way inside Han's validated 20–45° range.

- $p_{co} = 5\,\text{mm}$ — near Han's 35° test-plate pitch.

3.11.3 Hydraulic Diameter (D_h) and Sensitivity

$D_h = 4\,\text{mm}$. With $D_h = 4\,\text{mm}$, $p_{co} = 5\,\text{mm}$, $\beta = 30°$:

$$Ge_3 \approx 17{,}370, \qquad Ge_4 \approx -1.12$$

The -5.27 exponent makes p_{co}/D_h by far the most sensitive geometric input.

3.11.4 Channel Spacing (b)

$$b = \frac{D_h\,\phi}{2} = \frac{4\text{ mm} \times 1.17}{2} \approx 2.34\text{ mm}$$

3.11.5 Plate Width and Flow Length

$L_w = 200\,\text{mm}$; $L = 0.5\,\text{m}$ — both mid-range commercial placeholders pending vendor confirmation.

3.11.6 Channel Count (N_{cp}) and Flow Area (A_{flow})

Sizing to keep G_c inside Han's 13–34 $\text{kg}\,\text{m}^{-2}\text{s}^{-1}$ band (with $\dot{m} \approx 0.45\,\text{kg/s}$ and $G_{\text{target}} = 25\,\text{kg}\,\text{m}^{-2}\text{s}^{-1}$):

$$N_{cp} \approx \frac{0.45}{25 \times 4.68 \times 10^{-4}} \approx 38, \qquad A_{\text{flow}} = 38 \times 4.68 \times 10^{-4} \approx 0.0178\text{ m}^2$$

Back-check: $G_c = 0.45/0.0178 \approx 25.3\,\text{kg}\,\text{m}^{-2}\text{s}^{-1}$ ✓

3.12. Final Assigned Parameter Table

Table 3.5. Evaporator geometry block values, with derivation status.

Parameter	Value	SI	Basis	Status
D_h	4 mm	0.004 m	Midpoint of 3–5 mm range; sensitivity-checked	**ASSUMED**
$\Lambda\,(p_{co})$	5 mm	0.005 m	Near Han's 35° pitch	**ASSUMED**
β	30°	—	Mid-range 20–45°	**ASSUMED**
b	2.34 mm	0.00234 m	$= D_h\phi/2$ (Eq. 4)	**DERIVED**
L_w	200 mm	0.2 m	Mid-range commercial placeholder	**ASSUMED**
N_{cp}	≈38	38	Sized for $G_c \in [13, 34]\,\text{kg}\,\text{m}^{-2}\text{s}^{-1}$	**ASSUMED***
A_{flow}	$0.0178\,\text{m}^2$	$0.0178\,\text{m}^2$	$= N_{cp} \times b \times L_w$	**DERIVED**
$L\,(L_v)$	0.5 m	0.5 m	Placeholder in 0.3–0.6 m range	**ASSUMED**

*Recheck against confirmed $\dot{m}_r$ from compressor module.

Evaporation geometry factors: $Ge_3 \approx 17{,}370$, $Ge_4 \approx -1.12$ (Section 3.11.3).

3.13. Water-Side Pressure Drop: Martin (1996) Correlation

3.13.1 Why a Second Correlation Is Needed

The Han et al. (2003) correlation used in §3.11.3–3.11.4 is a *two-phase* friction factor formulated around the equivalent mass flux G_{Eq}, which blends liquid and vapour density across vapour quality. The chilled-water stream never changes phase, so G_{Eq} is undefined there and Han's correlation cannot be applied. The single-phase chevron friction correlation of **Martin (1996)** — already adopted for the economizer's two liquid streams (§5.7) — is used instead. The correlation, its Reynolds-dependent sub-factors, and the resulting Fanning-form pressure-drop equation are as given in Eqs. (Martin 1)–(Martin 3); they are not repeated here.

3.13.2 Shared Channel Geometry

The water side is assumed to share the same corrugation family (D_h, β, p_{co}, and hence b) as the refrigerant side of this plate (§3.11.3 –3.11.4), since both streams flow through the same brazed-plate pack. The water-side channel count $N_{cp,\mathrm{water}} = 24$ and plate width $L_w = 200\,\mathrm{mm}$ match the values used for the economizer's water stream, giving $A_{\mathrm{flow,water}} = N_{cp,\mathrm{water}}\, b\, L_w \approx 0.0112\,\mathrm{m}^2$.

3.13.3 Implementation

`calc_water_pressure_drop()` in `evaporator.py` calls the same `_martin_dp()` helper introduced for the economizer, evaluated at the mean chilled-water temperature (18 °C, i.e. the average of the 21/15 °C in/out design values) with $\dot{m}_{\mathrm{water}}$ from the water-side energy balance of §2.9. Because the stream is single-phase throughout, there is no zone splitting: the full plate length L_v is treated as one pass, exactly as in the economizer model.

Table 3.6. Evaporator water-side pressure drop (Martin, 1996), `evaporator.py` design point.

Quantity	Value
Water mass flow, $\dot{m}_{\mathrm{water}}$	$5.973\,\mathrm{kg/s}$
Flow area, $A_{\mathrm{flow,water}}$	$0.0112\,\mathrm{m}^2$
Channel mass flux, G	$531.8\,\mathrm{kg\,m^{-2}s^{-1}}$
Channel velocity, G/ρ	$0.53\,\mathrm{m/s}$
Reynolds number, Re	2,021
Friction factor, f (Eq. Martin 1)	0.1093
Pressure drop, $\Delta P_{\mathrm{water}}$ (Eq. Martin 3)	**7.73 kPa**

At $Re \approx 2{,}021$ the water side sits right at the laminar/turbulent transition used to switch between Eqs. (Martin 2a) and (Martin 2b); this is consistent with the comparable water-side Reynolds number found for the economizer (§5.7). A 7.73 kPa water-side loss is modest relative to the overall chilled-water loop and plays into chilled-water pump sizing alongside the piping diameters sized in Chapter 12.

Chapter 4

Condenser Brazed-Plate Heat Exchanger

Part A — Thermal & Mechanical Sizing and Model Selection

4.1. Design Basis

The condenser must reject both the refrigerating effect and the compressor work:

$$Q_{\text{cond}} = Q_{\text{evap}} + W_{\text{comp}} \approx 150\ \text{kW} + W_{\text{comp}}$$

Targeting COP $\geq$ 3.5: $W_{\text{comp}} \leq 42.9\,\text{kW}$, giving $Q_{\text{cond}} \lesssim 193\,\text{kW}$. A preliminary value of $\mathbf{Q}_{\text{cond}} = \mathbf{190\,kW}$ is adopted here, to be confirmed from the compressor module.

Glycol temperatures follow from the dry-cooler design point: at 35 °C outdoor with a 5 K approach, glycol leaves the dry cooler at 40 °C. A 6 K rise through the condenser gives glycol leaving at 46 °C. Condensing temperature is set to 52 °C, placing a 6 K minimum approach on the glycol-outlet side.

4.2. Thermal Sizing — LMTD Method

4.2.1 Three-Zone Temperature Profile

Unlike the evaporator, the condenser refrigerant passes through three distinct zones:

1. **Desuperheating** — superheated vapour cools from T_{in} to T_{sat} (single-phase).
2. **Condensing** — refrigerant condenses at constant T_{sat} (two-phase).
3. **Subcooling** — saturated liquid cools from T_{sat} to T_{out} (single-phase).

CoolProp-evaluated enthalpies at $p_{\text{in}} = 17.89\,\text{bar(a)}$:

The condensing zone dominates (81.5 %), justifying the single overall LMTD approximation below.

Table 4.1. Condenser design basis.

Parameter	Value
Application	Condenser — R-290 to 30 % propylene-glycol heat rejection
Condensing duty, Q_{cond}	190 kW (preliminary; to be confirmed from compressor module)
Refrigerant	Propane (R-290), HC, ASHRAE safety class A3
Condensing temperature, T_{sat}	52 °C
Condensing pressure, p_{in}	17.89 bar(a)
Compressor discharge temperature, T_{in}	≈72 °C ($T_{\text{sat}} + 20$ K)
Design subcooling	5 K (Section 4.2.3)
Refrigerant outlet temperature, T_{out}	47 °C
Secondary fluid	30 % propylene glycol, `INCOMP::MPG[0.30]`
Glycol inlet temp. (from dry cooler), $T_{\text{gl,in}}$	40 °C
Glycol outlet temp. (to dry cooler), $T_{\text{gl,out}}$	46 °C
Glycol mass flow rate	8.08 kg/s (≈28.8 m^3/h)
Glycol specific heat, $c_{p,\text{gl}}$	3918 J $kg^{-1}K^{-1}$ (CoolProp at 43 °C)
Assumed overall HTC, U	2,500 W $m^{-2}K^{-1}$ (refrigerant-glycol BPHE)

Table 4.2. Enthalpy states and zone heat fractions.

State	Temperature	Enthalpy	Zone fraction
Compressor discharge, h_{in}	72 °C	671.0 kJ/kg	—
Saturated vapour, h_g	52 °C	623.0 kJ/kg	Desuperheating: 14.0 %
Saturated liquid, h_f	52 °C	342.9 kJ/kg	Condensing: 81.5 %
Subcooled outlet, h_{out}	47 °C	327.6 kJ/kg	Subcooling: 4.5 %

4.2.2 Counter-Flow LMTD Approximation

$$\Delta T_{\text{lm}} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1/\Delta T_2)}$$

Table 4.3. Terminal temperature differences for single-LMTD approximation.

Terminal	Definition	Value
ΔT_1 (hot end)	$T_{\text{in}} - T_{\text{gl,out}} = 72 - 46$	26 °C
ΔT_2 (cold end)	$T_{\text{out}} - T_{\text{gl,in}} = 47 - 40$	7 °C

$$\Delta T_{\text{lm}} = \frac{26 - 7}{\ln(26/7)} = \frac{19}{1.313} = 14.5\ ^\circ\text{C}$$

$$A = \frac{Q}{U\,\Delta T_{\text{lm}}} = \frac{190{,}000}{2{,}500 \times 14.5} = 5.24\ \text{m}^2$$

Adding 20 % design margin gives a practical target area of approximately **6.3 m^2**.

4.2.3 Subcooling Justification

A design subcooling of 5 K is adopted because: (1) it prevents flash gas at the EEV inlet, which disrupts mass-flow calibration; (2) the project brief neglects liquid-line pressure drop so the full 5 K is delivered to the valve; (3) subcooling increases the refrigerating effect and improves COP by ≈1–2 % for R-290 at this level; and (4) 5 K is a common industry default with diminishing returns beyond this value.

4.2.4 Reasonableness of $U = 2{,}500\ \mathrm{W\,m^{-2}K^{-1}}$

For a refrigerant-to-glycol BPHE condenser, 30 % propylene glycol has higher viscosity and lower thermal conductivity than pure water, suppressing the secondary-side film coefficient. Typical U for refrigerant condensing against glycol falls in the range 2,000–3,500 $\mathrm{W\,m^{-2}K^{-1}}$ (vs. 2,500–4,500 for water). The value of 2,500 $\mathrm{W\,m^{-2}K^{-1}}$ is therefore reasonable and slightly conservative.

4.3. Design Pressure Determination

The condenser sits on the *high side* of the refrigerant circuit. At 52 °C condensing, R-290 pressure is 17.89 bar(a) / 16.87 bar(g) — already higher than the evaporator's operating pressure. The *standstill condition* still governs: at 55 °C, R-290 reaches 19.07 bar(a) / 18.06 bar(g), above the operating pressure. The required design pressure (>18 bar(g)) is identical to the evaporator; 45 bar standard ratings comfortably satisfy it.

4.4. Heat Exchanger Technology and Safety Considerations

4.4.1 Flammable Refrigerant (A3) and Glycol-Loop Considerations

The argument for double-wall construction is at least as strong here as for the evaporator. The glycol loop connects the BPHE to the *outdoor* dry cooler via exposed pipework. A single-wall leak would allow R-290 to migrate into the glycol loop and potentially accumulate near the dry-cooler enclosure. Under EN 378, ASHRAE 15, and IEC 60335-2-89, DW construction is required whenever an A3 refrigerant is separated from a secondary fluid that may carry a leak to a location where ignition sources or personnel may be present. DW construction is therefore adopted for the condenser.

4.5. Candidate Screening — Kelvion Brazed Plate Heat Exchanger Range

Identical screening logic to the evaporator: the Kelvion **GB-DW series** (45 bar, double-wall) is the clear candidate. The **GBH-DW 500H** is selected, providing the additional benefit of **spare-parts commonality** with the evaporator.

4.6. Detailed Comparison — Double-Wall (DW) Series

4.6.1 Estimating Area per Plate

Table 4.4. Estimated plate count using the $A \times B$ method.

Method	Area basis	Plates needed (500H)	Feasible?
$A \times B$	125 × 532 mm = 0.0665 m^2/plate, no factor	79 (bare) / 95 (margined)	Yes

The 95-plate margined estimate sits within the GBH-DW 500H's 100-plate limit.

4.7. Final Selection

Selected model: Kelvion GBH-DW 500H

- 45 bar rating exceeds the 18 bar(g) standstill requirement with large margin.
- DW construction appropriate for A3 refrigerant connected to the glycol/dry-cooler loop (Section 3.4.1).
- 95 plates for the 6.3 m^2 margined target — within the 100-plate model limit.
- Same model as the evaporator — simplified procurement and spare-parts management.

Part B — Refrigerant-Side Pressure-Drop Model

4.8. Role of the Pressure-Drop Model

`calc_pressure_drop()` in `condenser.py` computes the total frictional pressure drop across the three zones and sets `self.p_out = self.p_in - self.delta_p`. This sets the receiver/EEV inlet pressure for cycle closure. The method also provides `Q`, `UA`, `LMTD`, `m_dot_glycol`, and `cp_glycol` for the dry-cooler loop and overall cycle balance.

4.9. Source Correlation: Han, Lee & Kim (2003) — Condensation

4.9.1 Experimental Basis

This is the condensation companion to the evaporation paper used in Chapter 3. Both come from the same laboratory, same test rig, and same three BPHEs (chevron angles 20°, 35°, and 45°); published in *J. Korean Phys. Soc.*, Vol. 43, No. 1, July 2003 (pp. 66–73).

Experimental conditions:

- Fluids: R410A and R22, condensing direction (x: $0.9 \to 0.15$).
- Mass flux: 13–34 $\mathrm{kg\,m^{-2}s^{-1}}$; condensation temperatures: 20–30 °C.
- Heat flux: 4.7–5.3 $\mathrm{kW\,m^{-2}}$.
- Each BPHE: 4 thermal plates + 2 end plates = 5 channels; **2 refrigerant channels** (critical for Section 4.10.4).

Plate geometry: $L_w = 115\,\mathrm{mm}$, $L_v = 476\,\mathrm{mm}$, $D_p = 19.5\,\mathrm{mm}$; friction-factor correlation accuracy: ±15 % (r.m.s. deviation 10 %).

4.9.2 Geometric Nomenclature

Same symbols as Table 2.2 (both papers use the same test BPHEs); see Section 4.11.2 for the condensation-specific assignment.

4.9.3 Hydraulic Diameter

$$D_h = \frac{2b}{\phi}, \qquad \phi = 1.17 \tag{Eq. 1}$$

4.9.4 Quality and Equivalent Flow Parameters

$$\Delta x = x_{\text{in}} - x_{\text{out}} = \frac{Q_w}{\dot{m}_r\, i_{fg}} \tag{Eq. 6}$$

$$G_{\text{Eq}} = G_c\left[(1-x) + x\left(\frac{\rho_f}{\rho_g}\right)^{1/2}\right], \qquad G_c = \frac{\dot{m}}{N_{cp}\, b\, L_w} \tag{Eq. 27, 28}$$

$$Re_{\text{Eq}} = \frac{G_{\text{Eq}}\, D_h}{\mu_f} \tag{Eq. 26}$$

4.9.5 Two-Phase Friction Factor Correlation

$$f = Ge_3 \cdot Re_{\text{Eq}}^{Ge_4} \tag{Eq. 23}$$

$$Ge_3 = 3521.1 \times \left(\frac{p_{co}}{D_h}\right)^{+4.17} \times \left(\frac{\pi}{2} - \beta\right)^{-7.75} \tag{Eq. 24}$$

$$Ge_4 = -1.024 \times \left(\frac{p_{co}}{D_h}\right)^{+0.0925} \times \left(\frac{\pi}{2} - \beta\right)^{-1.3} \tag{Eq. 25}$$

Valid for $Re_{\text{Eq}} \in [300, 4{,}000]$ (explicitly stated by the authors; the evaporation paper does not state this limit).

4.9.6 Frictional Pressure Drop

$$\Delta P_{\text{fr}} = f \times \frac{L_v \times N}{D_h} \times \frac{G_{\text{Eq}}^2}{\rho_f} \tag{Eq. 12}$$

where $N = 2$ is the test-rig calibration constant (see Section 4.10.4).

4.9.7 Validity Ranges

- β: 20°–45°; Re_{Eq}: 300–4,000; G: 13–34 kg m^{-2}s^{-1}; x: 0.15–0.9; T_{cond}: 20–30 °C.
- Fluids: R410A and R22 — not R290. Transfer to propane is treated as reasonable but has not been independently validated.

4.9.8 Key Differences from the Evaporation Correlation

Table 4.5. Coefficient comparison: condensation vs. evaporation (Han et al., 2003).

Term	Condensation	Evaporation	Notable difference
Ge_3 pre-factor	3,521.1	64,710	Condensation ≈18× smaller
p_{co}/D_h exponent in Ge_3	**+4.17**	**−5.27**	**Opposite sign**
$(\pi/2-\beta)$ exponent in Ge_3	−7.75	−3.03	Steeper angle dependence
Ge_4 pre-factor	−1.024	−1.314	Condensation less negative
p_{co}/D_h exponent in Ge_4	+0.0925	−0.62	Opposite sign
Valid Re_{Eq} range	300–4,000	not stated	—
Evaluated at $\beta = 30^\circ$, $p_{co} = 5$ mm, $D_h = 4$ mm:			
Ge_3	6,246	17,361	Condensation ≈2.8× lower
Ge_4	−0.985	−1.120	—

The most significant difference is the **sign reversal on the p_{co}/D_h exponent in Ge_3**: for evaporation, larger p_{co}/D_h *reduces* friction; for condensation, it *increases* it — reflecting the different two-phase flow physics of a growing liquid film versus nucleation-dominated boiling. Net effect at our operating point: condensation friction factors are ≈2.8× lower than evaporation at the same Re_{Eq}.

4.10. Implementation in `condenser.py`

4.10.1 Three-Zone Structure

`calc_pressure_drop()` divides the path into three zones by enthalpy:

$$L_{\mathrm{dsh}} = L \cdot \frac{h_{\mathrm{in}} - h_g}{h_{\mathrm{in}} - h_{\mathrm{out}}}\ (14.0\ \%), \quad L_{\mathrm{cond}} = L \cdot \frac{h_g - h_f}{h_{\mathrm{in}} - h_{\mathrm{out}}}\ (81.5\ \%), \quad L_{\mathrm{sub}} = L \cdot \frac{h_f - h_{\mathrm{out}}}{h_{\mathrm{in}} - h_{\mathrm{out}}}\ (4.5\ \%)$$

4.10.2 Condensing-Zone Integration

The local pressure gradient is integrated across 200 quality steps ($x = 0 \to 1$) using `np.trapezoid`, capturing the continuous variation of G_{Eq}, Re_{Eq}, and f throughout the condensing zone.

4.10.3 Single-Phase Zones (Desuperheating and Subcooling)

Both single-phase zones use Blasius friction ($16/Re$ for $Re < 2000$; $0.079\,Re^{-0.25}$ for turbulent). Fluid properties are evaluated at the average enthalpy of each zone via CoolProp. Pressure drop:

$$\Delta P_{\text{zone}} = 2f\,\frac{G^2}{\rho}\,\frac{L_{\text{zone}}}{D_h}$$

4.10.4 Critical Note: Calibration Constant $N = 2$

In the Han et al. test rig, the BPHE had exactly 2 refrigerant channels ($N = 2$). The correlation was regressed against those measurements, so $N = 2$ in Eq. 12 is a *calibration constant* — it cannot be replaced by the design N_{cp}. The code uses `N_HAN_CAL = 2`, verified against Fig. 5 of the paper (4–7 kPa at $G_c = 34\,\mathrm{kg\,m^{-2}s^{-1}}$). Substituting the design $N_{cp} = 47$ would overpredict ΔP by a factor of 23.5×.

The physical interpretation: in a single-pass BPHE, port-to-port pressure drop is independent of the number of parallel channels when G_c is fixed. The design N_{cp} enters only through $G_c = \dot{m}/(N_{cp}\,b\,L_w)$.

4.11. Geometry Parameter Assignment for the GBH-DW Family

4.11.1 Available Reference Material

Same sources as the evaporator (Han et al. paper + Kelvion flyer). The flyer does not publish model-specific geometry. All values are estimated.

4.11.2 Chevron Angle, Corrugation Pitch, and Hydraulic Diameter

Same placeholders as the evaporator for consistency: $\beta = 30°$, $p_{co} = 5\,\mathrm{mm}$, $D_h = 4\,\mathrm{mm}$. Condensation geometry factors:

$$Ge_3 = 3521.1 \times (1.25)^{4.17} \times (\pi/3)^{-7.75} = 3521.1 \times 2.535 \times 0.699 \approx 6{,}246$$

$$Ge_4 = -1.024 \times (1.25)^{0.0925} \times (\pi/3)^{-1.3} = -1.024 \times 1.021 \times 0.942 \approx -0.985$$

4.11.3 Channel Spacing (b) and Plate Width (L_w)

$b = D_h\phi/2 = 2.34\,\mathrm{mm}$ (derived); $L_w = 200\,\mathrm{mm}$ (placeholder).

4.11.4 Channel Count (N_{cp}) and Flow Area (A_{flow})

With $Q_{\text{cond}} = 190\,\mathrm{kW}$ and $\Delta h = 343.4\,\mathrm{kJ/kg}$:

$$\dot{m}_r = \frac{190{,}000}{343{,}400} \approx 0.553\ \mathrm{kg/s} \qquad N_{cp} \approx \frac{0.553}{25 \times 4.68 \times 10^{-4}} \approx 47 \qquad A_{\text{flow}} \approx 0.0220\ \mathrm{m^2}$$

Back-check: $G_c = 0.553/0.0220 = 25.1\,\mathrm{kg\,m^{-2}s^{-1}}$ ✓. Note $N_{cp} = 47$ for the condenser vs. 38 for the evaporator, reflecting the higher refrigerant mass flow at the condenser duty.

4.12. Final Assigned Parameter Table

Table 4.6. Condenser geometry block values, with derivation status.

Parameter	Value	SI	Basis	Status
D_h	4 mm	0.004 m	Midpoint of 3–5 mm range; same as evaporator	**ASSUMED**
$\Lambda\,(p_{co})$	5 mm	0.005 m	Near Han's 35° pitch; same as evaporator	**ASSUMED**
β	30°	—	Mid-range 20–45°; same as evaporator	**ASSUMED**
b	2.34 mm	0.00234 m	$= D_h\phi/2$ (Eq. 1)	**DERIVED**
L_w	200 mm	0.2 m	Mid-range commercial placeholder	**ASSUMED**
N_{cp}	≈47	47	Sized for $G_c \in [13, 34]$ at $\dot{m}_r = 0.553\,\mathrm{kg/s}$	**ASSUMED***
A_{flow}	$0.0220\,\mathrm{m}^2$	$0.0220\,\mathrm{m}^2$	$= N_{cp} \times b \times L_w$	**DERIVED**
$L\,(L_v)$	0.5 m	0.5 m	Placeholder in 0.3–0.6 m range	**ASSUMED**

*Recheck N_{cp} and A_{flow} once real $\dot{m}_r$ is confirmed from the compressor module.

Condensation geometry factors: $Ge_3 \approx 6{,}246$, $Ge_4 \approx -0.985$ (Section 4.11.2). Compare with evaporator values: $Ge_3 \approx 17{,}361$, $Ge_4 \approx -1.120$.

4.13. Glycol-Side Pressure Drop: Martin (1996) Correlation

4.13.1 Why a Second Correlation Is Needed

As in the evaporator (§3.13), the Han et al. (2003) correlation used for the refrigerant side (§4.10.4) is a two-phase friction factor and does not apply to the glycol stream, which remains single-phase liquid throughout the condenser. The single-phase chevron friction correlation of **Martin (1996)** is used for the glycol side instead, on the same basis as the economizer (§5.7) and the evaporator's water side (§3.13); Eqs. (Martin 1)–(Martin 3) are not repeated here.

4.13.2 Shared Channel Geometry

The glycol side is assumed to share the refrigerant side's corrugation family ($D_h = 4\,\mathrm{mm}$, $\beta = 30°$, $p_{co} = 5\,\mathrm{mm}$, $b = 2.34\,\mathrm{mm}$, §4.11.2–4.10.4′s companions). The glycol-side channel count $N_{cp,\text{glycol}} = 24$ and plate width $L_w = 200\,\mathrm{mm}$ give $A_{\text{flow,glycol}} = N_{cp,\text{glycol}}\, b\, L_w \approx 0.0112\,\mathrm{m}^2$ — the same flow area as the evaporator's water side, since both use the same assumed channel count and geometry family pending vendor confirmation.

4.13.3 Implementation

`calc_glycol_pressure_drop()` in `condenser.py` calls the same `_martin_dp()` helper used by the economizer and the evaporator's water side, evaluated at the mean glycol temperature (43 °C, the average of the corrected 40/46 °C in/out design values — see §4.2.2) with $\dot{m}_{\text{glycol}} = 9.07\,\mathrm{kg/s}$, the dry-cooler-matched loop flow (Chapter 6), not the module's earlier internal demo value. As with the other single-phase streams in this report, there is no zone splitting: the glycol pass is treated as one sensible pass over the full plate length L_v.

Table 4.7. Condenser glycol-side pressure drop (Martin, 1996), corrected design point.

Quantity	**Value**
Glycol mass flow, $\dot{m}_{\mathrm{glycol}}$	9.07 kg/s
Flow area, $A_{\mathrm{flow,glycol}}$	$0.0112\,\mathrm{m}^2$
Channel mass flux, G	$807.5\,\mathrm{kg\,m^{-2}s^{-1}}$
Channel velocity, G/ρ	0.80 m/s
Reynolds number, Re	2,221
Friction factor, f (Eq. Martin 1)	0.1086
Pressure drop, $\Delta P_{\mathrm{glycol}}$ (Eq. Martin 3)	**17.50 kPa**

The glycol-side drop is larger than the evaporator's water-side drop (7.73 kPa, §3.13) despite the identical assumed flow area, because the condenser's glycol mass flow (9.07 kg/s) is over 50 % higher than the evaporator's water flow (5.973 kg/s) and glycol is both denser and more viscous than water at these temperatures — both raise G and, through G^2 in Eq. (Martin 3), the resulting ΔP. The glycol temperature range used here is the one the condenser actually runs at, cross-checked against the dry cooler's design point (§4.2.2) rather than assumed independently — the two must agree, since they are the same loop. It feeds directly into the glycol pump's duty point (Chapter 8) alongside the dry-cooler glycol-side loss (Chapter 6) and the economizer's free-cooling-mode glycol loss (§5.7).

Chapter 5

Free-Cooling Economizer (Waterside Economizer) Plate Heat Exchanger

Part A — Thermal & Mechanical Sizing and Model Selection

5.1. Design Basis and Role

The waterside economizer is a **glycol-to-chilled-water plate heat exchanger** installed *in parallel* with the mechanical chiller. When the outdoor air is cold enough, the dry cooler chills the intermediate 30 % propylene-glycol (PG) loop directly, and this exchanger transfers that "coolth" to the chilled water — cooling it from 21 °C to 15 °C **with the compressor switched off**. This is the free-cooling ("waterside-economizer") mode required by ASHRAE 90.1-2022 §6.5.1, which mandates that the loop be capable of meeting 100 % of the design cooling load when active.

5.1.1 The Defining Difference: Single-Phase on Both Sides

Unlike the evaporator (Chapter 3) and condenser (Chapter 4), the economizer carries **no refrigerant and undergoes no phase change**. Both streams are single-phase liquids — water on one side, 30 % PG on the other. Two consequences follow and they shape the entire chapter:

- **Thermal profile:** both streams are sensibly sloped (no isothermal boiling or condensing plateau), so the counter-flow LMTD uses the true terminal differences on both ends.
- **Pressure drop:** the Han, Lee & Kim (2003) *two-phase* correlation used in the previous two chapters is physically inapplicable. It is replaced by the single-phase chevron-plate friction correlation of **Martin (1996)** (Part B, §5.7), evaluated independently on each stream.

5.1.2 Free-Cooling Temperature Cascade

The economizer must be sized at the *most demanding* free-cooling condition: the highest outdoor air temperature (OAT) at which 100 % free cooling is still required, namely **OAT = 4 °C** (per the project's "full free cooling ≤ 4 °C" threshold). At colder OAT the glycol is colder, the approach is larger, and sizing is easier. The design cascade from ambient air to chilled-water supply is:

$$\underbrace{4\,^\circ\mathrm{C}}_{\text{OAT}} \xrightarrow{+5\,\mathrm{K}\ \text{dry-cooler approach}} \underbrace{9\,^\circ\mathrm{C}}_{\text{glycol supply}} \xrightarrow{\text{economizer}} \underbrace{15\,^\circ\mathrm{C}}_{\text{glycol return}} \quad \Big\| \quad \underbrace{21\,^\circ\mathrm{C}}_{\text{CHW return}} \xrightarrow{\text{economizer}} \underbrace{15\,^\circ\mathrm{C}}_{\text{CHW supply}}$$

The 5 K dry-cooler approach is consistent with the value adopted for chiller-mode heat rejection (35 °C air → 40 °C glycol) in Chapter 4. A 6 K glycol rise is chosen to mirror the 6 K chilled-water span, giving a balanced, symmetric exchanger.

Table 5.1. Economizer free-cooling design basis (100 % load, OAT = 4 °C).

Parameter	Value
Application	Waterside economizer — 30 % PG to chilled water, free cooling
Design duty, Q	150 kW (= 100 % design load; ASHRAE 90.1 §6.5.1)
Chilled water inlet / outlet	21 °C / 15 °C
Chilled-water mass flow	5.97 kg/s (≈21.5 m^3/h) — shared with evaporator loop
Glycol (from dry cooler) inlet / outlet	9 °C / 15 °C
Glycol mass flow	6.52 kg/s (≈22.8 m^3/h)
Secondary fluid	30 % propylene glycol, `INCOMP::MPG[0.30]`
Water c_p / glycol c_p	4186 / 3835 $\mathrm{J\,kg^{-1}K^{-1}}$ (CoolProp)
Design OAT (full free cooling)	4 °C DB; dry-cooler approach 5 K
Assumed overall HTC, U	3,500 $\mathrm{W\,m^{-2}K^{-1}}$ (glycol/water BPHE)

5.2. Thermal Sizing — LMTD Method

Both streams are single-phase and counter-flow, so:

$$\Delta T_{\text{lm}} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1/\Delta T_2)}$$

Table 5.2. Terminal temperature differences (both ends balanced by design).

Terminal	Definition	Value
ΔT_1 (hot end)	$T_{\text{w,in}} - T_{\text{gl,out}} = 21 - 15$	6 K
ΔT_2 (cold end)	$T_{\text{w,out}} - T_{\text{gl,in}} = 15 - 9$	6 K

With both terminals equal, the LMTD reduces to its limiting value $\Delta T_{\text{lm}} = 6.0\,\text{K}$:

$$UA = \frac{Q}{\Delta T_{\text{lm}}} = \frac{150{,}000}{6.0} = 25.0\ \text{kW K}^{-1}, \qquad A = \frac{Q}{U\,\Delta T_{\text{lm}}} = \frac{150{,}000}{3{,}500 \times 6.0} = 7.14\ \text{m}^2$$

Adding ≈20 % margin gives a practical target of approximately **8.6 $\mathbf{m^2}$**. This is the *largest* of the three plant heat exchangers (cf. 4.66 m^2 evaporator, 5.24 m^2 condenser) — the direct consequence of the small 6 K approach, which is the characteristic signature of a waterside economizer.

5.2.1 Reasonableness of $U = 3{,}500\ \mathrm{W\,m^{-2}K^{-1}}$

For a single-phase liquid-to-liquid BPHE, published overall coefficients fall in the 3,000–6,000 $\mathrm{W\,m^{-2}K^{-1}}$ range. The glycol side (30 % PG, higher viscosity and lower conductivity than water) suppresses its film coefficient, so a value at the lower-middle of the band is appropriate. At the design channel velocities of ≈0.43–0.45 m s^{-1} (Part B), $U = 3{,}500\,\mathrm{W\,m^{-2}K^{-1}}$ is a reasonable, slightly conservative preliminary assumption.

5.3. Design Pressure Determination

This is the one component in the plant that **never contacts refrigerant**. Both sides carry the low-pressure glycol/water loop, which is a pumped circuit operating at only a few bar. There is no R-290 standstill pressure to contain and no A3 flammable inventory to separate. The governing design pressure is therefore set by the hydraulic loop and pump shut-off head (typically ≤10 bar), *not* by the 18–20 bar(g) refrigerant standstill that dictated the evaporator and condenser. A standard **16 bar** plate-heat-exchanger rating is fully sufficient.

5.4. Heat Exchanger Technology and Selection

5.4.1 Double-Wall Construction is NOT Required Here

The evaporator and condenser adopted 45 bar *double-wall* (DW) Kelvion plates specifically because an A3 refrigerant (R-290) is separated from a secondary fluid. The economizer separates *glycol from water* — two benign secondary fluids, neither flammable. None of the EN 378 / ASHRAE 15 / IEC 60335 double-wall triggers apply. The economizer can therefore use a cheaper, **single-wall, standard-pressure** plate heat exchanger. This is a genuine cost and availability advantage and is the correct engineering call for this duty.

5.4.2 Why a Larger Frame Than the 500H

Two drivers push the economizer to a larger frame than the DW 500H used elsewhere:

1. **Area.** The margined target (≈8.6 m^2) exceeds what a single DW 500H can deliver: at ≈0.0665 m^2 per plate and a 100-plate limit its ceiling is ≈6.65 m^2. Meeting 8.6 m^2 would require two 500H units in parallel.

2. **Flow / ports.** The economizer passes the full chilled-water flow (21.5 m^3/h) *plus* a comparable glycol flow (22.8 m^3/h) — roughly double the per-unit throughput of the refrigerant-side exchangers. The 500H-class G1$\frac{1}{4}$/G2$\frac{1}{2}$ ports would run at excessive port velocity; a larger frame with G3 ports is needed.

Selected: a single-wall Kelvion GB-series plate HX on a larger (≈0.7 m tall, G3-port) frame, using the same chevron corrugation family as the evaporator/condenser (so plate technology and correlations carry over). A representative larger plate ($L_w = 0.25$ m, $L_v = 0.70$ m, ≈0.175 m^2/plate outer footprint) meets the 8.6 m^2 margined area in ≈**49 plates** — comfortably within a single frame. (If strict spare-parts commonality is preferred over cost, two DW 500H units in parallel are the fallback.)

5.5. Full vs. Partial Economizer Operation

The economizer is sized for *full* free cooling at OAT $\leq 4\,^\circ$C, but operates in three regimes set by the ASHRAE 90.1 thresholds:

- **OAT** $>$ **10 °C — mechanical cooling.** Glycol from the dry cooler is too warm to help; the economizer is bypassed and the chiller carries the full load.
- 4 $<$ **OAT** $\leq$ **10 °C — integrated (partial) free cooling.** The dry cooler delivers glycol between ≈9 and 15 °C. The economizer pre-cools the returning chilled water and the chiller trims the remainder, reducing compressor load and lift.
- **OAT** $\leq$ **4 °C — full free cooling.** Glycol reaches $\leq 9\,^\circ$C, the economizer carries the entire 150 kW, and the compressor is off. This is the sizing point of this chapter.

For the Champaign, IL (TMY3, ASHRAE climate zone 5A) location, a large fraction of the 8,760 annual hours sit below the 10 °C activation threshold, so the economizer contributes a substantial share of annual cooling and materially improves the facility PUE (evaluated in the seasonal-performance section of the main report).

5.5.1 Modelling the Partial Band: Rating, Not Sizing

The partial band cannot be evaluated with the LMTD method used to size this exchanger. Sizing is handed a duty and all four terminal temperatures and back-solves the UA required — it answers *"how big must it be?"* at one condition. In the partial band the hardware is already fixed and *both* outlet temperatures are unknown, so LMTD has nothing to stand on. The question is inverted: *"what does the exchanger we already bought deliver at 7 °C ambient?"* That is a rating problem, and it is solved with ε–NTU (`economizer.rate_economizer`):

$$\mathrm{NTU} = \frac{UA}{C_{\min}}, \qquad C_r = \frac{C_{\min}}{C_{\max}}, \qquad \dot{Q} = \varepsilon\, C_{\min}\,(T_{w,\mathrm{in}} - T_{g,\mathrm{in}})$$

This exchanger is deliberately *balanced* — the 6 K glycol rise mirrors the 6 K chilled-water span — so $C_{\mathrm{water}} \approx C_{\mathrm{glycol}}$ and $C_r \to 1$, where the general counter-flow effectiveness expression goes 0/0 and the correct limit form is required:

$$\varepsilon\Big|_{C_r=1} = \frac{\mathrm{NTU}}{1+\mathrm{NTU}}$$

With the installed $UA = 25.0\,\mathrm{kW/K}$ and $C_{\min} = 25.0\,\mathrm{kW/K}$, NTU $= 1.0$ and $\varepsilon = 0.50$ across the whole band (both are fixed hardware properties — only the inlet temperatures move).

> **Cross-check.** At OAT $= 4\,^\circ$C the glycol arrives at 9 °C and the rating model returns $\dot{Q} = 0.50 \times 25.0 \times (21 - 9) = \mathbf{150.0}\,\mathrm{kW}$ — *exactly* the duty the LMTD model was sized for. The two independent methods agree at their shared point, which both validates the rating model and guarantees the partial band joins full free cooling with no discontinuity at the 4 °C threshold.

The glycol supply is taken as OAT + 5 K throughout the band, this chapter's own dry-cooler approach basis. The economizer's share then falls linearly from 100 % of the load at 4 °C to **50 %**

at the 10 °C activation threshold, with the chiller trimming the remainder against a very low lift ($t_c \leq 20\,°\mathrm{C}$).

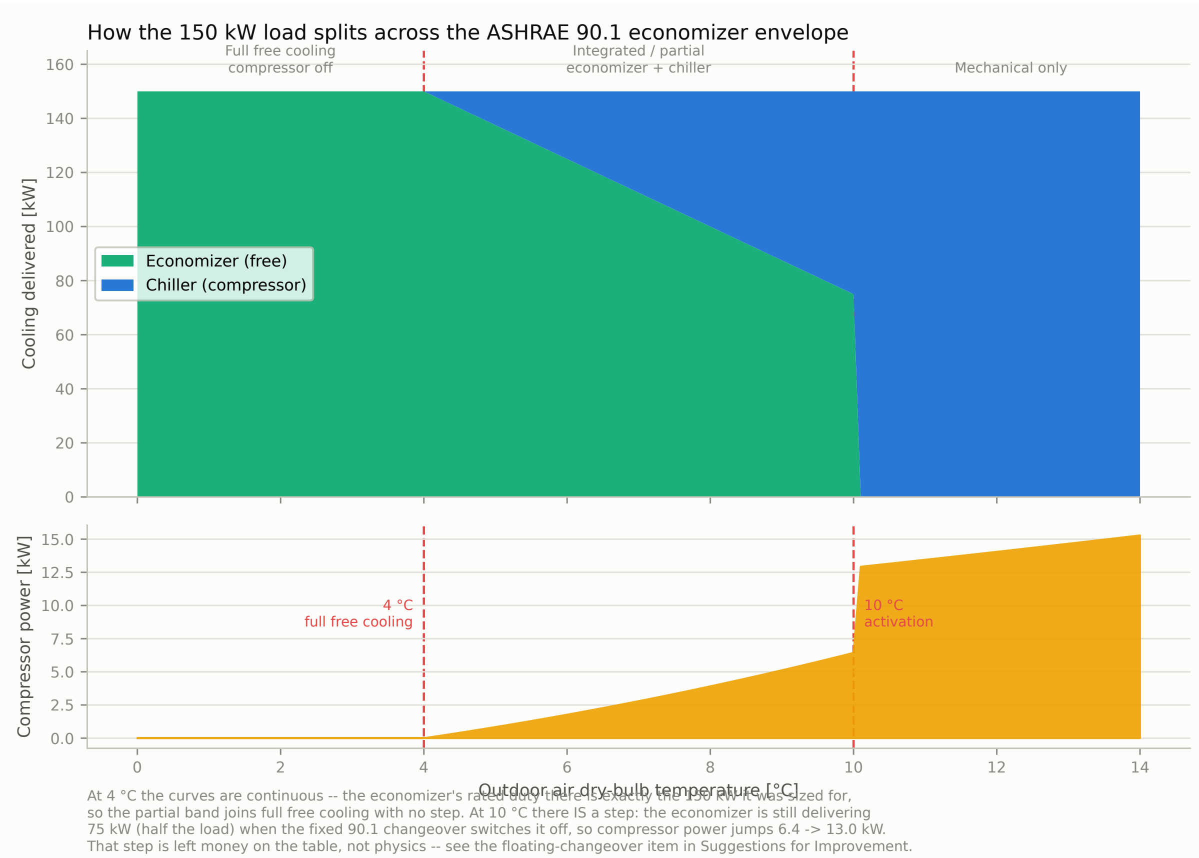


Figure 5.1. Load split and compressor power across the full ASHRAE 90.1 economizer envelope. The 4 °C boundary is continuous by construction; the step at 10 °C is the fixed changeover switching off an economizer that is still delivering half the load (§20.4).

Simplifications, stated. Two are worth naming. First, the glycol loop is treated as running at its free-cooling flow through the band, carrying both the economizer duty and the (small) condenser rejection to a single dry cooler; the loop's hot return follows from an energy balance rather than a resolved two-branch hydraulic model. Second, at the cold end of the band the chiller is asked for very little — about 6 kW at 4.5 °C, roughly 4 % of drive speed — which no VFD holds; the machine **cycles** there rather than modulating. Ideal cycling and ideal modulation give the same average power ($\dot{Q}/\mathrm{COP}$), so the reported energy stands, but real cycling losses are not modelled. They are bounded by the few kW of compressor duty involved (679 h/yr, 7.8 % of the year), so they cannot move the PUE materially.

Part B — Single-Phase Pressure-Drop Model

5.6. Role of the Pressure-Drop Model

`calc_pressure_drops()` in `economizer.py` computes the frictional pressure drop on *each* stream separately. Unlike the refrigerant-side drops of the evaporator and condenser (which feed back into the compressor suction/discharge state and hence the COP), the economizer drops set the **pump heads** of the chilled-water and glycol loops. Keeping them low is central to free cooling: any pump energy spent circulating the loops erodes the very saving that switching the compressor off is meant to deliver.

5.7. Source Correlation: Martin (1996) — Single-Phase Chevron Friction

5.7.1 Why Not Han, Lee & Kim (2003)?

The Han correlations are *two-phase* friction factors, regressed against evaporating / condensing refrigerant and formulated in terms of an equivalent mass flux G_{Eq} that blends liquid and vapour densities across vapour quality. With no vapour present in either economizer stream, G_{Eq} collapses and the correlation is out of its domain. A single-phase correlation is required.

5.7.2 The Martin (1996) Generalised Lévêque Correlation

Martin's correlation (H. Martin, "A theoretical approach to predict the performance of chevron-type plate heat exchangers," *Chem. Eng. Process.* 35 (1996) 301–310; also VDI Heat Atlas, Sec. N6) is the standard single-phase friction model for chevron plates. The Fanning-type friction factor f is:

$$\frac{1}{\sqrt{f}} = \frac{\cos\varphi}{\sqrt{0.045\tan\varphi + 0.09\sin\varphi + f_0/\cos\varphi}} + \frac{1-\cos\varphi}{\sqrt{3.8\,f_1}} \qquad \text{(Martin 1)}$$

with the smooth- and corrugated-channel sub-factors evaluated per Reynolds range:

$$Re < 2000: \quad f_0 = \frac{16}{Re}, \qquad f_1 = \frac{149}{Re} + 0.9625 \qquad \text{(Martin 2a)}$$

$$Re \geq 2000: \quad f_0 = \left(1.56\ln Re - 3.0\right)^{-2}, \qquad f_1 = \frac{9.75}{Re^{0.289}} \qquad \text{(Martin 2b)}$$

The frictional pressure drop then uses the same Fanning form already adopted for the single-phase zones of the evaporator/condenser:

$$\Delta P = 2\,f\,\frac{L_v}{D_h}\,\frac{G^2}{\rho}, \qquad G = \frac{\dot{m}}{N_{cp}\,b\,L_w} \qquad \text{(Martin 3)}$$

5.7.3 Chevron-Angle Convention

φ is the chevron angle measured from the *longitudinal (flow) axis*, so that a larger angle yields a harder plate and more friction. This is the same sense in which β enters Han's two-phase correlation (friction increases with β), so the project value $\beta = 30°$ is used **directly** as φ, with no $90° - \beta$ conversion. The code flags this on one line so it can be flipped to $(90° - \beta)$ if a vendor drawing defines the angle from the horizontal.

5.7.4 Validity

Equation (Martin 1) is valid across laminar and turbulent regimes ($Re \sim 1$ to $> 10^4$) and chevron angles up to 80°, and — being single-phase — is fluid-agnostic (no R-290-transfer caveat, unlike Han). At the design point the streams sit at $Re \approx 1{,}600$ (water) and ≈ 460 (glycol), well inside the correlation's range.

5.8. Implementation in `economizer.py`

5.8.1 Per-Stream Evaluation

The private helper `_martin_dp(m_dot, A_flow, fluid, T_avg)` evaluates Eqs. (Martin 1)–(Martin 3) for one stream: it pulls ρ and μ from CoolProp at the stream's mean temperature, forms the per-channel mass flux $G = \dot{m}/A_{\text{flow}}$ with $A_{\text{flow}} = N_{cp}\, b\, L_w$, computes Re, selects the laminar or turbulent branch, and returns ΔP. `calc_pressure_drops()` calls it twice — once for water, once for glycol — so each side gets its own G, Re, f and ΔP.

5.8.2 No Zone Splitting

Because neither stream changes phase, there is no boiling/condensing/desuperheating zone decomposition: each stream is a single sensible pass over the full plate length L_v. This makes the economizer model markedly simpler than the three-zone condenser model.

5.9. Geometry Parameter Assignment

The corrugation family (D_h, β, p_{co}, b) is inherited from the evaporator/condenser for consistency; the frame dimensions (L_w, L_v) and channel counts are scaled up for the larger economizer duty and its higher flows. Channel counts were chosen to land the channel velocity near $0.45\,\text{m}\,\text{s}^{-1}$ — high enough to support $U = 3{,}500\,\text{W}\,\text{m}^{-2}\text{K}^{-1}$, low enough to keep ΔP modest.

Table 5.3. Economizer geometry block values, with derivation status.

Parameter	Value	SI	Basis	Status
D_h	4 mm	0.004 m	Same corrugation family as evap/cond	**ASSUMED**
$\Lambda\,(p_{co})$	5 mm	0.005 m	Same as evap/cond	**ASSUMED**
$\beta\,(\varphi)$	30°	—	Used directly in Martin (§5.7)	**ASSUMED**
b	2.34 mm	0.00234 m	$= D_h\phi/2$, $\phi = 1.17$	**DERIVED**
L_w	250 mm	0.25 m	Larger frame (economizer flows)	**ASSUMED**
$L\,(L_v)$	700 mm	0.70 m	Larger frame	**ASSUMED**
N_{cp} (per side)	24	24	Sized for $v \approx 0.45\,\text{m}\,\text{s}^{-1}$	**ASSUMED**
A_{flow} (per side)	$0.0140\,\text{m}^2$	$0.0140\,\text{m}^2$	$= N_{cp}\, b\, L_w$	**DERIVED**

5.10. Results at the Free-Cooling Design Point

Table 5.4. Economizer computed results (`economizer.py`, design point).

Quantity	**Stream / basis**	**Water**	**Glycol (30 % PG)**
Mass flow $\dot{m}$	from Q, $\Delta T = 6\,\mathrm{K}$	5.97 kg/s	6.52 kg/s
Volume flow	at stream density	$21.5\,\mathrm{m^3/h}$	$22.8\,\mathrm{m^3/h}$
Channel mass flux G	$\dot{m}/A_{\text{flow}}$	$425\,\mathrm{kg\,m^{-2}s^{-1}}$	$464\,\mathrm{kg\,m^{-2}s^{-1}}$
Channel velocity	G/ρ	0.43 m/s	0.45 m/s
Reynolds number Re	$G\,D_h/\mu$	1,617	456
Friction factor f	Eq. (Martin 1)	0.1052	0.1420
Pressure drop ΔP	Eq. (Martin 3)	**6.67 kPa**	**10.43 kPa**
Duty Q		150.0 kW	
LMTD (counter-flow, balanced)		6.0 K	
UA		$25.0\,\mathrm{kW\,K^{-1}}$	
Area required ($U = 3{,}500$)		$7.14\,\mathrm{m^2}$ ($8.6\,\mathrm{m^2}$ margined)	
Plate count ($\approx 0.175\,\mathrm{m^2}$/plate, $\times 1.2$)		≈49 plates	

Both pressure drops are single-digit-to-low-teens kPa — deliberately modest, so the loop-pump penalty stays small and the free-cooling COP benefit is preserved. The exchanger meets the 100 % design-load free-cooling requirement of ASHRAE 90.1 §6.5.1 at OAT = 4 °C.

Chapter 6

Dry Cooler (Outdoor Air-Cooled Heat Rejection)

Part A — Thermal Sizing, Vendor Selection, and Fan Operation

6.1. Design Basis and Role

The dry cooler is the final heat-rejection step of the glycol loop: an outdoor, finned-tube, air-cooled heat exchanger with variable-speed axial fans that rejects heat from the 30 % propylene-glycol (PG) loop directly to ambient air. It operates at two qualitatively different duties depending on the mode of the plant:

- **Mechanical mode.** The compressor is running; the dry cooler rejects both the evaporator's refrigerating effect *and* the compressor's shaft work, received from the condenser (Chapter 4) as hot glycol at 46 °C.
- **Free-cooling mode.** The compressor is off; the dry cooler instead supplies cold glycol directly to the economizer (Chapter 5), rejecting only the bare 150 kW IT/cooling load with no compressor-work addition.

These are sized and checked as two *separate* operating points on the *same* physical coil — a single piece of hardware, evaluated twice.

6.1.1 Why a Different Method Than the Plate Heat Exchangers

The evaporator, condenser, and economizer (Chapters 3–5) are all single-pass, counter-flow, chevron-plate heat exchangers, sized by LMTD and evaluated with the Han, Lee & Kim (2003) two-phase or Martin (1996) single-phase chevron-plate friction correlations. None of that carries over to the dry cooler, because it is a fundamentally different exchanger type:

- **Flow arrangement.** Air crosses a finned round-tube bundle *perpendicular* to the glycol flow, over several tube rows — a **cross-flow** arrangement, not counter-current. A single LMTD systematically mis-predicts duty for this geometry.
- **Geometry.** Both plate-friction correlations are regressed against chevron *plates*; neither applies to air flowing over finned *round tubes*.
- **A fan, not a pump, drives the air side**, and it is variable-speed — a new degree of freedom (and a new energy term for the facility PUE) that the plate exchangers do not have.

This chapter therefore uses the **effectiveness–NTU method** (cross-flow, both fluids unmixed) for the thermal duty, **fan affinity laws** for the variable-speed fan power, and (Part B) a **vendor-anchored scaling method** for the glycol-side pressure drop rather than an independent friction correlation.

6.2. Governing Method: Cross-Flow Effectiveness–NTU

For a cross-flow exchanger with both fluids unmixed, the standard approximation (Incropera, *Fundamentals of Heat and Mass Transfer*) is:

$$\varepsilon = 1 - \exp\left\{\frac{1}{C_r}\, NTU^{0.22}\left[\exp\left(-C_r\, NTU^{0.78}\right) - 1\right]\right\}, \qquad C_r = \frac{C_{\min}}{C_{\max}} \tag{ε-NTU}$$

with $C = \dot{m}\, c_p$ the capacity rate of each stream. **Sizing** (mechanical-mode design point, §6.3) uses Eq. (ε-NTU) *backward*: the required ε is computed from the duty and terminal temperatures, and NTU — which cannot be isolated algebraically from Eq. (ε-NTU) — is obtained by numerical root-finding (`scipy.optimize.brentq`), after which $UA = NTU \cdot C_{\min}$. **Off-design checking** (free-cooling point, §6.4) uses the same equation *forward*: holding UA fixed (scaled for reduced air flow), a second root-finding search finds the air mass flow that reproduces a target duty at new terminal temperatures. Both uses are implemented in `dry_cooler.py` (`size_design_point()` and `predict_off_design()` respectively).

6.3. Mechanical-Mode Design Point

6.3.1 Required Duty

The dry cooler must reject both the IT/cooling load and the compressor's shaft work:

$$Q_{\text{cond}} = Q_{\text{evap}} + W_{\text{comp}} = 150\,\text{kW} + W_{\text{comp}}$$

Targeting the project's minimum COP requirement of 3.5 gives an upper bound on compressor work, $W_{\text{comp}} \le 150/3.5 \approx 42.9\,\text{kW}$, and hence $Q_{\text{cond}} \lesssim 193\,\text{kW}$ — rounded to a **190 kW** preliminary design duty (Chapter 4). **This number is provisional**: it is derived backward from the COP *requirement*, not from a completed compressor model, and must be reconfirmed once the compressor chapter (pending) supplies a real W_{comp} at the actual operating point.

6.3.2 Design-Point Inputs and Sizing Result

Note that the air-side rise of 5.56 K was *not* an initial free design choice: an early iteration of this model assumed a placeholder 10 K rise before a real vendor unit had been selected. Once the Kelvion selection (§6.5) returned an actual air outlet temperature, the model was re-run with the real 5.56 K value, which is what is reported here.

6.4. Free-Cooling Performance Check

Table 6.1. Dry cooler mechanical-mode design basis and `size_design_point()` result.

Parameter	**Value**	**Basis**
Target duty, Q	190 kW	COP-derived (provisional; §above)
Glycol inlet / outlet	46.0 / 40.0 °C	Condenser outlet (Ch. 4)
Ambient design DB	35.0 °C	Project spec
Secondary fluid	30 % propylene glycol	`INCOMP::MPG[0.30]`
Air-side temperature rise, ΔT_{air}	5.56 K	**DERIVED** — from vendor selection, §6.5
Glycol mass flow, $\dot{m}_{\text{gly}}$	9.07 kg/s (140.8 gpm)	Energy balance
Air mass flow, $\dot{m}_{\text{air}}$	38.09 kg/s (122,071 m^3/h)	Energy balance
C_{min} / C_{max}	35.54 / 38.35 kW K^{-1}	Glycol is C_{min}
C_r	0.927	$C_{\text{min}}/C_{\text{max}}$
Required effectiveness, ε	0.545	Eq. (ε-NTU), inverse
NTU	1.331	Root-found (`brentq`)
Required UA	**47.32 kW K^{-1}**	$= NTU \times C_{\text{min}}$

ASHRAE 90.1-2022 §6.5.1 requires the free-cooling path to carry 100 % of the design cooling load at the qualifying outdoor condition. Holding the *same* UA (scaled for reduced air flow via $h_{\text{air}} \sim \dot{m}_{\text{air}}^{0.6}$, a standard Reynolds-power approximation for finned tube banks with air-side resistance dominant), `predict_off_design()` solves for the air flow that delivers the free-cooling design duty:

Table 6.2. Free-cooling check: same physical coil, opposite operating point.

Parameter	**Value**
Target duty, Q	150 kW (bare IT load, no compressor)
Glycol inlet (from economizer)	15 °C
Ambient DB	4 °C (ASHRAE 90.1 full-free-cooling threshold)
Air mass flow required	20.75 kg/s (= 54.5 % of design air flow)
Fan speed required	≈54.5 %
Glycol leaving temperature (model)	10.70 °C

The coil sized for the hot/mechanical duty comfortably meets the cold/free-cooling duty at roughly half fan speed — confirming the ASHRAE 90.1 requirement is satisfied without a separate, dedicated free-cooling coil.

6.4.1 Open Item: Glycol Flow Assumption in Free-Cooling Mode

The model above holds the glycol mass flow *fixed* at the mechanical-mode design value (9.07 kg/s, a fixed-speed glycol pump assumption), so in free-cooling mode the glycol only cools by ≈4.2 K (15 → 10.7 °C) rather than the full 6 K span (15 → 9 °C) assumed in the economizer's own design basis (Chapter 5, §5.1.2), which uses a glycol flow of 6.52 kg/s. **These two chapters currently disagree on the free-cooling glycol flow rate**, and this should be reconciled before the plant design is finalised — either by specifying a variable-speed glycol pump (matching the economizer's 6.52 kg/s target in free-cooling mode and 9.07 kg/s in mechanical mode), or by re-running the dry cooler free-cooling check at the economizer's 6.52 kg/s flow instead of the fixed design value. This is flagged here rather than silently resolved, since the correct answer depends on a glycol-pump control-strategy decision that has not yet been made elsewhere in the report.

6.5. Vendor Selection: Kelvion Select RT

6.5.1 Selection Tool and Inputs

Unlike the evaporator and condenser (selected via Kelvion Select PHE for brazed plate heat exchangers), the dry cooler was selected via **Kelvion Select RT** (formerly "GPC"), Kelvion's configurator for condensers, dry coolers, gas coolers, and air coolers. Using the same vendor family as the plate exchangers keeps the plant on a single manufacturer for procurement and spares commonality.

Table 6.3. Kelvion Select RT capacity-requirement inputs (mechanical-mode design point).

Field	Value entered
Fluid	Propylene Glycol, 30.0 % vol
Capacity (active input)	620 MBH (≈182 kW condenser duty)
Fluid inlet / outlet	107.6 °F / 100.4 °F (= 42.0/38.0 °C)
Calculation mode	Fixed fluid temperatures
Ambient	95.0 °F (= 35.0 °C)
Altitude	699 ft (≈213 m, Champaign, IL elevation)

The glycol temperatures are the binding constraint, and they are not free: the 10 K air-to-refrigerant approach that the COP target requires (§14.1.1) allots only 3 K to the dry cooler and 4 K to the glycol rise, which fixes 42/38 °C against 35 °C air. **A 3 K approach on an air-cooled coil is demanding**, and it — not the headline duty — is what drives the size of the machine selected below.

6.5.2 Selection Notes

Two points in the selection process are worth recording for reproducibility. First, the Select RT capacity screen has two mutually exclusive driving inputs ("Capacity" and "Volume flow rate"); an initial run left the flow-rate radio active, silently returning undersized results (≈172–181 kW instead of the 190 kW target). Re-selecting the Capacity radio and re-running corrected this. Second, the tool defaults to Ethylene Glycol; this was corrected to Propylene Glycol (30 % vol) to match the project's secondary fluid.

6.5.3 Why Headline Capacity Is the Wrong Screening Criterion

Before the candidate table, one point governs how it must be read. **A dry cooler's catalog capacity is meaningless without the condition it is quoted at**, and the condition here is unusually hard.

Consider a representative `ULF` unit rated **213.2 kW** at a comfortable 46/40 °C glycol duty — an 11 K mean air-to-glycol difference against 35 °C air. Re-rate that same coil at this project's 42/38 °C requirement and the driving difference collapses to 7 K, a 36 % reduction. Since duty scales with it, the same physical machine delivers only ≈146 kW — *well short* of the ≈182 kW required, despite a headline rating 17 % above it:

Table 6.4. The same coil, two conditions. Capacity is not a property of the machine alone.

Condition	**Glycol**	**Mean ΔT to 35 °C air**	**Duty**
Comfortable rating	46 → 40 °C	11 K	213.2 kW
This project	42 → 38 °C	7 K (−36 %)	≈145.7 kW

There is no way to buy back the difference with air flow: pushing the glycol outlet to 40 °C — a 5 K approach — would pinch the condenser and undo the 10 K cascade the COP target depends on. The approach is thermodynamically forced by §14.1.1, so **the coil must simply be bigger**.

The condition-independent measure is **surface area**, and that is what the screening below is really about. Expect the selected unit to have a *lower* headline MBH than units quoted at easier conditions while being a substantially larger machine.

6.5.4 Candidate Comparison

Select RT returned a results table of qualifying `ULF` flatbed units at the 620 MBH / 42/38 °C duty. They were screened on the criteria that actually bind — capacity margin at the *real* condition, fan-speed headroom, and sound:

Table 6.5. Candidate screening at the 42/38 °C duty (620 MBH required; European decimal notation converted).

Type	**MBH (margin)**	ΔP	**Fan speed**	L_{pA}	**Note**
ULF-PA106Y4V-096Z110	639.5 (3 %)	4 psi	1100/1100	64 dB(A)	Low ΔP, but **no fan headroom**
ULF-PA107Y4V-091Z100	630.8 (2 %)	4 psi	1000/1100	67 dB(A)	Low ΔP, thin margin, loud
ULF-PA106K4V-091F095	**670.4 (8 %)**	8 psi	**950/1050**	**60 dB(A)**	**Selected**

ULF-PA106K4V-091F095 is selected. At **60 dB(A)** it is the quietest candidate by 4 dB, which matters for a data-centre site where the unit may sit near occupied space. It carries the best capacity margin (8 %) and, decisively, **10 % fan-speed headroom** (950 of 1050 rpm). That headroom was preferred over the lower-ΔP "Z110" variants precisely *because* those run 1100/1100 — flat out at the design point, with no reserve for a hotter-than-design hour, coil fouling, or a failed fan. The penalty accepted in exchange is a higher glycol-side pressure drop (8 psi vs. 4 psi), paid by the glycol pump (§14.4).

6.6. Selected Unit: Kelvion ULF-PA106K4V-091F095

Table 6.6. Selected unit — key specifications as reported by Kelvion Select RT.

Parameter	Value
Model	ULF-PA106K4V-091F095
Capacity	670.36 MBH = **196.5 kW** at glycol 42 / 38 °C, air 35 °C
Surface area	20,977 ft^2 = **1,949 m^2**
Air volume flow	93,782 cfm (=159,337 m^3/h); air outlet 101.7 °F ⇒ $\Delta T_{\text{air}} = 3.72$ K
Fans	6 × 0.910 m EC axial, 950 / 1050 rpm (10 % headroom)
Sound pressure, L_{pA} (33 ft)	60 dB(A)
Glycol-side pressure drop	8 psi = **55.16 kPa**
Coils × sections × circuits	1 × 2 × 29

With this unit the annual simulation runs **clean across all 8,760 hours** — no hour requires more than 100 % fan speed to hold its duty.

> **Open item — full datasheet not yet pulled.** The values above are what the Select RT *results table* reports. A full configured datasheet (dimensions, weight, fin spacing, materials, and above all **fan power**) has not been generated for this unit, which is why the fan power carried through the PUE is a fan-law estimate rather than a vendor figure (§6.13). This is the largest remaining vendor gap in the design and should be closed before ordering.

6.7. Model Validation Against a Vendor Datasheet

The ε-NTU model of §6.3 carries the entire annual simulation — every fan-speed and fan-power figure in this report comes out of it — so it is worth checking against Kelvion's own thermodynamic calculation rather than trusting it on faith.

The check uses a **full configured datasheet for a different unit of the same `ULF` family** (ULF-PA104Y4V-096Z100, rated 213.23 kW at 46/40 °C glycol). That unit is not the one selected — it cannot meet this project's 3 K approach (§6.5.3) — but that is irrelevant here: validating a model needs a complete, self-consistent vendor calculation to compare against, and this is the datasheet for which all streams are reported. Feeding its duty and its real air-side rise (5.56 K) into the model:

Agreement within ±2 % on *both* streams is a strong validation of the cross-flow ε-NTU approach for this application, given that the model uses no fitted constants beyond the standard Incropera correlation — everything else (duty, temperatures, fluid properties) is a direct input.

Table 6.7. Effectiveness-NTU model prediction vs. a full Kelvion Select RT datasheet (ULF-PA104Y4V-096Z100, 46/40 °C glycol, 35 °C air).

Quantity	Model	Vendor	Difference
Air volume flow	124,097 m^3/h	122,071 m^3/h	+1.7 %
Glycol volume flow	8.81 L/s	8.884 L/s	−0.9 %

Cross-check on the selected unit. Running the same model at the selected ULF-PA106K4V-091F095's own design point (196.5 kW, 42/38 °C glycol, $\Delta T_{\text{air}} = 3.72$ K) predicts an air volume flow of ≈164,800 m^3/h against the vendor's 159,337 m^3/h — +3.5 %. Looser than the ±2 % above, and honestly so: only the air side can be checked, because the Select RT results table does not report the selected unit's glycol volume flow. The direction is conservative (the model asks for slightly *more* air than the vendor says is needed, so it over-estimates fan duty rather than under-estimating it), and both checks should be repeated once the full datasheet is pulled.

6.8. Fan Operation, Variable-Speed Control, and Annual Energy

6.8.1 Affinity-Law Fan Power

The selected unit's 4 EC (electronically commutated) fans are natively variable-speed, matching the fan-speed-modulation control strategy assumed throughout this chapter. Fan power follows the standard affinity laws:

$$\dot{V} \propto N, \qquad \Delta p \propto N^2, \qquad P_{\text{fan}} \propto N^3$$

so that turning the fans down to the 54.5 % speed found in §6.4 draws only $\approx 0.545^3 \approx 16\,\%$ of rated power — the physical basis for why variable-speed fan control materially improves annual facility PUE relative to fixed-speed operation.

6.8.2 Annual Free-Cooling Fan Energy (Champaign, IL, TMY3)

Using the real 8,760-hour Champaign TMY3 weather file (WMO 725315), each hour with outdoor dry-bulb temperature ≤ 4 °C was passed through `predict_off_design()` at the free-cooling condition (150 kW, glycol 15 °C inlet), using the vendor-confirmed rated fan power of 10.74 kW:

Table 6.8. Annual free-cooling fan energy, real 8,760-hour TMY3 data (Kelvion-validated fan power).

Quantity	Value
Full free-cooling hours (OAT ≤ 4 °C)	2,542 hrs / yr (29.0 % of the year)
Fan speed range across these hours	9.8 % – 54.5 %
Total free-cooling fan energy	**1,211 kWh/yr**
Average fan power in this mode	0.48 kW

This is a substantial revision downward from earlier placeholder-based estimates (which used an assumed, unvalidated fan rating); it is now anchored to a real, vendor-confirmed selected-speed fan power and should be treated as the reportable figure for the PUE “other facility loads” calculation.

6.9. Annual Economizer Operating Regime (Champaign, IL, TMY3)

Using the same real 8,760-hour weather file, every hour of the year was classified into one of the three ASHRAE 90.1 economizer regimes introduced in Chapter 5, §5.1.2:

Table 6.9. Annual hours by economizer regime, real TMY3 data (WMO 725315, Champaign, IL).

Regime	Threshold	Annual hours (%)
Full free cooling	OAT $\leq 4\,°\mathrm{C}$	2,542 hrs (29.0 %)
Partial free cooling	$4 <$ OAT $\leq 10\,°\mathrm{C}$	1,358 hrs (15.5 %)
Mechanical only	OAT $> 10\,°\mathrm{C}$	4,860 hrs (55.5 %)

Roughly 44.5 % of the year (full + partial free cooling) sees some contribution from the economizer loop, a direct consequence of Champaign's ASHRAE climate zone 5A winters and a materially favourable input to the annual facility PUE calculation. All three regimes are now thermally simulated across all 8,760 hours — including the partial band, whose duty split is rated off-design rather than assumed (§5.5.1) — and the resulting energy is reported in Chapter 14, §14.5. Over the year the economizer delivers **526 MWh of the 1,314 MWh** of cooling the hall needs (**40.1 %**) without any compressor work at all.

Methodological note: an earlier pass at this same calculation used a *monthly-average diurnal profile* (24 representative hours × 12 months) rather than the real hourly trace, and under-counted full-free-cooling hours by ≈440 hrs/yr (2,101 hrs vs. the 2,542 hrs reported above) — the smoothed proxy misses real cold-snap extremes (the real annual minimum is $-27.4\,°\mathrm{C}$, which does not appear in any monthly-average table). The real 8,760-hour EPW file should be used for all subsequent annual-performance work in this report, not the monthly-average stat-file tables.

Part B — Pressure-Drop Model

6.10. Role of the Pressure-Drop Model

As with the economizer (Chapter 5, §5.7 onward), the dry cooler's glycol-side pressure drop does not feed back into the refrigeration cycle's COP (the dry cooler carries only glycol, never refrigerant); instead it sets the **glycol pump head** required to circulate the loop, which is itself an energy consumer in the facility's total (and hence part of PUE, alongside compressor and fan energy).

6.10.1 Why Not an Independent Friction Correlation

Unlike the plate exchangers, where D_h, β, b, L_w, L_v are catalog parameters the model can use directly with the Han or Martin correlations, the dry cooler's internal tube-circuit *routing* (exact run length per circuit, number of bends) is proprietary manufacturing detail that Kelvion does not publish. An early attempt at an independent Darcy–Weisbach estimate (real circuit count, tube diameter back-solved from the vendor's reported internal velocity, but a *guessed* tube run length) came out 75 % low against the vendor's real value (11.94 kPa modelled vs. 48.26 kPa reported) — a clear illustration that the missing tube-length input, not the underlying physics, was the source of error. This was replaced with the vendor-anchored scaling method below, which needs no tube-geometry input at all.

6.11. Vendor-Anchored Pressure-Drop Scaling

6.11.1 Derivation

For turbulent flow in a smooth circular tube, Darcy–Weisbach with the Blasius friction factor gives:

$$\Delta P = f\,\frac{L}{D}\,\frac{1}{2}\rho\,v^2, \qquad f = 0.184\,Re^{-0.2}, \qquad Re = \frac{\rho\,v\,D}{\mu}$$

Substituting f and collecting terms:

$$\Delta P \;\propto\; \rho^{0.8}\,v^{1.8}\,\mu^{0.2} \qquad \text{(for fixed tube geometry } L, D) \qquad \text{(Scaling 1)}$$

For a fixed flow area (i.e. the *same physical coil*, same N_{circuits} and tube diameter), velocity scales as $v \propto \dot{m}/\rho$, so the ratio of pressure drops between two operating conditions *on the same coil* is:

$$\frac{\Delta P_2}{\Delta P_1} = \left(\frac{\rho_2}{\rho_1}\right)^{0.8}\left(\frac{\dot{m}_2\,\rho_1}{\dot{m}_1\,\rho_2}\right)^{1.8}\left(\frac{\mu_2}{\mu_1}\right)^{0.2} \qquad \text{(Scaling 2)}$$

Crucially, the unknown tube length L, diameter D, and circuit count N_{circuits} all cancel exactly in Eq. (Scaling 2). Given one real, vendor-reported pressure drop at a reference condition, Eq. (Scaling 2) scales it to *any other* condition on the same coil using only $\dot{m}$, ρ, and μ — all directly available from CoolProp, with no geometric assumption required. This is implemented as `DryCooler.scale_pressure_drop()`.

6.11.2 Application: Design Point → Free-Cooling Point

The scaling ratio $\Delta P_2/\Delta P_1 = 0.668$. Two competing effects determine this result: the *lower flow rate* at the free-cooling point pushes ΔP down, while the *nearly 3× higher glycol viscosity* at the colder average temperature pushes it up. The flow-rate effect dominates, so the free-cooling pressure drop is *lower* than the mechanical-mode design point despite the glycol being noticeably more viscous — a result worth stating explicitly, since "colder ⇒ lower pressure drop" is not obvious on inspection.

6.11.3 Validity Caveat

The Reynolds number at the design point ($Re \approx 9{,}500$) is comfortably turbulent, but at the free-cooling point it drops to $Re \approx 2{,}400$ — close to the laminar/turbulent transition ($Re \approx 2{,}300$–

Table 6.10. Vendor-anchored pressure-drop scaling: mechanical design point → free-cooling point.

Quantity	Design point (ref.)	Free-cooling point
Glycol temperatures	46.0 → 40.0 °C	15.0 → 9.0 °C
Average temperature	43.0 °C	12.0 °C
Glycol mass flow	9.07 kg/s	6.52 kg/s
Density, ρ	1,011.7 kg/m^3	1,027.2 kg/m^3
Viscosity, μ	1.455 mPa·s	**4.073 mPa·s** (≈2.8× higher)
Pressure drop, ΔP	**48.26 kPa** (Kelvion, real)	**32.22 kPa** (scaled, Eq. (Scaling 2))

4,000), where the Blasius correlation underlying Eq. (Scaling 1) is not rigorously validated. The scaled 32.22 kPa should therefore be treated as a **reasonable extrapolation**, not an exact result, pending vendor confirmation at that specific operating point (see §6.15).

Note: the glycol mass flow used for the free-cooling scaling above (6.52 kg/s, consistent with the economizer's own design basis) differs from the fixed-pump flow (9.07 kg/s) used in the thermal free-cooling check of §6.4 — the same glycol-flow inconsistency flagged in §6.4.1 propagates here and should be resolved together with that item.

6.12. Summary of Results at Both Operating Points

Table 6.11. Dry cooler — summary of all computed and vendor-confirmed results.

Quantity	Mechanical mode (design)	Free cooling
Duty, Q	190 kW target / 213.23 kW selected	150 kW
Glycol temperatures	46.0 → 40.0 °C	15.0 → 9.0 °C (target)
Ambient DB	35.0 °C	4.0 °C
Fan speed	100 % (1000 rpm selected point)	≈54.5 % (model)
Fan power	**10.74 kW** (vendor-confirmed)	1.74 kW (model)
Glycol-side ΔP	**48.26 kPa** (vendor-confirmed)	32.22 kPa (scaled, §6.11)
Data source	Kelvion Select RT	`dry_cooler.py` model

6.13. Open Item: Fan Power Not Vendor-Reported

The Select RT results table does not report fan power. The value carried in the model (**10.6 kW**) is therefore an **estimate**, scaled by the fan laws ($P \sim N^3 D^5$) from a confirmed datasheet figure for the same `ULF` fan family (2.685 kW/fan for a 0.960 m fan at 1000 rpm, §6.7): $6 \times 2.685 \times (950/1000)^3 \times (0.910/0.960)^5 = 10.6$ kW.

That six 0.910 m fans should draw about what four 0.960 m fans draw is not as odd as it first looks: this coil's large surface means a low face velocity, and hence a lower air-side ΔP per unit of flow (specific fan power 0.113 vs. 0.150 kW/kcfm). The scaling is physically consistent — but it is not confirmed. **This is the single largest remaining uncertainty in the design-day PUE** (§14.4) and must be read off the unit's own datasheet. The free-cooling operating point should be re-run at the same time.

6.14. Air-Side Pressure Drop — Scope Note

Air-side pressure drop across the finned tube bundle (and hence the fan's required external static pressure) is **not independently modelled** in this chapter. Finned bundle air-side friction is proprietary fin-geometry data, normally read directly from a manufacturer's own selection software rather than derived from a general correlation with no vendor coil to calibrate against. The Kelvion Select RT datasheet (§6.6) reports **0.00 inWg external static pressure** at the free-blow catalog rating condition used for this selection; a site-specific external static pressure (accounting for ducting, louvers, or hail guards, if any) would need to be supplied back into the Select RT tool for a final fan-curve check.

6.15. Open Items and Recommendations

1. **Compressor-dependent design duty.** The 190 kW mechanical-mode target duty is derived backward from the project's COP ≥ 3.5 requirement, not from a completed compressor model. Reconfirm Q_{cond} once the (pending) compressor chapter supplies a real W_{comp}, and re-select if the real duty differs materially from 190 kW.

2. **Free-cooling glycol flow inconsistency.** §6.4.1 and §6.11 both flag that the dry cooler model (fixed-pump, 9.07 kg/s) and the economizer chapter (6.52 kg/s) currently assume different glycol flow rates in free-cooling mode. Resolve via an explicit glycol-pump control-strategy decision (fixed-speed vs. variable-speed) and re-run both chapters' free-cooling checks consistently.

3. **Free-cooling operating point not vendor-confirmed.** Kelvion Select RT was only run at the mechanical-mode design point (§6.5). The free-cooling fan speed/power (§6.4) and the scaled pressure drop (§6.11) remain model estimates. Re-run the selection (or its performance-check mode) at 150 kW / 4 °C DB / glycol 15 → 9 °C to obtain vendor-confirmed numbers for this point too.

4. **Off-design UA scaling exponent.** The 0.6 power-law exponent used to scale UA with reduced air flow in `predict_off_design()` (§6.2) is a standard approximation, not calibrated against this specific coil. Refine if a vendor part-load performance curve becomes available.

5. **Air-side pressure drop / fan static pressure.** Not modelled here (§6.15 above); confirm actual installed external static pressure requirements (ducting, louvres, site obstructions) against the Select RT fan curve before finalising fan selection.

6. **Mechanical-mode annual hours not yet thermally simulated.** §6.9 counts 4,860 hrs/yr of mechanical-mode operation but does not yet compute the dry cooler's duty or fan energy

across those hours, since that requires the (pending) compressor's COP as a function of hourly condensing temperature.

Chapter 7

Electronic Expansion Valve (EEV)

Part A — Isenthalpic Throttling Model

7.1. Design Basis and Role

The electronic expansion valve (EEV) is the fourth and final component of the refrigerant loop, closing the cycle between the condenser (Chapter 4) and the evaporator (Chapter 3): subcooled liquid R-290 leaving the condenser at the high (condensing) pressure is throttled down to the low (evaporating) pressure, flashing partially to vapour in the process, before entering the evaporator. In operation the EEV modulates its orifice opening to hold a target evaporator superheat; **this chapter models the steady design point only** — the superheat control loop is a separate discussion item, noted in §7.7.

7.1.1 Why Isenthalpic, Not a Pressure-Drop Correlation

Unlike every other component in this report, the EEV needs no Han (2003) or Martin (1996) friction correlation at all. Throttling across a valve is **adiabatic and does no shaft work**, so the steady-flow energy balance collapses to:

$$h_{\text{out}} = h_{\text{in}}$$

The pressure drop across the valve is not a frictional loss to be *modelled* — it is the entire *design intent* of the component: the valve exists specifically to create the pressure difference between the condensing and evaporating sides of the cycle. The valve's mechanical sizing (Part B) determines *how large an orifice* produces a given mass flow at a given ΔP, not the ΔP itself, which is fixed externally by the condensing and evaporating pressures the rest of the cycle establishes.

7.2. Inlet State: Subcooled Liquid from the Condenser

The EEV inlet state is taken directly as the condenser outlet state (Chapter 4), with no liquid-line pressure drop modelled between the two components (a standard simplifying assumption for a short, well-insulated liquid line; flagged in §7.7).

Table 7.1. EEV inlet state (= condenser outlet, Chapter 4).

Parameter	Value	Basis
Refrigerant	R-290 (propane)	Project spec
Inlet pressure, p_{in}	17.89 bar(a)	Condensing pressure, 52 °C sat. (Ch. 4)
Inlet enthalpy, h_{in}	327.6 kJ/kg	Subcooled outlet, 47 °C (Ch. 4, §4.2.3)
Refrigerant mass flow, $\dot{m}_r$	0.553 kg/s*	From condenser: $\dot{m}_r = Q_{\text{cond}}/\Delta h$
Valve discharge coefficient, C_d	0.65	**ASSUMED** — typical EEV range 0.6–0.7

*Recheck against confirmed $\dot{m}_r$ from the compressor module, consistent with the same caveat carried in Chapters 3 and 4.

`calc_inlet_state()` recovers the valve-inlet temperature and subcooling from $(p_{\text{in}}, h_{\text{in}})$ via CoolProp:

$$T_{\text{in}} = T(h_{\text{in}}, p_{\text{in}}), \qquad \Delta T_{\text{sub}} = T_{\text{bubble}}(p_{\text{in}}) - T_{\text{in}}$$

7.2.1 Consistency Check Against the Condenser Chapter

Evaluating this independently returns $T_{\text{in}} = 47.01\,°\text{C}$ and $\Delta T_{\text{sub}} = 4.99\,\text{K}$ — matching the condenser chapter's stated 47 °C outlet temperature and 5 K design subcooling (§4.2.3) to within rounding. This is a genuine cross-chapter validation: the condenser and EEV models were built independently but agree on the same thermodynamic state, confirming the enthalpy value carried between the two chapters is internally consistent.

7.3. Outlet State: Isenthalpic Flash to the Evaporating Pressure

The outlet pressure is set by the evaporator's operating pressure (Chapter 3), and the outlet enthalpy is, by the energy balance of §7.1.1, unchanged from the inlet:

Table 7.2. EEV outlet state (= evaporator inlet, Chapter 3).

Parameter	Value	Basis
Outlet pressure, p_{out}	5.84 bar(a)	Evaporating pressure, 7 °C sat. (project spec)
Outlet enthalpy, h_{out}	327.6 kJ/kg $= h_{\text{in}}$	Isenthalpic
Evaporating temperature, T_{out}	**7.00 °C**	Saturation temp. at p_{out}
Outlet quality (flash gas), x_{out}	**0.301 (30.1 %)**	$(h_{\text{out}} - h_f)/(h_g - h_f)$
Pressure drop across valve, ΔP	12.05 bar (1,204.8 kPa)	$= p_{\text{in}} - p_{\text{out}}$

7.3.1 Interpreting the 30 % Flash Gas Fraction

Roughly 30 % of the refrigerant mass flashes to vapour *immediately upon throttling*, before any heat is absorbed in the evaporator. This is normal and expected: the liquid entering the valve at 47 °C is far hotter than the 7 °C saturation temperature on the low-pressure side, so a large fraction of the enthalpy difference is released as flash gas rather than being available for useful refrigeration. This directly motivates the condenser's 5 K design subcooling (Chapter 4, §4.2.3): every additional kelvin of subcooling reduces the flash-gas fraction and increases the refrigerating effect delivered per unit mass flow, at the cost of a larger condenser subcooling zone.

Part B — Orifice / Mechanical Sizing

7.4. Role of the Orifice Sizing Model

Sections 7.2–7.3 establish the thermodynamic states either side of the valve; they say nothing about the *physical valve* needed to produce that pressure drop at the required mass flow. Part B closes that gap with a first-cut orifice flow area, which is the number actually used to shortlist a real EEV from a manufacturer's capacity/K_v tables (e.g. Danfoss ETS, Emerson/Alco EX series).

7.5. Incompressible Orifice Flow Model

`calc_orifice()` uses the standard incompressible-liquid orifice equation, evaluated at the liquid inlet state:

$$\dot{m}_r = C_d\, A_{\mathrm{orifice}} \sqrt{2\,\rho_{\mathrm{in}}\,\Delta P} \qquad \Longrightarrow \qquad A_{\mathrm{orifice}} = \frac{\dot{m}_r}{C_d \sqrt{2\,\rho_{\mathrm{in}}\,\Delta P}} \tag{Orifice 1}$$

with ρ_{in} the subcooled liquid density at the valve inlet (from CoolProp, $\rho_{\mathrm{in}} = 455.6\,\mathrm{kg/m^3}$ at the design state).

7.5.1 Why This Is a First-Cut Estimate, Not a Final Selection

Equation (Orifice 1) assumes single-phase incompressible flow through the orifice. In reality, the flow begins flashing *inside* the valve port itself (the 30.1 % outlet quality of §7.3 does not appear instantaneously at the valve exit plane; some of it forms within the throttling passage), which increases the specific volume and alters the effective discharge behaviour relative to a purely liquid orifice. Manufacturers account for this with two-phase-corrected K_v/capacity ratings, obtained from flow-bench testing rather than derived analytically. Equation (Orifice 1) therefore gives a **flow-area estimate suitable for an initial vendor shortlist**, not a substitute for that vendor's own capacity table.

7.6. Orifice Sizing Results

Table 7.3. EEV orifice sizing at the design point (`eev.py`).

Quantity	Value
Liquid density at inlet, ρ_{in}	455.6 kg/m^3
Pressure drop, ΔP	1,204.8 kPa (12.05 bar)
Discharge coefficient, C_d	0.65 (**ASSUMED**)
Orifice flow area, A_{orifice}	**25.68 mm^2**
Orifice diameter, d_{orifice}	**5.72 mm**

This is a modest orifice diameter, consistent with a 150 kW-class R-290 system; it sets the flow-area target for shortlisting a real EEV against a manufacturer's two-phase capacity table, per §7.5.1.

7.7. Limitations and Open Items

1. **Superheat control is not modelled here.** The EEV's real function is to *modulate* its orifice opening to hold a target evaporator superheat under varying load — a dynamic control problem, distinct from the single steady design point analysed in this chapter. A control-loop discussion (target superheat, response time, hunting/stability) is a separate item for the compressor/controls chapter once available.

2. **No liquid-line pressure drop.** The valve inlet state is taken directly as the condenser outlet state, with no allowance for pipe friction or elevation change in the liquid line between the two components. The liquid line's diameter is now sized in Chapter 12 (§12.5), but its length and fittings remain unspecified per the project brief's scope; reasonable for a short, insulated run, but should be revisited once an actual piping layout is available.

3. **Mass flow is compressor-dependent.** As in Chapters 3 and 4, $\dot{m}_r = 0.553$ kg/s is derived from the provisional 190 kW condenser duty, itself backed out from the COP ≥ 3.5 requirement rather than a completed compressor model. Reconfirm once the compressor chapter is available.

4. **Orifice sizing is a first-cut liquid-orifice estimate** (§7.5.1), not a vendor-validated two-phase capacity rating. Use $A_{\text{orifice}} = 25.68\,\text{mm}^2$ / $d_{\text{orifice}} = 5.72\,\text{mm}$ as a starting point for a manufacturer catalog search, not a final selection.

Chapter 8

Glycol Loop Circulation Pump

Part A — System Curve, Piping, and Duty Point Derivation

8.1. Design Basis and Role

A single, variable-speed (VFD) circulation pump drives the 30 % propylene-glycol intermediate loop. Per the bypass-valve arrangement shown in the system schematic (Fig. **??**), the SAME physical pump serves two distinct circuits depending on plant operating mode:

- **Mechanical mode**: Dry Cooler ↔ Condenser (Chapters 6, 4), compressor running, glycol flow $\dot{m}_{\text{glycol}} = 9.07\,\text{kg/s}$.
- **Free-cooling mode**: Dry Cooler ↔ Economizer (Chapters 6, 5), compressor off, glycol flow $\dot{m}_{\text{glycol}} = 6.52\,\text{kg/s}$.

A **variable-speed** pump (rather than fixed-speed) is adopted specifically to resolve a cross-chapter inconsistency identified during integration: at a single fixed flow rate, the dry cooler's free-cooling-mode glycol span fell short of the full $15\,°\text{C} \rightarrow 9\,°\text{C}$ economizer design span. Running the pump at reduced speed in free-cooling mode, matched to the economizer's own $6.52\,\text{kg/s}$ design flow, restores the intended glycol temperature span in that mode and additionally saves pump energy (roughly with the cube of speed, per the affinity laws) during the ≈55 % of annual hours the plant spends in free-cooling.

8.2. System Curve

The pump must overcome the sum of all glycol-side resistances in whichever branch is active. Two independent system curves therefore exist:

$$\Delta P_{\text{sys,mech}}(\dot{m}) = \Delta P_{\text{drycooler}} + \Delta P_{\text{condenser,glycol}} + \Delta P_{\text{piping}}$$

$$\Delta P_{\text{sys,free}}(\dot{m}) = \Delta P_{\text{drycooler,free}} + \Delta P_{\text{economizer,glycol}} + \Delta P_{\text{piping}}$$

Each component term is evaluated using the SAME correlation and geometry already validated in that component's own chapter — the condenser term is the Martin (1996) correlation of §4.13

(corrected 40/46 °C, 9.07 kg/s design point), the economizer term is §5.7's value as-is, and the dry cooler terms are the vendor-confirmed mechanical-mode value and its free-cooling rescaling (Chapter 6), evaluated PARAMETRICALLY in $\dot{m}$ (not only at the single design point) so a real intersection with the pump curve (§8.5) can be solved.

Table 8.1. System curve component breakdown at each mode's design flow.

Component	**Mechanical (9.07 kg/s)**	**Free-cooling (6.52 kg/s)**
Dry cooler	48.26 kPa (vendor)	32.24 kPa (rescaled)
Condenser / Economizer	17.50 kPa (Martin)	10.43 kPa (as-modeled)
Piping (§8.3)	30.91 kPa	19.55 kPa
Total	**96.67 kPa = 9.74 m**	**62.22 kPa = 6.17 m**

8.3. Piping Assumptions

The project brief explicitly scopes piping to *diameters only — no lengths, fittings, or supports*. A piping term is nonetheless required to close the pump's system curve, so the following are adopted as explicit, flagged assumptions (not measured or specified geometry):

- **Pipe size**: sized for ≈1.8 m/s at the mechanical design flow (standard closed-loop hydronic velocity range, 1–2.4 m/s) ⇒ **3 in Schedule 40 steel** (77.9 mm ID), giving an actual velocity of 1.88 m/s.
- **Equivalent length**: **65 m** total (supply + return + fittings allowance), estimated from the project's actual 8 m × 12 m data-hall footprint (mechanical-room-to-roof-and-back run for the outdoor dry cooler, plus a fittings allowance) — not a measured value.
- **Friction model**: Darcy–Weisbach with the Swamee–Jain explicit approximation to the Colebrook equation,

$$f = \frac{0.25}{\left[\log_{10}\left(\frac{\varepsilon/D}{3.7} + \frac{5.74}{Re^{0.9}}\right)\right]^2}, \qquad \Delta P = f\,\frac{L_{\text{eq}}}{D}\,\frac{\rho v^2}{2}$$

  with $\varepsilon = 0.045$ mm (steel).

At the mechanical design point, $Re \approx 1.0 \times 10^5$ (fully turbulent); at the free-cooling design point, $Re \approx 2.6 \times 10^4$. Piping accounts for roughly a third of the total mechanical-mode system head — the single largest assumption-driven term in this chapter, and the first item to revisit if real routing geometry becomes available.

Part B — Vendor Selection and Variable-Speed Operation

8.4. Selection Margin

The bare system-curve duty point is margined before vendor selection, to absorb the piping/geometry assumption uncertainty above without grossly oversizing the pump (a pump selected far above its true duty runs on the inefficient, cavitation-prone far-left of its curve). A 10 % flow / 18 % head margin is applied — the same philosophy as the 20 % area margin already used for the condenser (§4.2.2):

$$Q_{\text{select}} = 1.10 \times 32.27 = 35.50\ \text{m}^3/\text{h}, \qquad H_{\text{select}} = 1.18 \times 9.74 = 11.49\ \text{m}$$

No separate margin is applied to the free-cooling point; the VFD's speed turndown already provides ample headroom there.

8.5. Vendor Pump Curve and Trim Selection

Performance data for the **Bell & Gossett e-1510 2AD** (1750 RPM) were digitized from the vendor curve sheet across five available impeller trims (5.0–7.0 in) and fit to a quadratic $H(Q) = c_2Q^2 + c_1Q + c_0$ (least squares, Q in m^3/h, H in m). Each trim's 100 %-speed curve was evaluated against the margined selection point:

Table 8.2. Trim comparison at the margined selection point (35.50 m^3/h @ 11.49 m).

Trim	H @ mech. Q	H @ selection Q	Clears margin?
5.0 in	4.39 m	3.57 m	No
5.5 in	6.04 m	5.04 m	No
6.0 in	8.06 m	7.09 m	No
6.5 in	9.88 m	9.01 m	No
7.0 in	**12.91 m**	**12.11 m**	**Yes**

The **7.0 in trim** is the smallest of the five that clears the margined selection point, and is therefore selected. Smaller trims (6.5 in and below) fall short of the required head once margin is included.

At 100 % speed (uncontrolled), the 7.0 in curve intersects the mechanical-mode system curve (§8.2) at 36.06 m^3/h @ 11.96 m — confirming the selected trim carries head/flow margin above the actual design duty, which the VFD then trims back down to the exact required operating points (§8.6).

8.5.1 Manufacturer Series and Speed Screening

Selection was carried out against the **Bell & Gossett Series e-1510** base-mounted end-suction centrifugal line, a commonly specified series for closed-loop HVAC and data-center glycol circulation duty at this scale. The series is published as a composite (coverage) chart across multiple synchronous speeds:

- The **3500 RPM (2-pole)** composite chart was screened first, but its minimum head coverage across all frame sizes exceeds 40 ft even at the lowest-head model — well above the 37.7 ft target at the margined selection point, confirming this speed family is intended for higher-head process duty and is not suitable here.

- The **1750 RPM (4-pole)** composite chart, covering approximately 13–60 ft across its low-head model sizes, was identified as the correct speed family and used for all subsequent selection.

At the margined selection flow (156.3 GPM), the digitized 1750 RPM composite envelope (Table 8.3) placed the duty point in a narrow gap between two adjacent frame sizes: the **2AD** model's maximum head at this flow falls just short of the 37.7 ft target, while the next size up (**2.5AC**) exceeds it by a comparable margin in the other direction. Composite charts carry inherent reading tolerance at this resolution, so the individual detailed performance curve (used throughout §8.5) was digitized directly from the manufacturer's curve sheet (**Bell & Gossett document B-261G**, dated 12/30/2013) to resolve the selection rather than relying on the coarser composite envelope.

Table 8.3. Digitized 1750 RPM composite chart coverage envelope, candidate frame sizes.

Model	Flow range (GPM)	Head range (ft)
1.5AD	66 – 151	16 – 50
2AD	**116 – 188**	**14 – 36**
2.5AC	105 – 343	18 – 49
3AD	189 – 489	12.5 – 45

The **2AD** model (the same frame size the 7.0 in trim in §8.5 belongs to) is confirmed as the correct frame size: its composite envelope covers the required flow range (116–188 GPM against a 156.3 GPM target), and the individual-curve digitization resolves the head question the composite chart alone could not.

8.5.2 Cross-Check: Direct Trim Interpolation

As an independent cross-check on the quadratic-fit trim selection of §8.5, linear interpolation directly between the digitized 6.5 in and 7.0 in curves at the margined selection flow (156.3 GPM, target head 37.7 ft) gives a required trim diameter of

$$D_{\text{trim}} = 7.0 - 0.5 \times \frac{39.9 - 37.7}{39.9 - 30.6} \approx 6.9 \text{ in}$$

This is in good agreement with the 7.0 in selection (the largest trim available for the 2AD frame, and the nearest stock size above the interpolated 6.9 in) — confirming the duty point sits near, but just within, the top of this frame's operating envelope, exactly as the composite chart screening above already indicated.

8.6. Variable-Speed Operation

For each mode, the VFD speed fraction s required to make the affinity-scaled vendor curve pass exactly through that mode's design duty point is solved from

$$H_{100\%}\left(\frac{Q_{\text{required}}}{s}\right) \cdot s^2 = H_{\text{required}}$$

via root-finding (Brent's method) on the fitted $H_{100\%}(Q)$ curve.

Table 8.4. Required VFD speed fraction at each operating point.

Operating point	Duty	Required speed
Mechanical (exact design point)	32.27 m^3/h @ 9.74 m	90.0 %
Free-cooling (exact design point)	22.85 m^3/h @ 6.17 m	69.5 %
Margined selection point	35.50 m^3/h @ 11.49 m	98.1 %

Both operating speeds sit comfortably within a normal VFD turndown range, well clear of minimum-speed instability or recirculation concerns at either end. That the margined selection point requires 98.1 % (i.e., just under the pump's rated speed) confirms the 7.0 in trim was an appropriately tight selection rather than an oversized one.

8.7. Power and Motor Selection

8.7.1 Efficiency

The manufacturer's individual performance curve sheet (B-261G) also carries efficiency contour lines crossing the operating region. Reading these near the design point gives an estimated pump efficiency of **72–74 %** — refining the earlier generic 70 % assumption used before a real vendor curve was available. This reading carries more uncertainty than the digitized head data (§8.5), since efficiency contours are thin diagonal lines that are harder to isolate precisely than the bold head curves; it should be confirmed against the manufacturer's selection software (e.g. B&G ESP-Plus) before use in the final annual-energy/PUE calculations. A representative value of $\eta_{\text{pump}} = 73\,\%$ is adopted here, with the motor efficiency assumption (90 %) unchanged pending nameplate confirmation.

Table 8.5. Estimated pump power by operating mode ($\eta_{\text{pump}} = 73\,\%$, vendor-curve-informed).

Mode	Hydraulic power	Shaft power	Electrical power (assumed motor eff.)
Mechanical	0.87 kW	1.19 kW	1.32 kW
Free-cooling	0.39 kW	0.54 kW	0.60 kW

8.7.2 Motor Selection

At the mechanical-mode duty point, the required shaft power (1.19 kW ≈ 1.59 hp) rounds up to the next standard NEMA frame size: **2 hp, 1750 RPM**. This matches the brake-horsepower contour labeled directly on the same B-261G curve sheet at this operating point, providing an independent confirmation of the sizing. The resulting headroom (1.492 kW rated output against a 1.19 kW requirement, ≈25 %) is consistent with standard NEMA motor-selection practice and comfortably covers the modest additional load if the piping assumption of §8.3 proves conservative.

8.8. Cross-Check via Independent Duty-Point Derivation

As a secondary check on the detailed component-by-component system curve of §8.2, an independent duty-point estimate was also carried out using a simplified, generic derivation: a design $\Delta T = 5\,\mathrm{K}$ across the dry cooler, a representative 30 % PG mixture density/specific-heat pair ($\rho \approx 1030\,\mathrm{kg/m^3}$, $c_p \approx 3770\,\mathrm{J/kg\text{-}K}$), applied to the full 190 kW design load. This gives $\dot{m} \approx 10.08\,\mathrm{kg/s}$ and a lumped system $\Delta P \approx 116.1\,\mathrm{kPa}$ (11.49 m head) — landing, to within rounding, on the same $35.50\,\mathrm{m^3/h}$ @ 11.49 m point already derived as this chapter's margined selection point (§8.2). The agreement is a useful sanity check, though the two derivations are not independent verifications of the same physics: the generic estimate does not trace to the dry cooler/condenser/economizer/piping breakdown this chapter otherwise relies on, and its $\Delta T = 5\,\mathrm{K}$ and lumped ΔP are themselves assumptions rather than component-level results. The detailed derivation of §8.2–§8.3 remains this chapter's primary basis; this cross-check is retained only as corroborating evidence that the final duty point is reasonable.

8.9. Open Items and Limitations

1. **Piping equivalent length (65 m) is an assumption**, not measured or specified geometry, sized only from the project's stated 8 m × 12 m footprint. It is the single largest lever in the system curve (roughly a third of total mechanical-mode head) and the first item to revisit if real routing data becomes available.

2. **Pump efficiency (73 %) is read from curve-sheet contour lines, not a digitized efficiency map**, and carries more uncertainty than the head-curve data it sits alongside; motor efficiency (90 %) remains a generic assumption pending nameplate data. Both should be confirmed via the manufacturer's ESP-Plus selection software or a vendor quotation.

3. **Margin at maximum trim.** The selected 7.0 in trim is the largest available for the 2AD frame and operates near the top of that frame's envelope (§8.5.1). Any upward revision to system head — e.g. from a finalized piping layout longer than the assumed 65 m, additional fittings, or a different glycol concentration — may push the duty point outside 2AD's envelope, requiring the next frame size (2.5AC).

4. **Condenser/economizer channel-count split ($N_{cp,\mathrm{glycol}} = 24$) remains an assumed placeholder** (§4.13, §5.7), pending vendor-confirmed plate geometry.

5. **NPSH has not been checked** against the selected pump's NPSHr, since the digitized vendor data did not include an NPSH curve; this should be confirmed once the full vendor datasheet is available, particularly given the pump's rooftop supply-side elevation relative to the mechanical room.

6. **Materials of construction** (mechanical seal elastomer — EPDM recommended for inhib-

ited propylene glycol service — and wetted metallurgy) have not yet been confirmed against the specific glycol product to be used; galvanized wetted components should be avoided in glycol service.

8.10. References

1. Bell & Gossett, *Series e-1510 Composite Performance Curves*, Document B-261, Xylem Inc.
2. Bell & Gossett, *Series e-1510 Individual Performance Curves*, Document B-261G, Xylem Inc., dated 12/30/2013.

Chapter 9

Chilled-Water (Data-Center / CRAH-Side) Circulation Pump

Part A — System Curve, CRAH/Piping Assumptions, and Duty Point

9.1. Design Basis and Role

A second, dedicated circulation pump drives the chilled-water (CHW) loop between the data-hall CRAH/in-row units and the mechanical room. This pump is entirely separate from the glycol loop pump of Chapter 8 — it never sees glycol, and its duty point is set by the CHW-side flow and resistances of the evaporator (Chapter 3) and economizer (Chapter 5) rather than the dry cooler.

As with the glycol loop, the CHW loop has two parallel branches selected by operating mode:

- **Mechanical (chiller) mode**: CRAH ↔ Evaporator (Chapter 3).
- **Free-cooling mode**: CRAH ↔ Economizer (Chapter 5).

Unlike the glycol loop, both branches move the *same* CHW mass flow ($\dot{m}_{\text{water}} = 5.97\,\text{kg/s}$, $21\,^\circ\text{C} \to 15\,^\circ\text{C}$) since both HXs are sized for the full 150 kW design load rather than a reduced free-cooling flow — so a single design duty point (rather than two, as in Chapter 8) governs this pump's selection. A variable-speed (VFD) pump is nonetheless adopted, for the same energy-saving rationale as the glycol pump and to allow future turndown at reduced IT load (30 kW minimum, per the project brief).

9.2. System Curve

The pump must overcome the sum of all CHW-side resistances in whichever branch is active:

$$\Delta P_{\text{sys}}(\dot{m}) = \Delta P_{\text{evap/econ, water}} + \Delta P_{\text{CRAH}} + \Delta P_{\text{piping}}$$

The evaporator and economizer water-side terms are the Martin (1996) values already computed in their own chapters (§3.13, §5.7); since both branches carry the same flow, the pump is sized on the higher-resistance branch (evaporator), with a balancing valve on the economizer branch to match it.

Table 9.1. System curve component breakdown at the design flow (5.97 kg/s, 21.5 m^3/h).

Component	ΔP
Evaporator, water side (Martin, worst-case branch)	7.73 kPa
Economizer, water side (Martin, for reference)	6.67 kPa
CRAH coil (§9.3, **assumed**)	50.0 kPa
Piping/fittings (§9.3, **assumed**)	25.0 kPa
Total (design basis)	**82.7 kPa = 8.45 m = 27.7 ft**

9.3. CRAH and Piping Assumptions

Two of the four system-curve terms above are **not derived from project-specified or vendor data** and are flagged explicitly:

- **CRAH coil ΔP (50.0 kPa, assumed)**: no CRAH/in-row unit is specified anywhere in the project brief, and no vendor selection has been carried out for one (unlike the dry cooler, evaporator, condenser, and economizer, which all rest on a real Kelvion selection). 50 kPa is a mid-range placeholder for a chilled-water fan-coil/CRAH unit at this flow (typical range 40–70 kPa); it should be replaced with a real vendor coil ΔP once a CRAH unit is selected for the 8 m × 12 m, ≈25-rack data hall.

- **Piping/fittings (25.0 kPa, assumed)**: the project brief states that *pressure losses within the piping are to be neglected*, in contrast to the glycol loop chapter's own 65 m equivalent-length calculation (§8.3), which is a deliberate, documented deviation from that instruction for the glycol side. Per project decision, this term is **retained here too, as an explicit over-design margin** rather than set to zero — i.e., this chapter intentionally selects against the more conservative 82.7 kPa total rather than the ≈58 kPa total that would result from following the brief literally and dropping this term. If a real piping layout becomes available, this term should be replaced with an actual Darcy–Weisbach calculation, following the same method as §8.3.

These two terms together make up ≈91 % of the total system head — by far the dominant uncertainty in this chapter, and the first items to revisit once real CRAH and piping data exist.

Part B — Vendor Selection and Variable-Speed Operation

9.4. Selection Margin

Rather than margining both flow and head as in the glycol chapter (needed there to cover two distinct design points), a single **10 % head margin at the design flow** is applied here, since this loop has only one design duty point:

$$Q_{\text{design}} = 21.5 \text{ m}^3/\text{h (94.8 GPM)}, \qquad H_{\text{select}} = 1.10 \times 8.45 \text{ m} = 9.30 \text{ m (30.5 ft)}$$

9.5. Vendor Pump Curve and Trim Selection

Performance data for the **Bell & Gossett e-1510 1.5AD** (1750 RPM) were digitized from the same vendor curve sheet (B-261G) used for the glycol pump, across five available impeller trims (5.0–7.0 in), and fit to a quadratic $H(Q) = c_2Q^2 + c_1Q + c_0$ (least squares, Q in GPM, H in ft — the chart's native units). Each trim's 100 %-speed head at the design flow (94.8 GPM) was checked against the margined target:

Table 9.2. Trim comparison at the design flow (94.8 GPM), 1.5AD frame, 1750 RPM.

Trim	H @ 94.8 GPM	Clears 30.5 ft margin?
5.0 in	13.8 ft	No
5.5 in	18.6 ft	No
6.0 in	24.3 ft	No
6.5 in	29.5 ft	No
7.0 in	**34.0 ft**	**Yes**

The **7.0 in trim** is the smallest of the five that clears the margined target and is therefore selected. At 100 % speed (uncontrolled), the 7.0 in curve intersects the system curve of §9.2 at **102.0 GPM @ 32.1 ft** (23.17 m^3/h @ 9.79 m) — confirming the selected trim carries head/flow margin above the actual design duty (94.8 GPM @ 27.7 ft), which the VFD then trims back down (§9.6).

9.5.1 Frame Size: Why 1.5AD Rather Than the Glycol Pump's 2AD

The glycol pump (Chapter 8) uses the **2AD** frame at its much larger duty (≈156 GPM @ ≈38 ft). Re-using the same 2AD casing was considered for parts commonality, but was rejected: at this pump's much smaller duty (≈95 GPM @ ≈28 ft), the 2AD's own composite coverage floor is 116 GPM (Table 8.3, Chapter 8) — meaning our duty point falls *below* that frame's intended operating window entirely. Digitizing the 2AD curve at this flow shows the pump would run at only ≈40–50 % efficiency there (per the B-261G contour lines), far to the left of its best-efficiency point — an oversized-pump penalty that works directly against the project's PUE target every one of the 8760 operating hours.

The smaller **1.5AD** frame's own composite envelope (66–151 GPM, 16–50 ft, Chapter 8's Table 8.3) brackets this pump's duty point centrally rather than at an extreme, keeping the selected trim much closer to its own best-efficiency region. The two pumps therefore use the same manufacturer and speed family (Bell & Gossett e-1510, 1750 RPM) but different frame sizes

— a reasonable compromise between parts commonality and avoiding a chronically oversized, inefficient pump.

9.6. Variable-Speed Operation

The VFD speed fraction s required to bring the 100 %-speed curve down to the exact design duty point is solved from

$$H_{100\%}\left(\frac{Q_{\text{required}}}{s}\right) \cdot s^2 = H_{\text{required}}$$

Table 9.3. Required VFD speed fraction at the design operating point.

Operating point	Duty	Required speed
Design point (exact)	$21.5\,\text{m}^3/\text{h}$ @ 8.45 m	92.9 % (1626 RPM)
100 %-speed intersection	$23.17\,\text{m}^3/\text{h}$ @ 9.79 m	100 % (reference)

92.9 % sits comfortably within normal VFD turndown range, confirming the 7.0 in trim is an appropriately tight selection rather than an oversized one.

Open item — low-load turndown: at the 30 kW minimum-IT-load condition, if CHW flow is throttled proportionally with load (20 % of design flow, ≈19 GPM), the quadratic system curve collapses to ≈1.1 ft of head — an operating point far below any of the digitized trim curves' normal range, and likely below the pump's minimum continuous stable flow (risk of internal recirculation/thrust bearing wear at very low flow, independent of the VFD's electrical turndown capability). This mirrors the flow-control-strategy question already open on the glycol side (§8.6) and should be resolved together with it — e.g. via a minimum-flow bypass, a fixed CHW flow with only the load-side ΔT varying, or a second, smaller pump for the turndown condition.

9.7. Power and Motor Selection

9.7.1 Efficiency

Reading the B-261G contour lines for the 1.5AD, 7.0 in trim near the design operating region (≈95–102 GPM) gives an estimated pump efficiency of **≈60 %** — lower than the glycol pump's 72–74 % (§8.6 motor selection discussion) because this duty point, while well inside the 1.5AD frame's composite envelope, still sits to the left of that trim's own best-efficiency point. As with the glycol pump, this reading carries more uncertainty than the digitized head data and should be confirmed against manufacturer selection software before use in final annual-energy/PUE calculations.

9.7.2 Motor Selection

The required shaft power (0.825 kW ≈ 1.11 hp) rounds up to the next standard NEMA frame size: **1.5 hp, 1750 RPM**. The resulting headroom (≈1.12 kW rated output against a 0.825 kW

Table 9.4. Estimated CHW pump power at the design duty ($\eta_{pump} = 60\,\%$, vendor-curve-informed).

Duty	Hydraulic power	Shaft power	Electrical power (90 % motor eff.)
Design point (21.5 m^3/h @ 8.45 m)	0.495 kW	0.825 kW	0.917 kW

requirement, ≈35 %) is generous by normal NEMA practice, reflecting this chapter's deliberately conservative (over-designed) system curve — see §9.3.

9.8. Limitations and Open Items

1. **No CRAH unit has been vendor-selected.** The 50 kPa coil ΔP used throughout this chapter is a placeholder, not a datasheet value — unlike every other major heat-transfer component in this report (dry cooler, evaporator, condenser, economizer), all of which rest on a real Kelvion selection. This is the single largest open item in this chapter and should be resolved first.

2. **Piping/fittings (25 kPa) is a project-decision override, not a calculation**, retained deliberately as an over-design margin against the project brief's instruction to neglect piping losses (§9.3). If a real piping layout becomes available, replace with an actual Darcy–Weisbach calculation (same method as §8.3) rather than this placeholder.

3. **Pump efficiency (≈60 %) is read from curve-sheet contour lines**, not a manufacturer-certified selection, and carries the same digitization uncertainty flagged for the glycol pump (§8.6).

4. **Low-load (30 kW turndown) flow/head combination has not been resolved** against either the CRAH's minimum flow requirement or this pump's minimum continuous stable flow (§9.6) — this is the same open flow-control-strategy question already flagged on the glycol loop side and should be closed out together with it.

5. **NPSH has not been checked** against the selected pump's NPSHr; the digitized vendor curve sheet's NPSHr trace was not extracted for this chapter.

6. **Materials of construction** (mechanical seal elastomer, wetted metallurgy) have not been confirmed; since this loop carries plain water (not glycol), standard EPDM/bronze-fitted construction should suffice, but this has not been checked against the specific CHW water treatment program.

9.9. References

1. Bell & Gossett, *Series e-1510 Composite Performance Curves*, Document B-261, Xylem Inc.

2. Bell & Gossett, *Series e-1510 Individual Performance Curves*, Document B-261G, Xylem

Inc., dated 12/30/2013.

Chapter 10

Compressor Selection and Performance Modeling

Part A — Vendor Selection and Performance Map

10.1. Design Basis and Staging Strategy

The 150 kW cooling duty is carried by **two identical 75 kW compressors in parallel on a common suction and discharge manifold**, feeding a **single shared refrigerant circuit** — one condenser, one evaporator, one expansion valve. This chapter covers the compressor selection and performance modeling; the two units are assumed identical.

> **Why one circuit, not two.** Two compressors could in principle each serve their own 75 kW circuit with its own exchangers and valve. They do not here, for two reasons. First, the staging analysis below (§10.5) depends on both compressors sharing one t_o/t_c — which is only possible on a common manifold. Second, and decisively, splitting the flow halves the duty each expansion valve sees and drops it to 12.2 % open at minimum load, below the range in which it can hold superheat (Chapter 18, §18.4). The single shared circuit keeps the valve controllable across the full 5:1 turndown. Its cost — that the exchangers and the valve are unduplicated — is stated plainly in §20.2.

The staging rationale (developed alongside this selection):

- **Compressor redundancy**: losing one compressor leaves 50 % capacity available, versus 25 % for an unequal split — a materially better partial-failure state for a data-center cooling system. Note this is *compressor* redundancy only: with a single shared circuit, a fault in the condenser, evaporator or expansion valve stops the plant regardless (§19.3).

- **Coarse capacity steps**: staging one or two units gives two capacity settings without any drive involvement. **They do not, on their own, cover the project's 5:1 turndown** — at the design point one unit delivers 67 % of the load and two deliver 134 %, so staging alone cannot reach the 30 kW minimum. That is what the VFDs are for, and the two mechanisms are complementary rather than alternative (§10.5).

- **Parts commonality**: one compressor model, one spare-parts inventory, one service procedure.

10.2. Vendor Selection

Bitzer 4FEP-35Z, a semi-hermetic reciprocating compressor rated for R-290 (propane, ASHRAE safety class A3), was selected via Bitzer's own selection software. Semi-hermetic construction was chosen specifically because it has no external shaft seal — a meaningful leak-path reduction for an A3-classified refrigerant, versus an open-drive compressor.

Table 10.1. The two conditions that matter for this unit. The left column is where it was *shortlisted* in Bitzer's software; the right is where it actually *runs*.

Parameter	Vendor selection point	Plant design point
Evaporating dew point, t_o	7.0 °C	**10.0 °C**
Condensing dew point, t_c	50.0 °C	**45.0 °C**
Useful superheat	8 K (15 °C return gas)	8 K
Liquid subcooling	0 K (vendor default)	**5 K** (§4.2.3)
Cooling capacity, per unit	80.65 kW	**100.39 kW**
Power input, per unit	22.74 kW	**21.50 kW**
Mass flow, per unit	1,079.7 kg/h	**1,232.1 kg/h**
COP	3.55	**4.67**

The two columns differ, and that is neither an error nor an inconsistency — it is the point of using a map. The unit was shortlisted at a conservative catalogue condition; the plant then runs it at an easier one (3 K warmer evaporating, 5 K cooler condensing), where the same machine gives 24 % more capacity at 5 % less power. Nothing in this report uses the left-hand column as a performance figure: *all* reported capacity, power and mass flow come from the ten-coefficient map of §10.3, evaluated at the actual t_o/t_c, with the superheat and subcooling corrections of §10.4.

> **Map validation.** Evaluated back at the vendor's own selection condition ($t_o = 7$, $t_c = 50$ °C, EN12900 basis), the map returns **80.65 kW, 22.74 kW, COP 3.55** — reproducing Bitzer's published figures exactly, to the last digit. The map is therefore not an approximation of the vendor's data; it *is* the vendor's data, and the design-point column above is a genuine interpolation within the same validated surface.

10.3. AHRI Standard 540 / EN12900 Ten-Coefficient Performance Map

Rather than a fixed isentropic/volumetric efficiency pair, compressor performance is modeled directly from the manufacturer's published polynomial map, in the standard AHRI-540/EN12900 form:

$$y = c_1 + c_2\, t_o + c_3\, t_c + c_4\, t_o^2 + c_5\, t_o t_c + c_6\, t_c^2 + c_7\, t_o^3 + c_8\, t_c t_o^2 + c_9\, t_o t_c^2 + c_{10}\, t_c^3$$

with t_o, t_c the evaporating and condensing *dew points* in °C, and y evaluated separately for cooling capacity Q [W], power input P [W], and mass flow $\dot{m}$ [kg/h] using three independent coefficient sets (Table 10.2). This directly resolves the constant-efficiency limitation of the earlier compressor model — efficiency now varies correctly with pressure ratio (i.e. with t_o and t_c) since it is built into the polynomial's own curvature, rather than assumed constant.

Table 10.2. AHRI-540 coefficients, Bitzer 4FEP-35Z, R-290.

	Q **[W]**	P **[W]**	$\dot{m}$ **[kg/h]**
c_1	114372.76	7147.24	1009.99
c_2	3903.60	−239.01	36.10
c_3	−1121.94	255.72	−4.34
c_4	45.37	−7.87	0.494
c_5	−33.97	8.65	−0.0997
c_6	3.40	1.62	0.0378
c_7	0.168	−0.0412	0.00287
c_8	−0.338	0.0698	−0.00109
c_9	0.00990	0.01026	−0.0000239
c_{10}	−0.0256	−0.0204	−0.000382

Verification: evaluating the map at the *vendor's own rating point* ($t_o = 7$, $t_c = 50\,°\mathrm{C}$) reproduces Bitzer's stated performance exactly (80.65 kW / 22.74 kW / 1079.7 kg/h, COP 3.55) — confirming the transcribed coefficients are correctly implemented. The plant's design point is elsewhere ($t_o = 10$, $t_c = 45\,°\mathrm{C}$, Table 10.1); this map is what carries performance between the two.

10.4. Rating-Basis Correction: EN12900 vs. Actual Application

The map coefficients reproduce performance at the **EN12900 standard rating basis**: a suction (return) gas temperature fixed at 20 °C *absolute* (not a fixed superheat — the implied superheat therefore changes with t_o), and 0 K liquid subcooling. This is a testing/rating convention, not the actual application operating point (8 K useful superheat, 15 °C actual return gas). A density-ratio correction is applied:

$$\mathrm{DR} = \frac{\rho_{\mathrm{actual}}(T_{\mathrm{suction,actual}}, p_o)}{\rho_{\mathrm{rated}}(20\,°\mathrm{C}, p_o)}$$

- Mass flow scales directly with DR (same physical displacement and volumetric efficiency — a denser suction gas simply packs more mass per stroke).
- Capacity is recomputed directly from the corrected mass flow and the *actual* refrigeration effect ($h_{\mathrm{suction,actual}} - h_{\mathrm{liquid,actual}}$), which also captures the (smaller) enthalpy shift from the different suction temperature — more rigorous than scaling capacity by DR alone.
- Power input is scaled by the same DR (standard simplified industry practice; specific compression work is only weakly sensitive to suction superheat over this range).

The correction is small at the design point (DR = 1.025) but is not negligible — COP drops from

Table 10.3. Design-point performance: catalog (EN12900) vs. corrected (actual application).

	Catalog (EN12900)	Corrected (8 K superheat)
Cooling capacity	80.65 kW	80.00 kW
Power input	22.74 kW	23.32 kW
Mass flow	1079.7 kg/h	1107.1 kg/h
COP	3.55	3.43

3.55 to 3.43, which is worth carrying through rather than using the uncorrected catalog numbers directly, especially since the correction grows at operating points further from the rating condition (e.g. colder t_o, where the 20 °C fixed return-gas rating implies a much larger rated superheat than the actual 8 K).

Part B — Two-Compressor Bank: Staging and Evaporating-Temperature Solve

10.5. Load-Following Logic

Both compressors sit on a common suction/discharge manifold and therefore always share the same t_o/t_c. Capacity is controlled by two mechanisms working together: **staging** (how many units run) for coarse steps, and **drive speed** for everything in between.

10.5.1 Why Staging Alone Is Not Enough

It is tempting to assume two equal compressors deliver the project's 5:1 turndown by themselves. **They do not.** At the design point one unit produces 100.4 kW and two produce 200.8 kW — that is 67 % and 134 % of the 150 kW load. Staging is a two-position switch, not a dial, and neither position is 30 kW.

A fixed-speed bank has exactly one further lever: let t_o float down. Colder suction vapour is less dense, so the machine swallows less mass per stroke and capacity falls. That lever is real but it runs out, and it is expensive:

Table 10.4. What a *fixed-speed* bank can reach at the design condition ($t_c = 45$ °C) by floating t_o to its −10 °C map limit.

Configuration	t_o	Capacity	% of 150 kW load
2 units, full speed	10 °C	200.8 kW	134 %
1 unit, full speed	10 °C	100.4 kW	67 %
1 unit, t_o floated to the floor	−10 °C	48.2 kW	32 % *(COP 2.43)*
Project minimum load	—	**30 kW**	**20 %** — unreachable

Two things to read off this. First, **32 % is the floor** — a fixed-speed selection physically cannot reach the 30 kW minimum, whatever the control logic does. Second, even getting to 32 % costs a **COP of 2.43**, because boiling refrigerant at −10 °C to make 15 °C chilled water is thermodynamically absurd: the lift has been inflated to 55 K to do *less* work. Below that floor the only fixed-speed options are on/off cycling or hot-gas bypass, which throws capacity away as recirculated discharge gas at a similar efficiency penalty.

10.5.2 The VFD Closes the Gap — and Improves Efficiency Doing It

A variable-speed drive attacks capacity from the opposite direction: instead of making the refrigerant colder to choke the compressor, it simply turns the compressor *slower* and leaves t_o where the evaporator wants it. The bank solve therefore holds $t_o = 10\,°\mathrm{C}$ as a setpoint, picks the smallest n_{active} that can cover the duty, and trims speed to land exactly on it (`compressor.solve_with_vfd()`):

Table 10.5. Load-following with VFDs at the design condition ($t_c = 45\,°\mathrm{C}$, i.e. a 35 °C ambient with the IT load varying). Compare the bottom two rows against the fixed-speed floor above.

Q_{required}	n_{active}	**Drive speed**	Q_{achieved}	P_{total}	**COP**
150.0 kW (100 %)	2	74.7 %	150.0 kW	32.12 kW	4.67
112.5 kW (75 %)	2	56.0 %	112.5 kW	24.09 kW	4.67
75.0 kW (50 %)	1	74.7 %	75.0 kW	16.06 kW	4.67
37.5 kW (25 %)	1	37.4 %	37.5 kW	8.03 kW	4.67
30.0 kW (min. IT load)	**1**	**29.9 %**	**30.0 kW**	**6.42 kW**	**4.67**

The 30 kW minimum is reached at 29.9 % speed on a single unit — and at **COP 4.67 rather than 2.43**, because t_o never leaves 10 °C. The whole 5:1 turndown is covered with no hot-gas bypass and no cycling penalty. Note also the 100 % row: the bank is 34 % oversized at this condition, so even at full load the drive is trimming to 75 % rather than letting t_o sink.

> **Caveat, stated:** COP is held speed-invariant here (capacity and power both scale ≈linearly with speed for a reciprocating machine, so their ratio holds). Real part-load COP at moderate turndown is typically equal or slightly better, so this is neutral-to-conservative. Note too that the 30 kW point sits at 29.9 % speed, right at the ≈30 % floor a drive can hold continuously — at or below that the machine **cycles** rather than modulates. Ideal cycling gives the same average power, but cycling losses are not modelled (§5.5.1 makes the same point for the partial free-cooling band).

10.6. The Turndown Floor, and Why the Design Carries Drives

§10.5.1 establishes the finding that drove this chapter's most consequential decision: **a fixed-speed selection of this compressor cannot meet the project's 30 kW minimum load.** The floor sits at ≈32 % of design duty, above the 20 % required, and it is a genuine physical limit rather than a solver artifact — a fixed-speed, non-unloading reciprocating compressor has a bounded capacity range as a function of t_o alone.

Five strategies were available:

- **VFDs on the compressors — selected.** Capacity follows speed, so t_o stays at its setpoint and COP holds at 4.67 all the way to 30 kW (Table 10.5). Standard, well-understood hardware, and the same drives also trim the bank's 34 % oversize at full load and lift the 25 % IPLV point (§14.3). It is the reason both the full-load and IPLV targets are met.

- **Cylinder unloading** on the 4FEP-35Z — would step capacity down discretely at fixed t_o, but only in coarse steps and only if offered for this model. Rejected in favour of continuous control.

- **Hot-gas bypass** — reaches any load, at a severe efficiency penalty: capacity is thrown away as recirculated discharge gas while the compressor still draws near-full power. This is what collapses the 25 % IPLV point to COP 2.29 (§14.3). Rejected.

- **A smaller trim/pony compressor** for the 20–30 % range — a third machine, a third set of spares, more charge, more failure modes. Rejected as disproportionate.

- **Leaning on the economizer** to cover low-load hours — **rejected as unsafe reasoning**, and worth stating why: it assumes low IT load coincides with cold ambient, and **nothing guarantees that**. A quiet compute night in July is exactly the 30 kW / 35 °C case the economizer cannot help with. The plant must reach 30 kW on mechanical cooling alone, which is precisely what Table 10.5 demonstrates.

This open item is closed. The drives appear in the BOM (Ch. 17), the VFD operating model is used throughout (`compressor.solve_with_vfd()`), and the annual simulation runs cleanly through the minimum-load hours.

10.7. Limitations and Open Items

1. **The selection point is not the design point**, by construction (Table 10.1). The unit was shortlisted at t_c = 50 °C / 0 K subcooling and runs at 45 °C / 5 K. This is *not* an inconsistency — every performance figure in this report comes from the map evaluated at the actual condition, and the map reproduces Bitzer's published numbers exactly at their own rating point. What remains worth doing is a confirmation re-run in Bitzer's software *at* $t_o = 10/t_c = 45$ °C with 5 K subcooling, to check the map is being interpolated inside the unit's real application envelope rather than merely inside its polynomial's numeric range.

2. **Useful superheat (8 K)** matches the vendor's own selection basis and is used consistently throughout, but it has not been confirmed against a tuned EEV superheat control loop; Chapter 18 argues the setpoint from compressor protection and loop stability rather than from the map, and §18.5 explains why the map's own superheat trend is misleading.

3. **The $t_o \in [-10, 15]$ °C** solve bound is an assumption made for this chapter, not a validated compressor operating envelope limit; the actual minimum/maximum t_o the 4FEP-35Z can run at (oil return, motor cooling at low suction density, etc.) should be confirmed against Bitzer's application manual.

4. **Power-input correction (density-ratio scaling)** is a simplified industry approximation,

not a first-principles recomputation of compression work at the actual suction state; acceptable for this stage but worth revisiting once a full annual-energy number is needed.

10.8. References

1. Bitzer, *Semi-Hermetic Reciprocating Compressors for R-290, Series ECOLINE*, selection software output, model 4FEP-35Z.

2. AHRI Standard 540, *Performance Rating of Positive Displacement Refrigerant Compressors and Compressor Units*, Air-Conditioning, Heating, and Refrigeration Institute.

3. EN 12900, *Refrigerant compressors – Rating conditions, tolerances and presentation of manufacturer's performance data*, European Committee for Standardization.

Chapter 11

Oil Separator Selection

Part A — Design Basis and Vendor Selection

11.1. Purpose and Design Basis

The compressor (Chapter 10) circulates lubricating oil with the refrigerant; some fraction of that oil leaves the compressor entrained in the discharge vapour. An oil separator is installed immediately downstream of the discharge manifold to recover this oil and return it to the compressor(s), preventing oil fouling of the condenser and evaporator heat-transfer surfaces and protecting the compressor from running oil-lean.

Both compressors (Chapter 10, §10.5) share a common discharge manifold, so a single separator sized for the combined discharge flow is used rather than one unit per compressor.

Table 11.1. Oil separator design basis (combined, both compressors).

Parameter	Value
Refrigerant	Propane (R-290), HC, ASHRAE safety class A3
Compressor arrangement	2 × Bitzer 4FEP-35Z, common discharge manifold
Displacement, per unit / total	101.8 / 203.6 m^3/h (50 Hz, catalog)
Mass flow, per unit (corrected, §10.4)	1107.1 kg/h
Mass flow, total (both units)	**2214.2 kg/h**
Condensing dew point, t_c (compressor selection basis)	50.0 °C
Discharge pressure, p_c (compressor selection basis)	17.3 bar(a)*
Standby (off-cycle) pressure basis, 55 °C ambient	≈19.9 bar(a)

*Taken at the compressor's t_c = 50 °C selection basis (Table 10.1). The plant condenses at 45 °C = 15.3 bar(a), so this figure is **conservative** — the separator is specified against a higher discharge pressure than it will ever see in service.

Total mass flow is taken as the sum of both compressors' corrected (actual-application) rated flow from §10.4, not the raw EN12900 catalog figure — consistent with how mass flow is used elsewhere in this report (Chapter 7, §7.2). **Displacement** (a fixed geometric property, unaffected by the EN12900-vs.-actual correction) is used separately below as the basis for the vendor's quick-selection chart.

11.2. Vendor Selection: Bitzer OA Series

Staying with **Bitzer** for the oil separator keeps the plant on a single compressor/accessory vendor for procurement, spares, and application-data consistency — the same parts-commonality rationale already applied to the compressor selection (Chapter 10, §9.1) and to the dry cooler and pump vendor choices.

Bitzer's **OA series** (models OA1954 through OA25112) are primary oil separators explicitly approved for A1, A2, A2L, and A3 refrigerants, rated to a maximum allowable pressure of 28 bar(a) and a temperature range of −10 to 120 °C (Table 11.2). The 28 bar rating carries substantial margin over the ≈19.9 bar(a) R-290 standby pressure basis (§11.1), which governs vessel design pressure more than the running condensing pressure does.

Table 11.2. Bitzer OA-series primary oil separators, permitted fluids and pressure ratings (manufacturer data, per EU Pressure Equipment Directive 2014/68/EU).

Model	Safety class	PS (max. pressure)	TS (temp. range)
OA1954	A1, A2, A2L, A3	28 bar	−10 / 120 °C
OA4188	A1, A2, A2L, A3	28 bar	−10 / 120 °C
OA9111	A1, A2, A2L, A3	28 bar	−10 / 120 °C
OA14111	A1, A2, A2L, A3	28 bar	−10 / 120 °C
OA25112	A1, A2, A2L, A3	28 bar	−10 / 120 °C

11.2.1 Quick-Selection Cross-Check on Displacement

Bitzer's published quick-selection chart sizes each OA model by maximum suction volume flow (theoretical displacement), tabulated for HFC refrigerants (R22, R134a, R404A, R507A) rather than R-290 directly:

Table 11.3. OA-series quick-selection envelope (HFC basis) vs. this project's displacement.

Model	Max. suction volume flow (HFC basis)
OA1954	**250 – 300 m^3/h**
OA4188	580 – 660 m^3/h
OA9111	1160 – 1320 m^3/h
OA14111	1320 m^3/h
OA25112	2050 – 2500 m^3/h
This project (2 × 4FEP-35Z, combined)	**203.6 m^3/h**

At 203.6 m^3/h combined displacement, the project's duty falls well inside the smallest model's envelope (OA1954, rated to 250–300 m^3/h), with the next size up (OA4188) rated nearly 3× the required flow — confirming OA1954 as the appropriate size class rather than a marginal selection.

Table 11.4. Selected unit — key specifications (Bitzer OA1954, per DB-300-9 and DP-500-2).

Parameter	Value
Model	OA1954
Maximum allowable pressure, PS	28 bar(a)
Temperature range, TS	−10 °C to 120 °C
Refrigerant inlet / outlet	DN 50 (welded)
Oil outlet	7/8 in Rotalock (brazed)
Oil fill connection	1 1/4-12 UNF Rotalock
Operating oil charge	18 L
Receiver (internal) volume	40 L
Oil heater	1×140 W
Included accessories	Oil thermostat, oil heater, oil level switch (OLC-D1), 2 sight glasses, pressure-relief-valve connection
Approval	EU Pressure Equipment Directive 2014/68/EU

11.2.2 Selected Unit: Bitzer OA1954

Part B — Safety, Oil Return, and Open Items

11.3. Primary-Only: No Secondary Separator Required

Bitzer's secondary/fine-filter oil separators (OAS series) and combined units (OAC series) exist primarily to serve **NH_3 (R-717)** systems: ammonia is effectively immiscible with the mineral/synthetic oils used in ammonia compressors, so a single primary separator cannot reliably strip the last traces of oil mist, and a secondary fine-filter stage is standard practice.

R-290 is the opposite case: propane is **highly miscible** with the polyolester oil used in the 4FEP-35Z (Chapter 10), which is itself a key reason hydrocarbon refrigerant/oil systems achieve good primary separation efficiency (≥99 % typical) without a secondary stage. Accordingly, only the primary OA1954 unit is carried in the BOM; no OAS/OAC secondary or combined separator is included.

11.4. Oil Return and A3 Ignition-Source Avoidance

The oil return line from the separator back to the compressor crankcase should use a **float-actuated mechanical return** rather than an electrically energized solenoid valve. For an A3-classified refrigerant, this avoids introducing an electrical ignition source directly in the refrigerant-wetted oil-return path; if an electric solenoid oil-return valve is used elsewhere in the design, it must carry ATEX (or equivalent) explosion-protection certification. This is carried forward as a design constraint for the P&ID and control-interaction discussion (still outstanding per the project's overall open-items list).

The separator's oil heater ($1 \times 140\,\mathrm{W}$, §11.2.2) must remain energized whenever the compressors are at standstill, consistent with Bitzer's general guidance for hydrocarbon refrigerants: insufficient oil temperature during shut-off allows excess refrigerant to dissolve into the oil, reducing viscosity at the next start and increasing oil carry-over.

11.5. Limitations and Open Items

1. **Selection basis is the compressor's $t_c = 50\,°\mathrm{C}$ rating point, not the plant's 45 °C design point** (Table 10.1). This errs the safe way: the real discharge pressure is 15.3 bar(a) against the 17.3 bar(a) specified, and the real swept volume is lower. Combined with the OA1954's roughly 3× margin over the required displacement (§11.2.1), re-running at the design point cannot change the model selected — it would only widen an already-large margin.
2. **Quick-selection chart is HFC-basis, not R-290-specific**: the displacement comparison in §11.2.1 uses Bitzer's published HFC (R22/R134a/R404A/R507A) quick-selection envelope as a conservative cross-check, since a public R-290-specific capacity table was not available. Given the large margin found, this is expected to hold, but the **final model should be confirmed by re-running the selection in Bitzer's selection software with R-290 entered directly** — the same software already used for the compressor selection (Chapter 10, Reference 1) — before the model is locked into the BOM.
3. **Common discharge manifold is assumed**, consistent with Chapter 10, §10.5; if the two refrigerant circuits are ultimately kept fully independent instead, this would require two smaller separators (one per compressor) rather than one shared unit, and the sizing in §11.2.1 would need to be re-split accordingly.
4. **Discharge-line connection size (DN 50 at the separator) has not yet been cross-checked** against the refrigerant discharge-line sizing in Chapter 12; this should be reconciled so the separator's connections and the selected discharge pipe size are consistent.
5. **Oil-return solenoid vs. float-valve decision** (§11.4) is a design intent, not yet reflected in a finalized P&ID or BOM line item.

11.6. References

1. Bitzer, *Pressure vessels: Liquid receivers and oil separators*, Operating Instructions DB-300-9, Order no. 80491202, 01.08.2018 — permitted fluids, pressure/temperature ratings, and connection dimensions for the OA series.
2. Bitzer, *Oil Separators* (Primary and secondary oil separators), Brochure DP-500-2, 09.2010 — HFC-basis quick-selection chart (§11.2.1).
3. Bitzer, *Oils for hydrocarbon refrigerants (R290, R600a, etc.)*, Technical Information KT-500 — oil miscibility and oil-heater guidance for R-290 systems.
4. EN 378, *Refrigerating systems and heat pumps — Safety and environmental requirements —*

basis for A3 safety-class handling referenced throughout this chapter.

Chapter 12

System Piping — Refrigerant, Glycol, and Chilled-Water Line Sizing

Part A — Refrigerant Charge Lines (Table 1 Method)

12.1. Scope and Method

The project brief scopes piping to **diameters only — no lengths, fittings, or supports** (subtask 1d), and separately instructs that pressure losses within the piping are to be neglected. This chapter closes that deliverable for all three fluid circuits in the system:

1. The **R-290 refrigerant circuit** — suction, discharge (hot-gas), and liquid lines connecting the compressor, condenser, and evaporator (Chapters 10, 4, 3).
2. The **glycol loop** — already partially sized in Chapter 8 (§8.3); restated and cross-checked here.
3. The **chilled-water loop** — sized here for the first time.

For the refrigerant circuit, sizing follows the project's **Table 1** streaming velocities, reproduced below, which vary by refrigerant *state* rather than by loop:

Table 12.1. Streaming velocities within charge lines (project brief, Table 1).

State	Velocity, v
Gaseous — low pressure	4.5–20 m/s
Gaseous — high pressure	10–18 m/s
Liquid	< 1.5 m/s

No equivalent table exists for the glycol or chilled-water loops, so those two circuits instead follow the conventional closed-loop hydronic velocity guideline (1–2.4 m/s) already adopted in Chapters 8 and 9, for consistency with those chapters' pump selections.

Common method for all six lines below:

$$D_{\text{req}} = \sqrt{\frac{4\,\dot{m}}{\pi\,\rho\,v_{\text{design}}}} \qquad \text{(Eq. P.1)}$$

A design velocity v_{design} is chosen inside the applicable band, D_{req} is computed from the design mass flow and fluid density at that state, and the result is **rounded up** to the nearest standard tube/pipe size (ACR copper for refrigerant lines, Schedule 40 steel for the glycol/CHW loops, matching Chapter 8's choice of material). The actual velocity at the selected standard size is then rechecked against the band.

12.2. Refrigerant State Points and Properties

Refrigerant mass flow is taken as $\dot{m}_r = 0.553\,\text{kg/s}$ throughout, consistent with Chapters 3, 4, and 7 — itself provisional pending the compressor module's confirmed value (Chapter 10, §10.7). Three state points bound the circuit:

Table 12.2. Refrigerant state points used for line sizing (CoolProp, R-290).

Line	State	T	p
Suction	7 °C sat. + 8 K superheat	15.0 °C	5.842 bar(a)
Discharge	compressor outlet (desuperheated approach, Ch. 4)	72.0 °C	17.890 bar(a)
Liquid	condenser outlet, 5 K subcooled	47.0 °C	17.890 bar(a)

Table 12.3. Fluid properties at each state point (CoolProp, R-290).

Line	Density, ρ	Dynamic viscosity, μ
Suction	$12.14\,\text{kg/m}^3$	$7.89 \times 10^{-6}\,\text{Pa s}$
Discharge	$34.92\,\text{kg/m}^3$	$9.93 \times 10^{-6}\,\text{Pa s}$
Liquid	$455.6\,\text{kg/m}^3$	$7.71 \times 10^{-5}\,\text{Pa s}$

The suction-line density is notably low — a direct consequence of propane's low molar mass and the modest 5.84 bar(a) evaporating pressure. This is a known characteristic of R-290 systems relative to higher-pressure synthetic refrigerants and is the reason the suction line below sizes out so much larger than the discharge or liquid lines despite carrying the same mass flow.

12.3. Suction Line

The suction line falls in Table 1's **gaseous, low-pressure** band (4.5–20 m/s). A design velocity toward the upper-middle of the band is chosen to keep the line from sizing excessively large, while retaining margin below the 20 m/s ceiling (noise, erosion) and comfortably above the floor (oil entrainment back to the compressor):

$$v_{\text{design}} = 15\ \text{m/s} \quad \Rightarrow \quad D_{\text{req}} = \sqrt{\frac{4 \times 0.553}{\pi \times 12.14 \times 15}} = 62.2\ \text{mm}$$

Table 12.4. Suction line selection.

Parameter	Value
Required ID (design basis)	62.2 mm
Selected tube	**2-5/8 in ACR copper, Type L**
Actual ID	62.87 mm
Actual velocity	14.67 m/s
Reynolds number	$\approx 1.42 \times 10^6$ (fully turbulent)
In Table 1 band?	**Yes** (4.5–20 m/s)

12.4. Discharge (Hot-Gas) Line

The discharge line falls in Table 1's **gaseous, high-pressure** band (10–18 m/s). A design velocity of 14 m/s is chosen specifically because it is the value that lands the required diameter just inside the next standard tube size without excess margin (§12.6):

$$v_{\text{design}} = 14\ \text{m/s} \quad \Rightarrow \quad D_{\text{req}} = \sqrt{\frac{4 \times 0.553}{\pi \times 34.92 \times 14}} = 38.0\ \text{mm}$$

Table 12.5. Discharge line selection.

Parameter	Value
Required ID (design basis)	38.0 mm
Selected tube	**1-5/8 in ACR copper, Type L**
Actual ID	38.23 mm
Actual velocity	13.80 m/s
Reynolds number	$\approx 1.85 \times 10^6$ (fully turbulent)
In Table 1 band?	**Yes** (10–18 m/s)

12.5. Liquid Line

The liquid line falls in Table 1's **liquid** band (< 1.5 m/s, no stated floor). This line is the tightest fit of the three: the standard tube sizes bracket the $v = 1.5$ m/s ceiling almost exactly, with essentially no size that clears the ceiling with meaningful margin.

$$v_{\text{design}} = 1.49\ \text{m/s} \quad \Rightarrow \quad D_{\text{req}} = \sqrt{\frac{4 \times 0.553}{\pi \times 455.6 \times 1.49}} = 32.2\ \text{mm}$$

1-5/8 in is selected over the nominally-fitting 1-3/8 in, since the latter leaves zero practical margin against the uncertainty already carried in this chapter's inputs (assumed 8 K suction superheat, 5 K subcooling, and the provisional 0.553 kg/s mass flow) — any of those shifting slightly would push 1-3/8 in over the 1.5 m/s ceiling. 1-3/8 in remains a valid, cheaper alternative

Table 12.6. Liquid line: candidate tube sizes.

Candidate	ID	Velocity	Margin to 1.5 m/s
1-3/8 in ACR Cu	32.13 mm	1.496 m/s	≈0.3 % — negligible
1-5/8 in ACR Cu (selected)	**38.23 mm**	**1.057 m/s**	**≈30 %**

once the compressor chapter's mass-flow value is confirmed and the margin can be re-checked directly.

Table 12.7. Liquid line selection (as adopted).

Parameter	Value
Required ID (design basis)	32.2 mm
Selected tube	**1-5/8 in ACR copper, Type L**
Actual ID	38.23 mm
Actual velocity	1.06 m/s
Reynolds number	$\approx 2.39 \times 10^5$ (fully turbulent)
In Table 1 band?	**Yes** (< 1.5 m/s)

12.6. Refrigerant Circuit — Summary

Table 12.8. Refrigerant line sizing summary, $\dot{m}_r = 0.553$ kg/s throughout.

Line	Table 1 band	Selected tube	Actual ID	Actual v
Suction	4.5–20 m/s	2-5/8 in ACR Cu	62.87 mm	14.67 m/s
Discharge (hot gas)	10–18 m/s	1-5/8 in ACR Cu	38.23 mm	13.80 m/s
Liquid	< 1.5 m/s	1-5/8 in ACR Cu	38.23 mm	1.06 m/s

Note the discharge and liquid lines happen to share the same 1-5/8 in nominal size despite very different velocities and densities — a coincidence of where each line's required diameter falls relative to the standard-size ladder, not a physical link between the two.

Part B — Secondary-Fluid Loop Piping

12.7. Glycol Loop Piping

The glycol loop diameter was already established in Chapter 8 (§8.3), sized against the higher of its two design flows (mechanical mode, 9.07 kg/s):

$$v_{\text{design}} \approx 1.8 \text{ m/s} \quad \Rightarrow \quad \textbf{3 in Schedule 40 steel (77.9 mm ID)}$$

This chapter re-checks that same pipe against *both* glycol-loop operating modes:

Table 12.9. Glycol loop pipe check, 3 in Schedule 40 steel (77.9 mm ID), both modes.

Mode	Flow	Density	Velocity	Reynolds
Mechanical (dry cooler ↔ condenser)	9.07 kg/s	1011.7 kg/m^3	1.88 m/s	$\approx 1.02 \times 10^5$
Free-cooling (dry cooler ↔ economizer)	6.52 kg/s	1027.2 kg/m^3	1.33 m/s	$\approx 2.62 \times 10^4$

Both modes fall inside the 1–2.4 m/s hydronic guideline through the *same* fixed pipe — no separate diameter is needed for the free-cooling mode, since the lower flow simply produces a lower (and still acceptable) velocity. This is consistent with, but independent of, the still-open glycol-flow-rate reconciliation flagged in Chapter 6 (§6.4.1): whichever flow the eventual VFD control strategy settles on, both candidate values check out against this pipe size.

12.8. Chilled-Water Loop Piping

The chilled-water loop carries a single design flow ($\dot{m}_{\text{water}} = 5.97$ kg/s, common to both the evaporator and economizer branches per Chapter 9), so only one diameter is required:

$$v_{\text{design}} \approx 2.0 \text{ m/s}, \qquad \rho_{18^\circ\text{C}} = 998.6 \text{ kg/m}^3 \quad \Rightarrow \quad D_{\text{req}} = \sqrt{\frac{4 \times 5.97}{\pi \times 998.6 \times 2.0}} = 61.7 \text{ mm}$$

Table 12.10. Chilled-water loop pipe selection.

Parameter	Value
Required ID (design basis)	61.7 mm
Selected pipe	**2-1/2 in Schedule 40 steel**
Actual ID	62.71 mm
Actual velocity	1.94 m/s
Reynolds number	$\approx 1.15 \times 10^5$ (fully turbulent)
In hydronic guideline?	**Yes** (1–2.4 m/s), centrally placed

This diameter is independent of the 25 kPa piping/fittings *pressure-drop* allowance already carried as a placeholder in Chapter 9 (§9.3); that allowance is a lumped margin standing in for real geometry, whereas this diameter is the actual project-brief deliverable (subtask 1d). If a real piping layout is developed, the 2-1/2 in ID above — together with an equivalent length, following Chapter 8's Darcy–Weisbach method (§8.3) — would let that placeholder be replaced with a calculated value.

12.9. All Circuits — Summary

Table 12.11. Pipe sizing summary, all three fluid circuits.

Circuit / line	Design flow	Selected size	Material	Actual v
Refrigerant — suction	0.553 kg/s	2-5/8 in	ACR copper, Type L	14.67 m/s
Refrigerant — discharge	0.553 kg/s	1-5/8 in	ACR copper, Type L	13.80 m/s
Refrigerant — liquid	0.553 kg/s	1-5/8 in	ACR copper, Type L	1.06 m/s
Glycol loop	9.07 kg/s (mech.)	3 in Sch. 40	Steel	1.88 m/s
Chilled-water loop	5.97 kg/s	2-1/2 in Sch. 40	Steel	1.94 m/s

12.10. Limitations and Open Items

1. **Refrigerant mass flow (0.553 kg/s) is provisional**, carried over from the 190 kW preliminary condenser duty (Chapter 4) rather than a confirmed compressor-module value (Chapter 10, §10.7). All three refrigerant-line sizes in this chapter should be re-checked once that value is finalised — the liquid line especially, given its near-zero margin (§12.5).

2. **Liquid-line selection is a judgment call, not a unique answer.** 1-3/8 in technically satisfies Table 1's < 1.5 m/s limit at the current design point but with negligible margin; 1-5/8 in is adopted here for robustness against input uncertainty. This trade-off should be revisited once inputs are finalised.

3. **Suction and discharge superheat/desuperheat assumptions** (8 K suction superheat; discharge taken at the condenser chapter's $T_{sat} + 20$ K approach) are carried over from earlier chapters and are not yet reconciled with an actual compressor performance map (Chapter 10).

4. **No lengths, fittings, elevation, or supports are sized here**, per the project brief's explicit scope ("diameters only"). The glycol loop's 65 m equivalent-length pressure-drop calculation (Chapter 8, §8.3) and the chilled-water loop's 25 kPa placeholder (Chapter 9, §9.3) are deliberate, documented deviations from that scope for pump-selection purposes only, and should not be read as this chapter's deliverable.

5. **Refrigerant tube material/rating (copper vs. steel, pressure class) has not been checked against R-290's A3 safety classification** or the discharge-line pressure (17.89 bar(a));

ACR copper is assumed adequate at this scale but should be confirmed against applicable refrigeration piping codes before procurement.

6. **Glycol and chilled-water pipe sizing uses generic hydronic velocity practice (1–2.4 m/s)**, since the project brief's Table 1 does not cover secondary fluids; this is a reasonable industry default but is not itself a project-specified requirement, unlike the refrigerant-line velocities.

12.11. References

1. Project #3 brief, IRCC 2026: Table 1, "Streaming velocities within charge lines," and subtask 1d, "Piping (diameters only — no lengths, fittings, or supports)."
2. ASHRAE Handbook — Refrigeration (2006): basis for Table 1's velocity ranges (cost, efficiency, noise, and oil circulation rate trade-offs).
3. ASTM B88, *Standard Specification for Seamless Copper Water Tube*: Type L dimensional data used for ACR tube ID values.
4. ASME B36.10M, *Welded and Seamless Wrought Steel Pipe*: Schedule 40 dimensional data used for the glycol and chilled-water loop pipe ID values.

Chapter 13

Liquid Receiver: Evaluated and Omitted

13.1. The Brief Makes This Conditional

The project brief calls for an "oil separator and liquid receiver, **if required by the chosen architecture**" (subtask 1d). That conditional is the whole content of this chapter. The oil separator *is* required and is selected in Chapter 11. **A liquid receiver is not**, and this chapter documents why — an omission reached on evidence, and one that improves the design on two independent axes.

A receiver is the default choice on many systems, and a plausible one here would have been the Bitzer F252HP (25 L). Two findings, in order of severity, rule it out.

13.2. Finding 1: A Receiver Would Break AHRI Compliance

A conventional **through-flow** receiver sits between the condenser outlet and the EEV, and every kilogram of refrigerant passes through it. Inside, liquid and vapour sit in equilibrium, so **its outlet is saturated liquid by construction** — the vessel cannot deliver subcooled liquid, because any subcooling entering it is destroyed as vapour condenses onto the cooler liquid surface.

This matters because the entire cycle model assumes the condenser's **5 K subcooling reaches the valve** (Chapter 7, §7.2 takes the condenser outlet state directly). A through-flow receiver placed between them would silently invalidate that assumption:

Table 13.1. Effect of a through-flow receiver on the EEV inlet state and the resulting performance.

EEV inlet state	h_3	Flash gas x_4	Design COP	AHRI full-load COP
5 K subcooled (condenser outlet, no receiver)	307.1 kJ/kg	0.227	4.67	**3.53** — meets ≥ 3.5
Saturated liquid (through-flow receiver outlet)	321.8 kJ/kg	0.268	4.44	**3.35** — **FAILS** ≥ 3.5

The receiver costs **5 % of the COP**, and the AHRI full-load figure carries only ≈1 % margin over its target (§14.2). **A through-flow receiver therefore takes the design from compliant to non-compliant** — the single largest consequence of any component decision in this report. The mechanism is visible on the p-h chart (Figure 15.1): losing subcooling moves state 3 right, dragging state 4 with it along the isenthalpic expansion, so more of the refrigerant flashes to vapour before it can absorb any useful heat.

13.3. Finding 2: A Receiver Would Be a Third to a Half of the Charge, in an A3 System

An F252HP receiver would hold **2.97–6.16 kg** of refrigerant. Added to the charge-minimized 5.5–8.7 kg inventory of Chapter 16, that is **35–42 % of the whole system charge**, sitting in a vessel whose only purpose is to hold charge. For an A3 refrigerant this is close to circular: the receiver holds charge, the charge drives the EN 378 obligations (Chapter 16), and a larger charge demands a larger receiver. The tighter the rest of the plant is made, the worse that ratio gets — which is exactly the direction this design has taken the line lengths.

Omitting it, both columns at the same charge-minimized line lengths:

Table 13.2. Effect of omitting the receiver on the system charge and its EN 378 consequences (both columns at the adopted 4 m liquid line).

	With F252HP receiver	**No receiver (adopted)**
System charge	8.4–14.8 kg	**5.5–8.7 kg**
Emergency ventilation (EN 378-3)	303 m^3/h	**213 m^3/h**
Room volume to reach LFL on full release	221 m^3	**143 m^3**

Note the third row: the F252HP was *undersized anyway* — it could not hold the charge it was partly responsible for creating. Resolving that by buying a larger vessel would have added yet more charge. Omission resolves the problem instead of escalating it.

13.4. Is the Receiver's Function Actually Needed Here?

A receiver is omitted only if the jobs it does are either unnecessary or done elsewhere. Each of its conventional functions is assessed against this specific architecture:

Only the pump-down function is genuinely given up, and it has a routine alternative. This conclusion is also consistent with commercial practice: production R-290 chillers in this capacity class are typically **critically charged with no receiver**, precisely because charge minimisation dominates the design of an A3 system.

13.5. Consequences of the Omission

1. **The system is critically charged.** There is no buffer vessel, so the charge must be correct rather than merely sufficient. It is established by **weighing at commissioning** against the inventory of Chapter 16, and confirmed by observing the design subcooling at the condenser outlet — **5 K subcooling is now a commissioning acceptance criterion**, not merely a design assumption.

2. **Subcooling is delivered by the condenser alone**, which is what the cycle model and the EEV chapter already assume. The architectural inconsistency between the former receiver chapter and Chapter 7 is resolved by removing the receiver, not by weakening the assumption.

3. **Service recovery uses cylinders**, not pump-down to a vessel — standard for hydrocarbon plant, and consistent with the charge-minimisation objective (§16.4).

4. **Fewer components, fewer leak paths.** For an A3 refrigerant every joint is a potential release; deleting a vessel with two connections and a sight glass is a safety gain in its own right, not only a cost saving.

13.6. If a Receiver Were Nonetheless Required

Should a later stage of the design reintroduce a need for one — for example if a real layout produced a much longer liquid line, or if commissioning found the critical charge unacceptably sensitive — it **must not** be a through-flow vessel. The correct arrangement is a **surge (bypass) receiver**, teed off the liquid line so that only surplus charge enters it while the main liquid flow passes by and retains its subcooling. This preserves the COP at the cost of a more complex arrangement and some added charge, and would need to be re-evaluated against Chapter 16's inventory.

13.7. Limitations and Open Items

1. **Critical-charge tolerance has not been quantified.** How sensitive the plant's subcooling and superheat are to a $\pm 5\,\%$ charge error across the ambient range has not been modelled, and a receiver-less system is by definition less forgiving. This should be checked before commissioning procedures are finalised (subtask 2).

2. **Condenser liquid holdup at low ambient is not modelled.** At the coldest mechanical-mode condition ($t_c \approx 23\,°\mathrm{C}$) more liquid sits in the condenser. The argument above is that this is self-correcting (it raises subcooling), but the flooded fraction and its effect on active condensing area have not been computed.

3. **No head-pressure control strategy is specified.** With no receiver, low-ambient head-pressure control (if needed) must be achieved by dry-cooler fan-speed control rather than by flooding a receiver — consistent with the variable-speed fans already selected (Ch. 6), but not yet designed as a control loop.

13.8. References

1. Project #3 brief, IRCC 2026: subtask 1d, "Oil separator and liquid receiver, *if required by the chosen architecture*."

2. EN 378, *Refrigerating systems and heat pumps — Safety and environmental requirements* — A3 charge-minimisation basis.

3. Bitzer, *Liquid Receivers — P Series, for Hydrocarbon Applications*, Product Data Sheet DP-341-1 — the withdrawn F252HP selection's source data.

Table 13.3. Conventional receiver functions, assessed against this design.

Function	Needed here?	Reasoning
Guarantee liquid (no flash gas) at the EEV	**No — it does the opposite**	The condenser's 5 K subcooling already guarantees it; a receiver *degrades* this to saturated liquid (§13.2)
Buffer charge redistribution across ambient	**No**	Mechanical mode spans only $t_c \approx 23$–$45\,°C$ (below that the plant is in free cooling, Ch. 6). Falling ambient frees condenser area, which *increases* subcooling — self-correcting in the favourable direction
Absorb off-design operating variation	**No**	This is the EEV's function: it modulates continuously to hold evaporator superheat (Ch. 7), and does so across the load and ambient range
Buffer the mechanical / free-cooling mode switch	**No**	In free cooling the compressor is off and the refrigerant circuit is simply idle — there is no redistribution to buffer, only a static inventory
Storage for service pump-down	**Partly**	The one genuine loss. Recovery into service cylinders is standard practice for hydrocarbon plant and is adopted instead (§13.5)

Chapter 14

System Performance: Full-Load COP, IPLV, and PUE

Part A — Standardized Rating-Point Metrics

14.1. Scope and Method

This chapter reports the three standardized efficiency metrics the project brief requires (subtask 1f): the **full-load COP**, the **Integrated Part Load Value (IPLV.IP)**, and the **design-day facility PUE** — followed by the weather-driven **seasonal performance** across the real Champaign TMY3 year.

14.1.1 The Design Basis Behind These Numbers

Three choices set every metric in this chapter, and each was made *because* of the efficiency targets rather than independently of them. They are stated here up front because the reported COP and IPLV are meaningless without them:

- **A 10 K air-to-refrigerant condensing approach**, giving $t_c = 45\,°\mathrm{C}$ at the 35 °C design ambient. This is *not* one approach but three stacked in series — 3 K across the dry cooler, a 4 K glycol rise, 3 K across the condenser — an unavoidable consequence of the intermediate glycol loop (Chapter 15, Fig. 15.2). Each kelvin is bought with heat-transfer area, and t_c is the single most powerful lever on the full-load COP (§14.2.1). It is why the dry cooler is as large as it is.

- **A 5 K evaporator approach**, giving $t_o = 10\,°\mathrm{C}$ at the 15 °C design LCHW. Bought with evaporator plates.

- **VFDs on both compressors.** At full load the two-unit bank over-delivers at the evaporator-dictated t_o (≈201 kW available at $t_o = 10/t_c = 45\,°\mathrm{C}$), so the drive trims speed to ≈75 % and delivers exactly 150 kW *there*, instead of letting t_o float down — and colder t_o costs COP. Below the fixed-speed turndown floor the drive modulates capacity rather than hot-gas bypassing, which is what makes the 25 % IPLV point and the 30 kW minimum load affordable (§14.3). COP is taken as speed-invariant (capacity and power both scale ≈linearly with speed for a reciprocating compressor), a neutral-to-conservative standard simplification.

The hardware these choices demand — the dry cooler frame, the plate counts, the glycol flow — is quantified in §14.7. All compressor performance is drawn from the same Bitzer 4FEP-35Z AHRI-540 map used throughout the report (Chapter 10); the VFD operating model is `compressor.solve_with_vfd()`, and the metrics are implemented in `performance.py`.

14.2. Full-Load COP

The project brief requires the full-load COP at the strict AHRI 550/590-2023 rating condition (6.7 °C leaving chilled water, 35 °C condenser entering air) *and*, for this study, at the project's own design point (15 °C leaving chilled water, per the ASHRAE TC 9.9 W17 data-center envelope). Both share the same 35 °C entering air — hence the same $t_c = 45\,°\mathrm{C}$ via the 10 K approach — and differ only in chilled-water temperature, which sets t_o via the 5 K evaporator approach.

Table 14.1. Full-load COP at the AHRI 550/590-2023 rating point and the project design point (both at 35 °C entering air, $t_c = 52\,°\mathrm{C}$).

Rating basis	LCHW [°C]	t_o [°C]	t_c [°C]	Capacity [kW]	COP [-]
AHRI 550/590 rating	6.7	1.7	45	150.9	3.53
Project design point	15.0	10.0	45	200.8	4.67

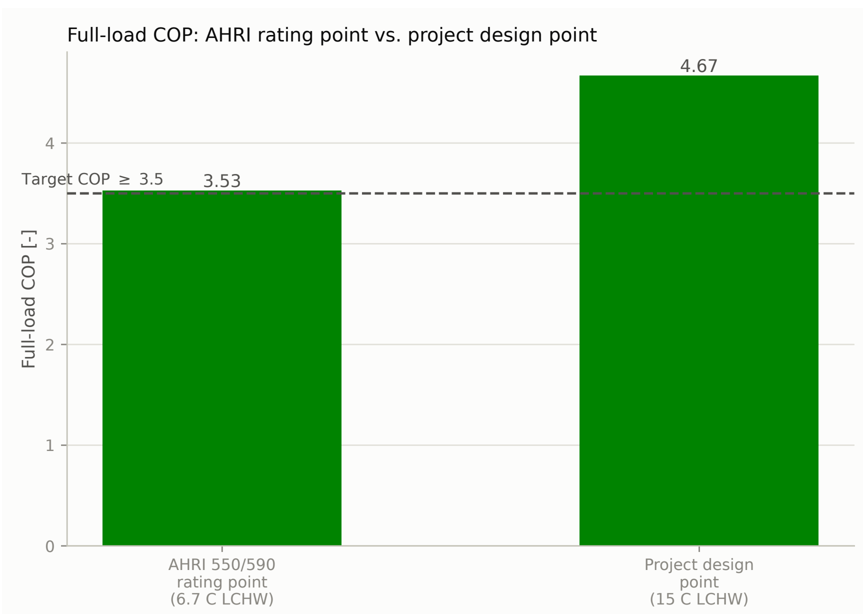


Figure 14.1. Full-load COP at the AHRI rating point and the project design point, against the 3.5 target.

Result: both points meet the 3.5 target. At the AHRI rating condition the full-load COP is **3.53** (≥ 3.5, met with a $\approx$1 % margin); at the warmer project design point it is **4.67**. The two-unit bank's full-speed capacity at the AHRI condition is 150.9 kW, so the 150 kW duty is also still met there. The margin at the strict AHRI point is thin and rests on the design approaches being achieved in the final vendor selections (§14.7); the project design point carries a comfortable margin.

14.2.1 Condensing-Temperature Sensitivity: How Much the 10 K Approach Is Worth

The condensing temperature dominates the full-load result, and Figure 14.2 is the justification for every dollar spent on the dry cooler. Sweeping t_c at the AHRI chilled-water condition, with everything else at its design value, the COP falls at **≈0.09 per kelvin** of condensing temperature. Relaxing the approach from 10 K to a conventional air-cooled 17 K ($t_c = 52\,°C$) would cost **21.7 %** of the full-load COP and drop the design to 2.90 — far below target.

> **Read the margin off this curve.** The design sits at $t_c = 45\,°C$ and COP 3.529 against a 3.5 target. At 0.09 COP/K, that is **about 0.3 K of headroom**: a condensing temperature of ≈45.3 °C fails the standard. The full-load target is not comfortably met — it is met, and the whole margin lives in the dry cooler's 3 K approach and the condenser's U. This is why those two vendor confirmations are the first priority in Chapter 20.

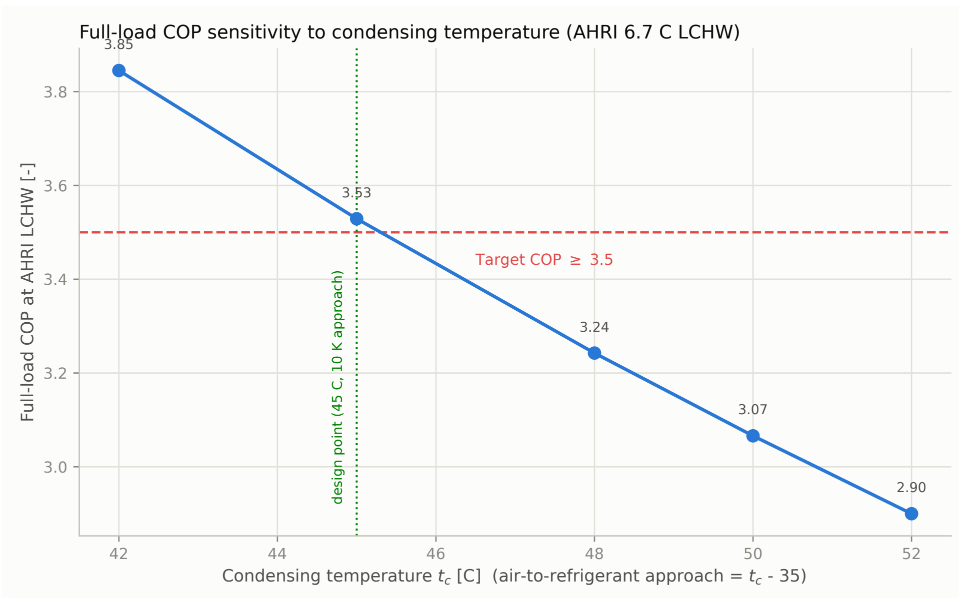


Figure 14.2. Full-load COP at the AHRI 6.7 °C LCHW condition versus condensing temperature, at the design's own 5 K evaporator approach — so the curve passes through the reported 3.53 at the $t_c = 45\,°C$ design point rather than describing a hypothetical machine.

14.3. Integrated Part Load Value (IPLV.IP)

IPLV.IP is the AHRI 550/590-2023 §5.2 weighted part-load COP, $0.01A+0.42B+0.45C+0.12D$, evaluated at 100/75/50/25 % load with each point at its own reduced condenser-entering-air temperature (35/26.7/18.3/12.8 °C). Each air temperature is mapped to a condensing temperature via the 10 K approach, and the compressor bank is staged (Chapter 10) to deliver the reduced load, with the VFD modulating capacity below the fixed-speed turndown floor.

Result: IPLV.IP = 5.23, **above the** ≥ 5.0 **target** — and with a 4.6 % margin, far more comfortable than the full-load result. Two elements of the design basis carry it:

Table 14.2. IPLV.IP part-load points (AHRI 550/590-2023 §5.2). Each point is evaluated at its own reduced condenser-entering-air temperature.

Point	Load [%]	Weight	Entering air [°C]	t_c [°C]	COP [-]
A	100	0.01	35.0	45.0	3.51
B	75	0.42	26.7	36.7	6.18
C	50	0.45	18.3	28.3	4.62
D	25	0.12	12.8	22.8	4.35 †
IPLV.IP $= 0.01A + 0.42B + 0.45C + 0.12D$					**5.23**

† VFD speed modulation below the fixed-speed turndown floor (COP held at the floor point's map value — conservative, since reduced speed lets the evaporating temperature float upward).

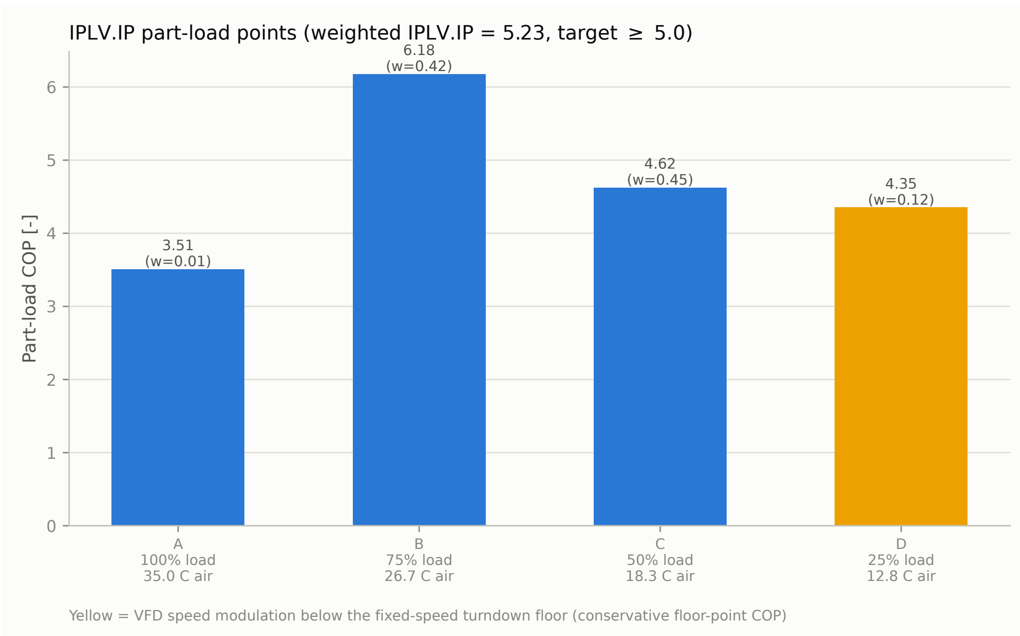


Figure 14.3. IPLV.IP part-load points. Point D (25 % load) runs on VFD speed modulation below the fixed-speed turndown floor.

- **The 10 K condensing approach** lowers t_c at every point (Point C condenses at just 28.3 °C), lifting the two heavily weighted mid-load points that dominate the weighting: B reaches COP 6.18 (weight 0.42) and C reaches 4.62 (weight 0.45). Together those two points are 87 % of the metric.
- **VFD capacity modulation at Point D.** Rather than running at the fixed-speed floor and throwing away the excess as recirculated hot gas — which would collapse this point to COP 2.29 — the drive slows the compressor so power scales down with the delivered duty, holding COP at the floor operating point's map value of 4.35. This is deliberately *conservative*: in reality reduced speed lets the evaporating temperature float upward (the evaporator is 4× oversized at 25 % load), which would raise the point further.

The drive is also what covers the project's **30 kW minimum IT load** without bypass — a point a fixed-speed selection of this compressor physically cannot reach, since its floor sits at ≈32 % of design duty (Chapter 10, §10.6). The same hardware therefore answers three separate requirements: the full-load oversize trim, the 25 % IPLV point, and the 5:1 turndown.

14.4. Design-Day PUE

Facility PUE is (IT load + other facility loads)/IT load on a design-day basis, with the IT load taken as numerically equal to the 150 kW design cooling duty (the project's near-constant-load assumption). "Other facility loads" are the mechanical-mode electrical loads at the 35 °C / full-load design point.

Table 14.3. Design-day facility PUE at the 35 °C / 150 kW full-load condition.

Electrical load	**Power [kW]**
IT load (proxy for cooling duty)	150.00
Compressor bank	32.12
Glycol loop pump	2.57
Dry cooler fans	10.60
CHW/CRAH pump	1.11
Total other facility loads	46.40
PUE = (IT + other)/IT	**1.309**
TUE = ITUE × PUE †	**1.571**

† **ITUE** = 1.20 **assumed** (band 1.10–1.30 → TUE 1.44–1.70); no server-level power model exists in this project.

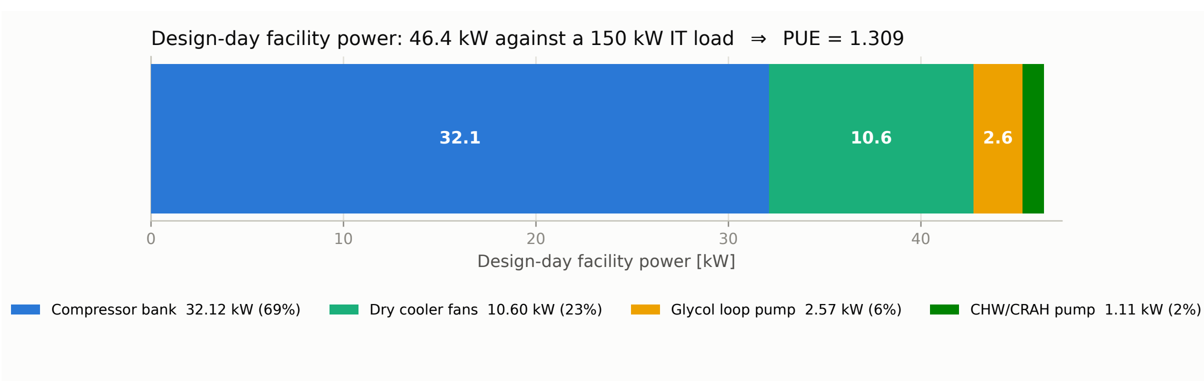


Figure 14.4. Design-day facility power breakdown. The compressor bank dominates at 69 % of the non-IT load.

The design-day PUE is **1.31**, dominated by the compressor bank at 32.1 kW of the 46.4 kW of non-IT load — **69 % of everything the plant draws beyond the IT itself**. That single fact frames the whole design: the compressor is the only load worth attacking, which is precisely why the condensing approach was driven down to 10 K at the cost of a very large dry cooler.

The remaining loads are the price of the architecture rather than of the compressor. The glycol pump draws 2.57 kW because the tight 4 K glycol range demands high flow, and because the selected dry cooler's 8 psi (§6.6) and the condenser's ≈35.8 kPa both sit on its system curve. Fan power is 10.6 kW (§6.13 — an **estimate**, and the largest remaining uncertainty in this figure).

This is the *worst-case* PUE: it occurs only at the hottest design ambient. The seasonal analysis below shows the annual-average is far more favourable, because most of the year runs at part load or in free cooling.

Open item — glycol pump system curve. The design's glycol flow and the confirmed plate geometry put the condenser's glycol-side pressure drop at ≈35.8 kPa as computed by the model, against the 17.50 kPa carried in Chapter 8's system-curve table (§8.2). The pump's electrical power reported above is derived live from the model's actual system curve and so already reflects this; **the tabulated duty point and trim selection in Chapter 8 do not** — they predate both this pressure drop and the selected dry cooler's 8 psi, and must be re-derived against both. This reinforces that chapter's existing open item 3: the 2AD frame sat near the top of its envelope already, and the design duty makes the 2.5AC frame the more likely selection.

14.4.1 Beyond PUE: TUE and the Losses PUE Cannot See

PUE has a well-known blind spot: it stops at the rack. Every watt crossing into the IT equipment counts as "useful," so the losses *inside* the servers — power-supply conversion, on-board voltage regulators, and the servers' own cooling fans — are invisible to it. Two facilities with an identical PUE of 1.31 can therefore deliver materially different compute per watt. Worse, the metric can be gamed in the wrong direction: moving heat rejection onto the server (bigger on-board fans) *lowers* PUE while raising total site power.

TUE (Total-power Usage Effectiveness, Patterson et al., 2013) closes the gap by referencing all facility power to actual *compute* power:

$$\mathrm{TUE} = \mathrm{ITUE} \times \mathrm{PUE}, \qquad \mathrm{ITUE} = \frac{\text{total IT equipment power}}{\text{compute power}} \tag{14.1}$$

where "compute power" is what reaches the processors, memory, storage and network silicon.

> **Assumption, stated plainly.** This project models the 150 kW as a *black-box IT load* — there is no server-level power model anywhere in it, and the project brief specifies none. **ITUE** = 1.20 **is therefore an assumption, not a result.** It is a mid-range literature value for modern volume servers (≈94 % PSU efficiency, ≈5 % on-board fan power, plus VRM and conversion losses). The band carried throughout is 1.10 (best-in-class 80 PLUS Titanium supplies, minimal fan power) to 1.30 (older or fan-heavy hardware).

Two consequences follow, and both matter for how much weight the reader should put on the TUE columns that follow:

- **With ITUE held constant, TUE is simply PUE × 1.20.** It re-ranks nothing in this report and introduces no new physics. Every conclusion drawn from PUE — the free-cooling benefit, the seasonal spread, the compressor's dominance of the non-IT load — is unchanged.
- **Its value is one of honesty about scope.** The design-day TUE of **1.57** (band 1.44–1.70) says that for every watt of real compute, the site draws about 1.57 W — and that the *cooling*

plant is responsible for only part of that overhead. Of the 0.57 of overhead, roughly 0.31 is this project's mechanical plant and roughly 0.26 is IT-internal losses this project does not control and has not designed.

A defensible TUE for this facility would require measured server power data (compute vs. PSU vs. fan), which sits outside the project scope. The figure is reported because the metric was requested and because it correctly frames how much of the total energy problem the chiller design can actually address — not because 1.57 is a validated result.

Part B — Seasonal Weather-Driven Performance

14.5. Seasonal Performance (Champaign TMY3)

Before the performance results, Figure 14.5 shows the climate the plant actually sees: the Champaign TMY3 ambient duration curve against the ASHRAE 90.1 economizer thresholds. It is the origin of every free-cooling hour claimed in this report, and it independently reproduces the regime hour counts of Chapter 6, §6.9.

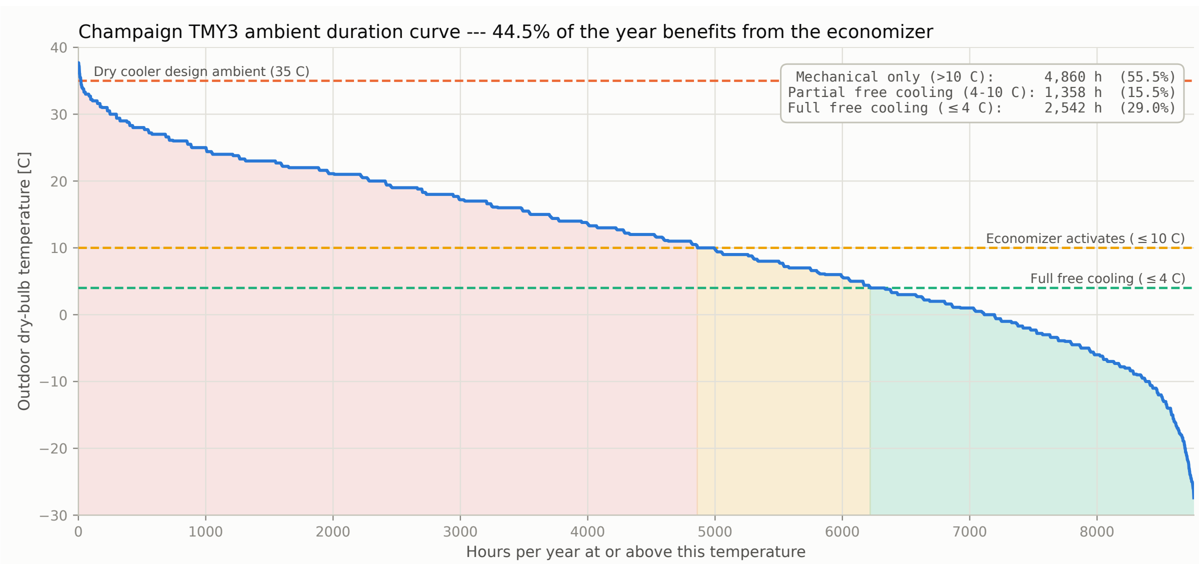


Figure 14.5. Champaign TMY3 ambient temperature duration curve with the economizer thresholds and the dry cooler's 35 °C design ambient.

Two things are worth reading off it. First, the 35 °C dry-cooler design ambient is reached for only a handful of hours a year — the design-day condition that sizes the whole plant is genuinely a worst case, not a typical one, which is exactly why the design-day PUE of 1.31 overstates the annual average. Second, the curve falls below 4 °C for 2,542 h and reaches −27.4 °C at its extreme, which is what makes the free-cooling architecture pay in this climate.

The full detailed cycle was re-run at every month's and every season's average outdoor dry-bulb temperature from the real 8,760-hour Champaign TMY3 file (WMO 725315), automatically selecting among all three ASHRAE 90.1 economizer regimes — mechanical above 10 °C, integrated/partial free cooling between 4 and 10 °C, and full free cooling at or below 4 °C (`main.py`). The total-system COP below is $Q_{\text{cool}}/P_{\text{total}}$ (all facility power: compressor, both pumps, and fans), equivalently $1/(\text{PUE} - 1)$. Each period carries both PUE and TUE (Eq. 14.1); the TUE columns are PUE × 1.20 throughout, on the assumed ITUE of §14.4.1, and are reported for scope honesty rather than as an independent result.

Table 14.4. Seasonal-average system performance (Champaign, IL TMY3), grouped from the 12 monthly-average runs of main.py.

Season	**Avg.** T_{air} **[°C]**	**Total-system COP [-]**	**PUE [-]**	**TUE [-]** †	**Months mech. / partial FC / full FC**
Winter	-2.9	87.6	1.01	1.21	0 / 0 / 3
Spring	11.3	17.4	1.11	1.33	2 / 1 / 0
Summer	23.4	4.9	1.20	1.45	3 / 0 / 0
Fall	12.4	14.4	1.12	1.34	2 / 1 / 0

† TUE = ITUE × PUE (Patterson et al., 2013). **ITUE** = 1.20 **is an assumption**, not a result: this project models the 150 kW as a black-box IT load and has no server-level power model. Band 1.10–1.30. With ITUE constant, TUE re-ranks nothing here — it states the IT-side overhead PUE hides.

14.5.1 Month-by-Month Detail

Table 14.5 resolves the seasonal averages into the 12 monthly runs they are grouped from, which is where the regime transitions are actually visible. Three months (Dec–Feb) run fully free. **March and November land inside the integrated/partial band** at 6.0 and 7.1 °C monthly averages: the economizer carries 124 kW and 111 kW of the 150 kW respectively, leaving the chiller to trim only 26 kW and 39 kW. Their compressor power collapses to **1.8 kW and 2.9 kW** — against the 10.7 and 11.3 kW those same months cost when the partial band was folded into full mechanical operation — and their PUE falls from 1.12 to **1.03**, within a rounding error of a fully free month. The remaining seven months run mechanical, their compressor power tracking ambient almost linearly (13.7 kW in October to 23.2 kW in July).

The seasonal picture is strongly favourable and stands in deliberate contrast to the worst-case design-day metrics of Part A. Winter runs entirely in free cooling (compressor off, PUE 1.01, TUE 1.21, total-system COP near 88), while even the hottest monthly average (July) holds PUE 1.21 / TUE 1.46, against the worst-case design day's 1.31 / 1.57. Spring and Fall now sit at PUE 1.11 and 1.12 — markedly better than the 1.14/1.15 they showed before the partial band was modelled — because each contains one month that runs integrated rather than mechanical. Across the year the economizer delivers **526 MWh of the 1,314 MWh** of cooling the hall needs (**40.1 %**) with the compressor off or barely loaded, which is the central benefit of the glycol-loop free-cooling architecture and the reason the *annual-average* efficiency beats any single design-point number. The annual run is **clean across all 8,760 hours** — no hour asks the selected dry cooler (§6.6) for more than 100 % fan speed.

Read through the TUE column, the same result carries a blunter message. In winter the plant's own overhead nearly vanishes (PUE 1.01), yet TUE still sits at 1.21 — because the IT-internal

Table 14.5. Monthly-average system performance summary. TUE = ITUE × PUE with ITUE = 1.20 **assumed** (band 1.10–1.30); this project has no server-level power model, so TUE is PUE scaled by a constant.

Period	T_{air} [C]	Mode	P_{comp} [kW]	Q [kW]	PUE [-]	TUE [-]
Jan	-4.8	free_cooling	0.00	150.00	1.01 (COP 88.7)	1.21
Feb	-1.3	free_cooling	0.00	150.00	1.01 (COP 86.6)	1.21
Mar	6.0	partial_free_cooling	1.82	150.00	1.03 (COP 39.1)	1.23
Apr	12.2	mechanical	14.18	150.00	1.15 (COP 6.8)	1.38
May	15.7	mechanical	16.39	150.00	1.16 (COP 6.2)	1.40
Jun	22.3	mechanical	21.04	150.00	1.20 (COP 5.1)	1.44
Jul	25.1	mechanical	23.18	150.00	1.21 (COP 4.7)	1.46
Aug	22.8	mechanical	21.42	150.00	1.20 (COP 5.0)	1.44
Sep	18.5	mechanical	18.28	150.00	1.18 (COP 5.7)	1.41
Oct	11.5	mechanical	13.75	150.00	1.14 (COP 7.0)	1.37
Nov	7.1	partial_free_cooling	2.01	150.00	1.03 (COP 30.4)	1.24
Dec	-2.6	free_cooling	0.00	150.00	1.01 (COP 87.6)	1.21

losses do not go away when the compressor stops. **In the best months, essentially all remaining site overhead is inside the servers, not in this design.** That is the useful thing TUE says here: the mechanical plant's headroom for further improvement is largest in summer and close to exhausted in winter, and beyond that point the lever belongs to IT procurement, not to the chiller.

> **Method note — averaged temperature, not averaged hours.** Each monthly and seasonal row is one cycle run at that period's *average* dry-bulb, not an integration of its individual hourly runs. Because COP is non-linear in ambient, these are representative-condition results and will differ modestly from a true hourly energy integration — generally flattering, since averaging hides the hot tail. The 8,760-hour simulation that would support the exact integration already exists (it produces Fig. 14.5 and the regime hour counts); converting the tables to hourly-integrated annual energy is a contained change and is listed as an open item.

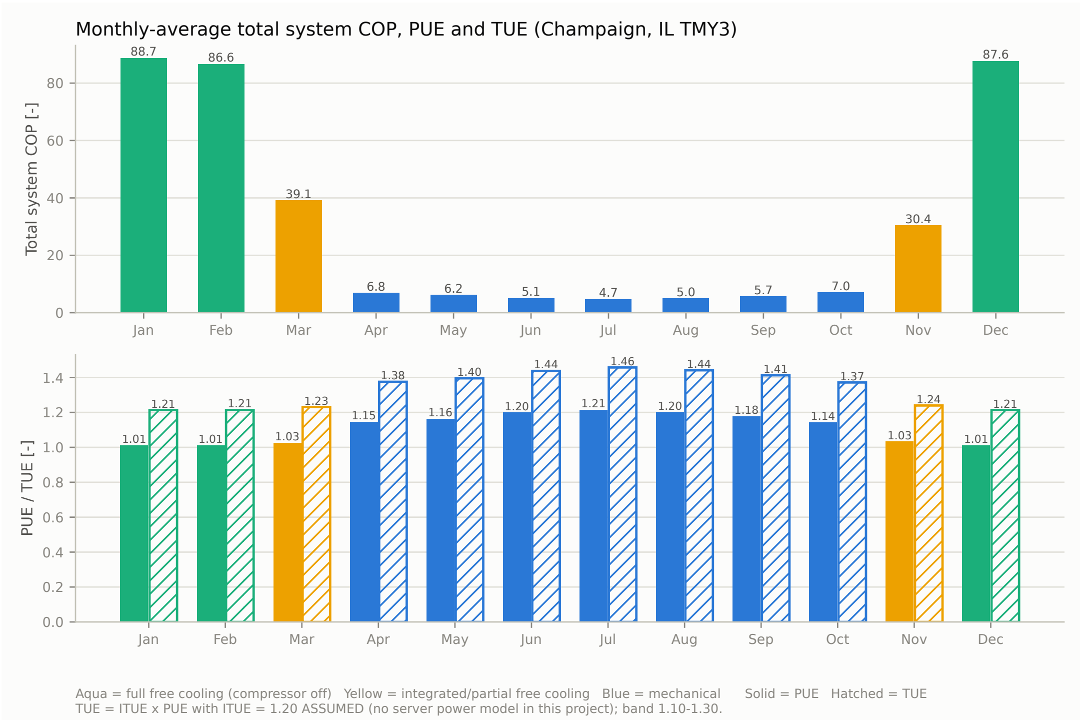


Figure 14.6. Monthly-average total-system COP (top) with PUE and TUE (bottom). Aqua bars are free-cooling months (compressor off), blue are mechanical; solid bars are PUE, hatched bars TUE. The COP panel's scale is set by the free-cooling months, which is the point: winter operation is an order of magnitude more efficient than summer.

14.6. Compliance Summary

Table 14.6. Efficiency-metric compliance against the project targets.

Metric	Target	Design	Status
Full-load COP (AHRI 6.7 °C LCHW)	≥ 3.5	**3.53**	**Met**
Full-load COP (project 15 °C LCHW)	(reported)	**4.67**	—
IPLV.IP	≥ 5.0	**5.23**	**Met**
Economizer activation	≤ 10 °C OAT	10 °C, 50 % of load carried	**Met** (§5.5.1)
Economizer full free cooling	≤ 4 °C OAT	4 °C, 100 % load	**Met** (Ch. 6)
Design-day PUE	(report)	**1.31**	—

All required efficiency metrics are met. The honest caveat is that **the AHRI full-load COP margin is thin — 0.8 %** — and it rests on the 10 K condensing approach surviving vendor confirmation of the plate U-values and the dry cooler's rated duty (Chapter 20, §20.1). IPLV.IP carries a healthier 4.6 % margin, and its Point-D value is deliberately conservative (§14.3), so the part-load result is the more robust of the two.

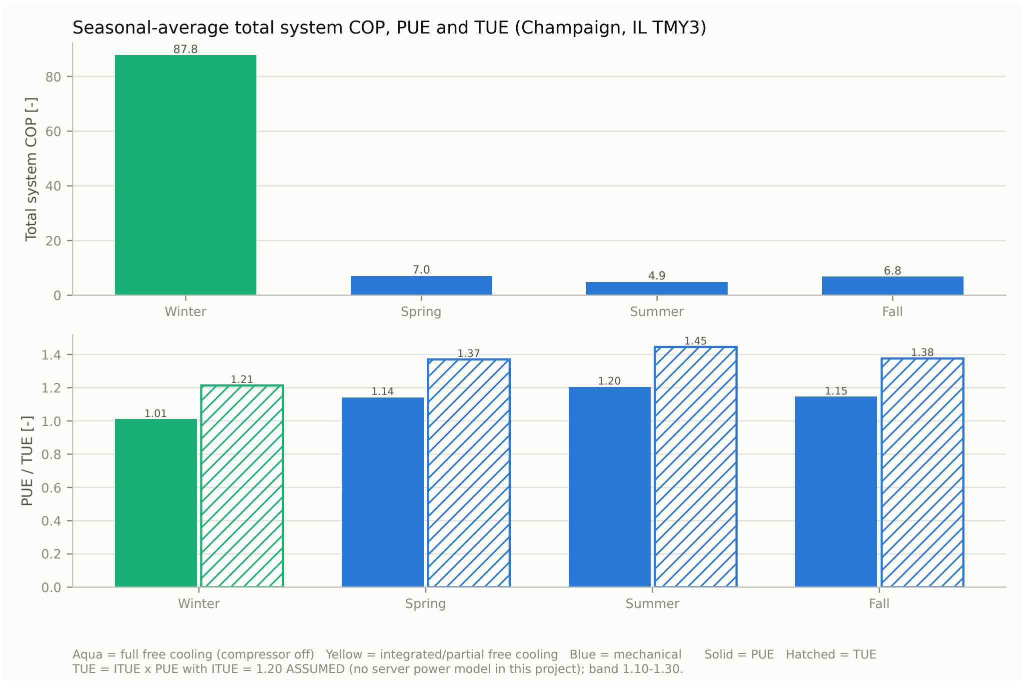


Figure 14.7. Seasonal-average total-system COP, PUE and TUE (same encoding as Fig. 14.6).

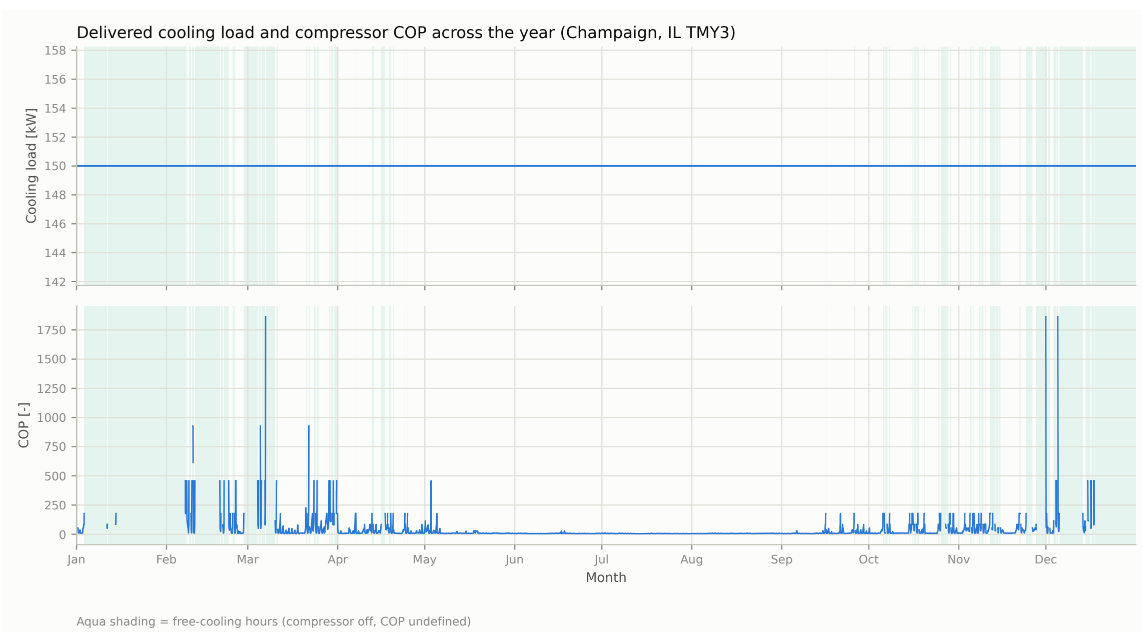


Figure 14.8. Delivered cooling load and compressor COP across all 8,760 hours; aqua shading marks free-cooling hours.

14.7. The Hardware These Targets Demand

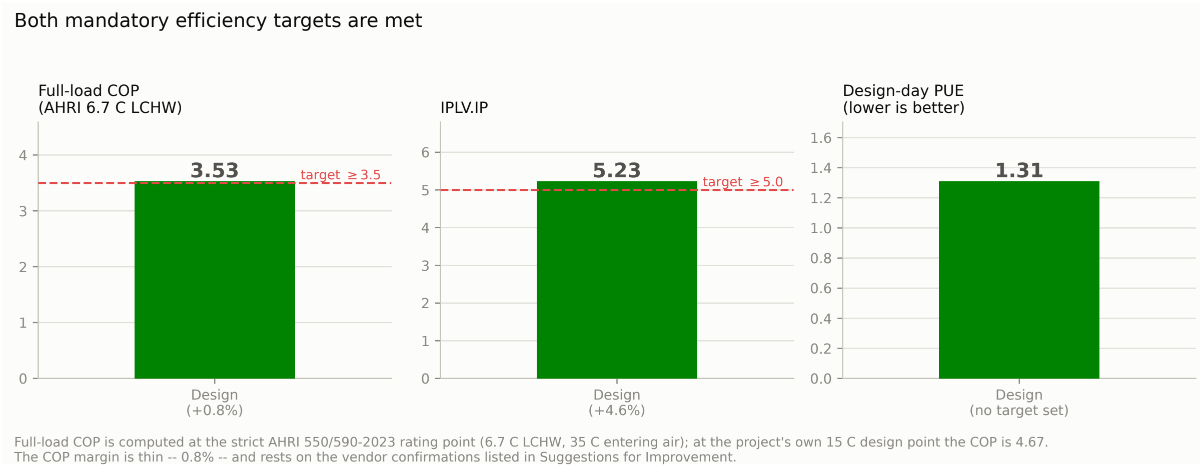


Figure 14.9. The design against the two mandatory efficiency targets, with the design-day PUE it achieves. The COP margin is thin; the IPLV margin is comfortable.

The efficiency targets are bought with heat-transfer area and drives. This section makes that cost explicit, because it is the dominant capital consequence of the whole design.

14.7.1 Overall Heat-Transfer Coefficients: the Double-Wall Derate

The design's approaches require $UA = 19.7\,\mathrm{kW/K}$ on the evaporator and $20.2\,\mathrm{kW/K}$ on the condenser. That UA must be bought as *area* (plates) or as *coefficient* (U), so the U adopted decides whether the exchangers fit their frame — which makes the following correction load-bearing rather than a detail.

> **The published U bands do not apply to this hardware.** The bands cited in the component chapters (§2.2.1: 2,500–4,500 for a water/propane BPHE; §3.2.4: 2,000–3,500 for refrigerant/glycol) are published for **generic, single-wall** plate exchangers. This design uses **double-wall** plates throughout, adopted for A3 leak safety (§2.4.1, §3.4.1) — and a double wall interposes a second plate plus an interstitial contact resistance directly in the heat-transfer path. The correct U is therefore *lower* than the published band, not somewhere within it. It is an easy error to make in the opposite direction.

A resistance-network check anchors the single-wall starting point: for the evaporator, $1/U = 1/h_w + t/k + 1/h_r$ with a water film $h_w \approx 6{,}000$, a 0.4 mm stainless wall, and a propane boiling film $h_r \approx 7{,}000\,\mathrm{W\,m^{-2}K^{-1}}$ returns $U \approx 2{,}990$ — essentially the 3,000 the band would suggest. Published double-wall derates fall in the 10–30 % range; a mid-range **20 % derate** is adopted, giving:

Both adopted values sit *below* their published band. The 20 % derate is itself an estimate from a 10–30 % literature range and is the first vendor confirmation to close (§20.1): Kelvion should be asked for the DW derate *relative to the equivalent single-wall plate.*

Table 14.7. Overall heat-transfer coefficient assumptions, derated for double-wall construction.

Exchanger	Published single-wall band	Adopted (DW)	Basis
Evaporator (water / R-290)	2,500–4,500	**2,400**	Resistance-network single-wall value (≈2,990), ×0.80 double-wall derate
Condenser (R-290 / 30 % PG)	2,000–3,500	**2,000**	§3.2.4's single-wall value, ×0.80 double-wall derate

Vendor plate geometry

The plate counts rest on confirmed **Kelvion HP DW 500H** geometry: a heat-transfer area of **195 × 600 mm = 0.117 m^2 per plate** in a frame accepting up to **120 plates**. This matters more than it might appear — the real plate is 76 % larger than the $A \times B$ estimate used before vendor data was available (§2.6.1, §3.6.1), and that margin is exactly what lets both exchangers absorb the tight approaches *and* the double-wall derate inside a standard frame (§14.7).

These remain assumptions, not vendor results. U is an *outcome* of real plate geometry, channel velocity, and the specific double-wall construction, not a free parameter, and the 20 % derate is an estimate spanning a 10–30 % literature range rather than a datasheet figure. Both coefficients must be confirmed in **Kelvion Select PHE** before procurement — the decisive question being the *double-wall derate relative to the equivalent single-wall plate.* The same selection run should also report the resulting channel pressure drops, which feed pump sizing (Chapters 8, 9). The sensitivity is documented in §14.7: U can fall to ≈1,700 W m^{-2}K^{-1} on either exchanger — a further ≈30 % below the values adopted here — before the 120-plate frame binds.

14.7.2 Component by Component

Both plate exchangers remain within the selected Kelvion HP DW 500H frame, with 36 and 16 plates of headroom respectively, so the capital impact concentrates almost entirely in the dry cooler and the glycol pump.

That the plate exchangers fit so comfortably is a direct consequence of the confirmed vendor plate being 76 % larger than the $A \times B$ placeholder used before it was available — enough to absorb both the tight approaches and the double-wall derate. The corresponding **sensitivity margin is wide**: U would have to fall to ≈1,685 (evaporator) or ≈1,729 W m^{-2}K^{-1} (condenser) — roughly 30 % below the already double-wall-derated values adopted here, and below the floor of even the single-wall published bands — before the 120-plate frame becomes binding. The plate-exchanger sizing is therefore robust to a substantially worse-than-expected vendor answer, which is the opposite of the knife-edge the earlier optimistic-U treatment implied.

Table 14.8. The hardware the design basis demands. Plate exchangers are sized by LMTD at the double-wall-corrected U values of Table 14.7 against the confirmed Kelvion HP DW 500H plate ($0.117\,\mathrm{m}^2$, 120-plate frame); the dry cooler via the ε-NTU model of Chapter 6. Plate counts include the report's standard 20 % margin.

Component	As designed	Consequence
Evaporator BPHE	LMTD 7.6 K, UA 19.7 kW/K, U 2,400 (DW), $8.21\,\mathrm{m}^2$, ≈**84 plates**	**Fits the HP DW 500H frame** with ≈36 plates spare, despite the 5 K approach *and* the double-wall derate
Condenser BPHE	LMTD 9.0 K, UA 20.2 kW/K, U 2,000 (DW), $10.12\,\mathrm{m}^2$, ≈**104 plates**	**Fits the frame** with ≈16 plates spare
Dry cooler	**Kelvion ULF-PA106K4V-091F095**, $1{,}949\,\mathrm{m}^2$, $UA \approx 73.7$ kW/K, 60 dB(A), 10 % fan headroom	**The dominant capital item of the whole plant.** The 3 K approach is what makes it this large (§6.5.3); the fan headroom is what keeps all 8,760 hours off the fan ceiling
Glycol loop	11.6 kg/s at a 4 K range, condenser ΔP ≈35.8 kPa, dry cooler 8 psi	The tight 4 K range demands high flow, so pump electrical power is 2.57 kW. System curve and trim in Ch. 8 predate these numbers — re-derive against the 2.5AC frame (§14.4, Ch. 8 open item 3)
Compressors	2 × 4FEP-35Z with VFDs	Standard drive addition; delivers the 5:1 turndown to the 30 kW minimum load without hot-gas bypass

Chapter 15

Cycle Representation: p-h Chart and Temperature Cascade

15.1. Pressure-Enthalpy Chart of the Designed Cycle

Figure 15.1 is the project's p-h chart deliverable (subtask 1h): the designed cycle plotted on the R-290 saturation dome at the **project design point** — 150 kW at the 35 °C design ambient. Every state point is evaluated from the same Bitzer AHRI-540 map and CoolProp properties the rest of the report uses (`cycle_figures.py`), so the chart cannot drift from the model.

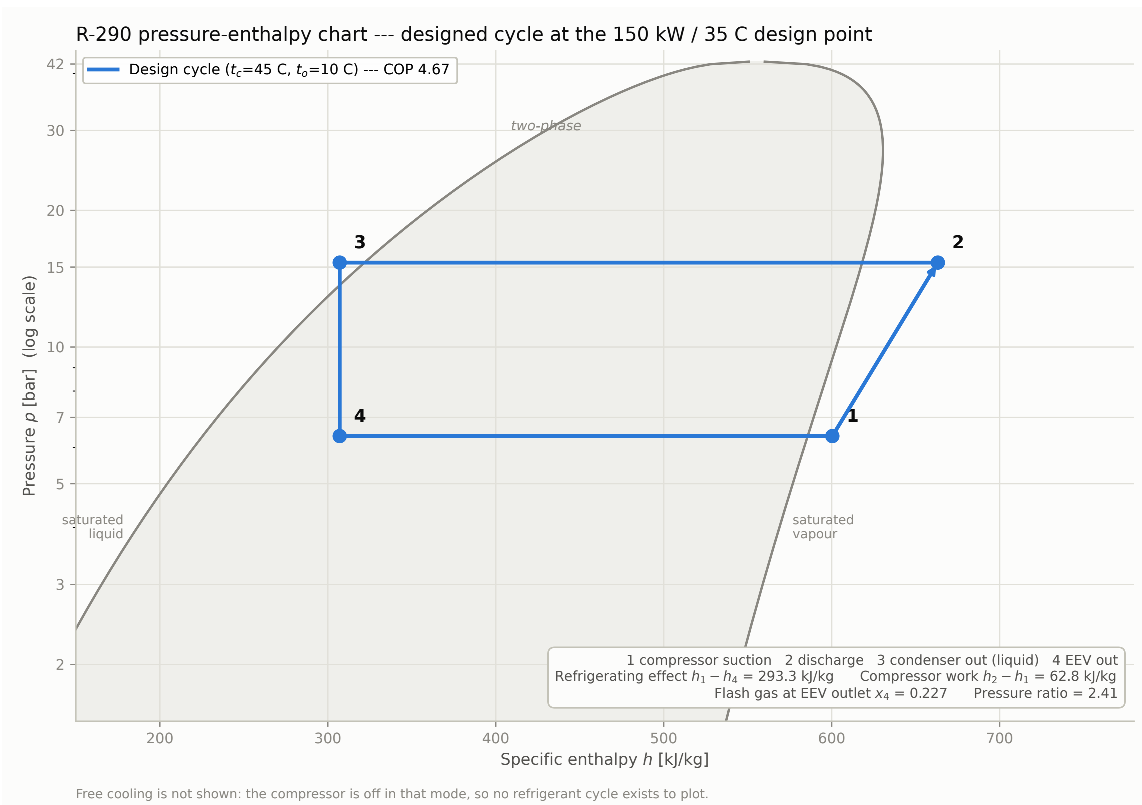


Figure 15.1. R-290 p-h chart of the designed cycle at the 150 kW / 35 °C design point.

15.1.1 Why the Low Condensing Temperature Pays Twice

The chart is also the clearest place to see *why* this design spends so much dry cooler on holding t_c down to 45 °C. The COP is the ratio of the horizontal 4 → 1 leg (refrigerating effect) to the

Table 15.1. Cycle state points at the design point ($t_o = 10$, $t_c = 45\,^\circ$C, 8 K superheat, 5 K subcooling).

State	Description	T [°C]	p [bar]	h [kJ/kg]
1	Compressor suction (evaporator out, 8 K superheat)	18.0	6.37	600.4
2	Compressor discharge	64.8	15.34	663.2
3	Condenser out (subcooled liquid, 5 K)	40.0	15.34	307.1
4	EEV out / evaporator in (isenthalpic, $x = 0.227$)	10.0	6.37	307.1

$1 \rightarrow 2$ leg (compressor work), and raising t_c damages *both*. Re-solving the same cycle — same compressor, same t_o, same superheat and subcooling — with the condensing approach relaxed to a conventional air-cooled 17 K ($t_c = 52\,^\circ$C):

Table 15.2. The same cycle at two condensing temperatures. Only t_c changes; the loss is compounded, not linear.

	$t_c = 52\,^\circ$**C (17 K approach)**	$t_c = 45\,^\circ$**C (this design)**	**Change**
Compressor work $h_2 - h_1$	72.4 kJ/kg	62.8 kJ/kg	−13 % (smaller lift)
Refrigerating effect $h_1 - h_4$	272.8 kJ/kg	293.3 kJ/kg	+8 % (less flash gas)
Flash gas at EEV outlet x_4	0.284	0.227	−20 %
Pressure ratio	2.81	2.41	−14 %
COP $= (h_1 - h_4)/(h_2 - h_1)$	**3.77**	**4.67**	+**24** %

The work reduction is the expected consequence of a smaller lift. The *second* gain is less obvious and is visible only on the chart: lowering t_c moves state 3 left (the liquid leaves the condenser at 40 rather than 47 °C), and because expansion is isenthalpic, state 4 moves left with it. Less refrigerant flashes to vapour across the valve (22.7 % against 28.4 %), so more of it arrives in the evaporator as liquid actually able to absorb heat.

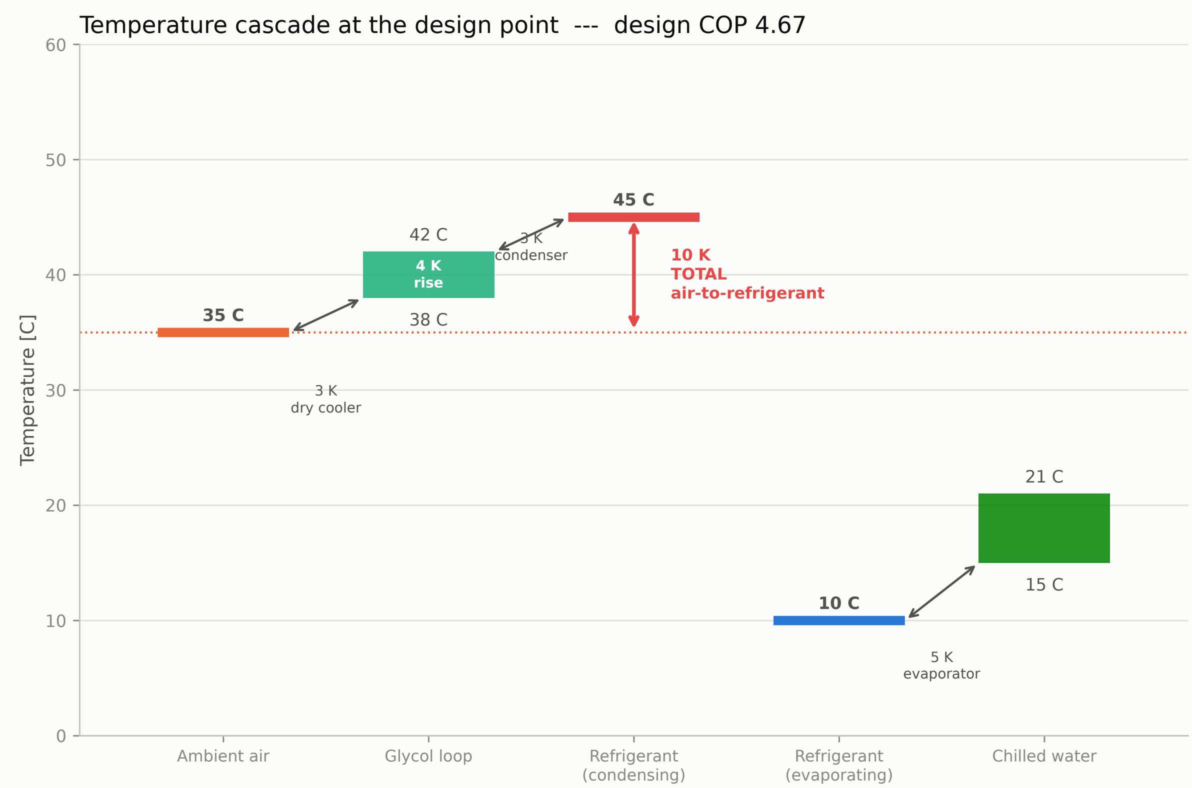
> A lower condensing temperature therefore pays **twice** — once in compressor work, once in refrigerating effect — which is why the 24 % COP gain is nearly double the 13 % work saving alone. This compounding is the entire economic case for the 1,949 m^2 dry cooler.

State 2 lies well right of the saturated-vapour line, confirming the discharge is genuinely superheated (64.8 °C against a 45 °C saturation temperature) — the desuperheating zone the condenser's three-zone model accounts for (§15.3).

Free cooling is not plotted: the compressor is off in that mode and the economizer carries the load on the waterside, so no refrigerant cycle exists to draw.

15.2. Temperature Cascade

Figure 15.2 shows why the condensing temperature — and therefore the full-load COP — is what it is. It is the single most load-bearing diagram in the report.



The intermediate glycol loop stacks three approaches in series between ambient air and the refrigerant --- 3 K across the dry cooler, a 4 K glycol rise, 3 K across the condenser. Their sum is the 10 K that sets t_c, and t_c sets the COP. This is the price of keeping an A3 refrigerant out of the data hall, and the reason the dry cooler is sized as generously as it is.

Figure 15.2. Temperature cascade from ambient air to chilled water at the design point. The condensing temperature is set by three approaches stacked in series.

The intermediate glycol loop — the architecture the project brief prescribes — places **three approaches in series** between ambient air and the condensing refrigerant: the dry cooler's approach, the glycol temperature rise, and the condenser's own approach. A direct air-cooled condenser would pay only one. This is the structural reason the full-load COP is harder to reach here than the ASHRAE 90.1 *air-cooled chiller* table (from which the $\geq$3.5 target is drawn) implicitly assumes, and it is the cost side of the trade that buys the free-cooling economizer and the winter PUE of 1.01.

Each of the three is therefore driven as tight as the hardware allows: **3 K** at the dry cooler, a

4 K glycol rise, **3 K** at the condenser. Their sum is the 10 K air-to-refrigerant approach that fixes $t_c = 45\,°C$. On the cold side the evaporator approach is held to **5 K**, lifting t_o to 10 °C. The lift is squeezed from both ends, and §15.1.1 shows what each kelvin is worth.

> None of these three is generous, and that is the point: at 0.09 COP/K (§14.2.1) and only ≈0.3 K of margin to the target, **there is no slack anywhere in this cascade**. Any one of them slipping by half a kelvin — a fouled coil, an optimistic plate U, a warmer glycol loop — puts the full-load COP below 3.5.

15.3. Heat-Exchanger Temperature Profiles

Figure 15.3 plots temperature against duty through both refrigerant-side exchangers at the design point — the picture behind the LMTD sizing of Chapters 3 and 4, and where each exchanger actually pinches.

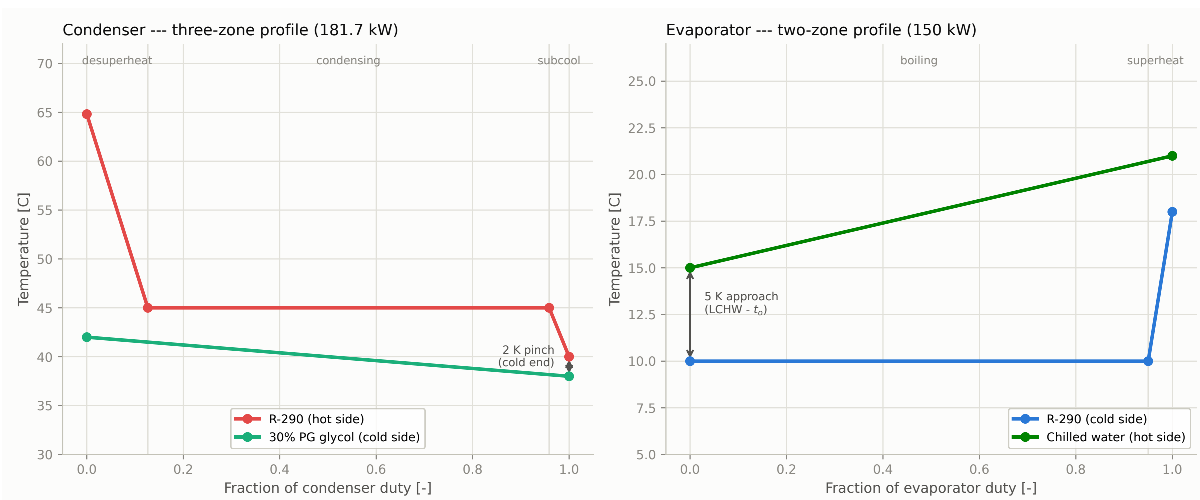


Figure 15.3. Counter-flow temperature profiles at the design point: condenser (three-zone) and evaporator (two-zone).

The **condenser** shows the three zones its model decomposes: a steep desuperheating leg (64.8 → 45 °C) over the first ≈13 % of duty, the dominant isothermal condensing plateau at 45 °C, and a short subcooling tail to 40 °C. Its tightest point is the **cold end**, where the 40 °C subcooled liquid leaves against 38 °C entering glycol — a 2 K pinch. This is precisely the constraint that forces the dry cooler's 3 K approach (§6.5.3): warmer glycol would close this gap to zero and the exchanger would stop working, regardless of how much area it had.

The **evaporator** boils isothermally at 10 °C over ≈95 % of its duty, then superheats to 18 °C in a short tail. Its approach is set at the cold end, where the 15 °C leaving chilled water meets the 10 °C evaporating refrigerant — the 5 K approach, bought with the plate count of §14.7.

Chapter 16

Refrigerant Charge Analysis and Safety Measures

Part A — Charge Inventory

16.1. Why This Chapter Is a Condition, Not an Addendum

The project brief admits a refrigerant of safety class A3 *only* “**with charge analysis**” (subtask 1a). R-290 is A3, so this chapter is a **condition on the working-fluid selection itself** (Chapter 2) rather than an optional extra: without it, the R-290 choice is not substantiated. It also carries subtask 1i (safety measures), which the per-component chapters have until now addressed only piecemeal.

16.2. Charge Inventory

The inventory below is built component by component from the same geometry and CoolProp properties the rest of the model uses (`charge_analysis.py`), at the design point. It is reported as a **range**, not a point value: refrigerant redistributes with operating mode, and the void fractions in the two two-phase exchangers are genuinely uncertain. Safety systems are sized on the *upper* bound.

Table 16.1. R-290 charge inventory at the design point (4 m liquid line, no receiver). Ranges reflect two-phase void fraction and oil-dissolved refrigerant.

Component	**Internal volume**	**Charge (low)**	**Charge (high)**
Condenser (2-phase)	12.9 L	1.26 kg	2.90 kg
Liquid line (4 m)	3.6 L	1.69 kg	1.69 kg
Oil separator (40 L vessel)	40.0 L	1.37 kg	1.37 kg
Compressors (2 x, oil-dissolved)	—	0.40 kg	1.20 kg
Evaporator (2-phase)	10.4 L	0.40 kg	1.19 kg
Discharge line (5 m)	5.7 L	0.17 kg	0.17 kg
Suction line (4 m)	12.4 L	0.16 kg	0.16 kg
Total system charge	—	**5.5 kg**	**8.7 kg**

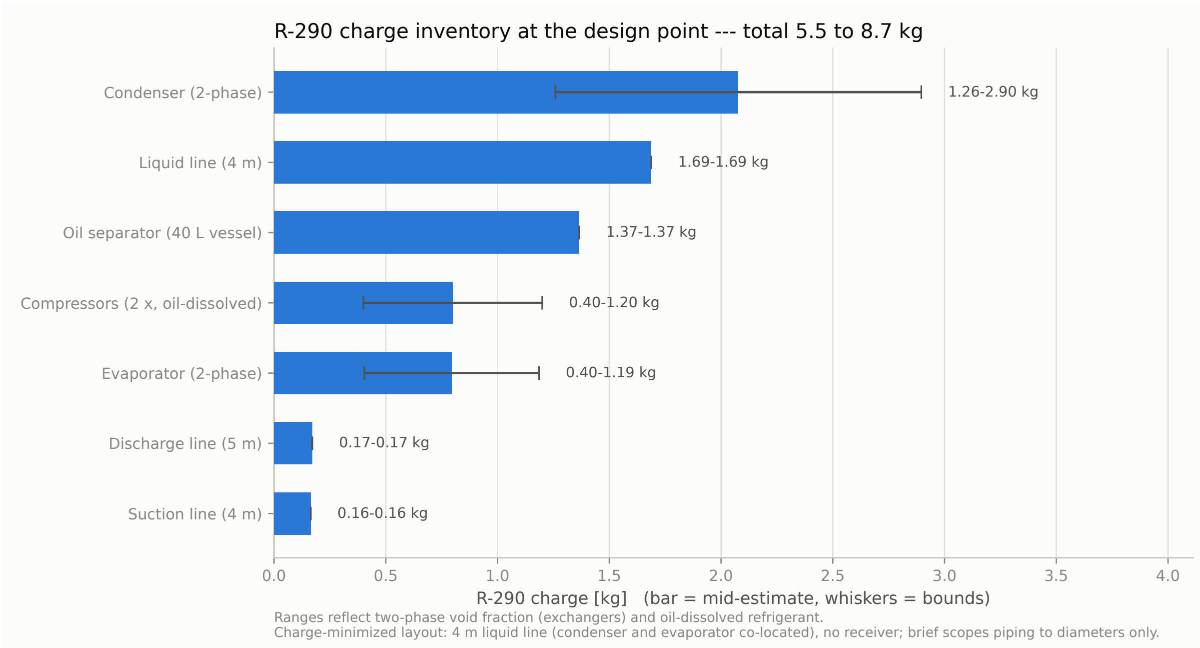


Figure 16.1. R-290 charge inventory at the design point (4 m liquid line, no receiver). Whiskers show the bounds from two-phase void fraction and oil-dissolved refrigerant.

16.2.1 What Dominates the Inventory

The total comes to **5.5–8.7 kg**, and it is worth naming which items carry it, because the answer has been *engineered* rather than merely observed. In an early layout the liquid line dominated the whole inventory — it is the only line holding dense subcooled liquid, and at a loose 15 m it alone would be 6.3 kg, half the system. So the plant is deliberately laid out to make that line short: co-locating the condenser and evaporator holds the liquid line to **4 m and 1.69 kg** (§16.2.2).

With that done, no single item dominates any more. The largest is now the **condenser two-phase holdup (1.7–2.9 kg)**, followed by the liquid line (1.69 kg) and the *oil separator*'s 40 L of discharge vapour (1.37 kg). The suction and discharge lines, being low-density vapour, together hold only **0.3 kg** despite their length — which is exactly why the layout effort is spent on the liquid line and not on them.

Had a **liquid receiver** been fitted, the inventory would read **8.4–14.8 kg** — and because the rest of the plant is now so lean, that receiver would be a *third to a half* of the whole charge (§13.3), making every EN 378 obligation in Part B correspondingly heavier. Omitting it is the single largest charge-reduction decision in the design; minimizing the liquid line is the second.

16.2.2 The Liquid Line: the Lever, and Where It Bottoms Out

The liquid line is the one line whose length matters to charge, at **0.42 kg/m** of dense subcooled liquid — roughly 25 times the suction line's contribution per metre. The project brief scopes piping to diameters only (§12.1), so its length is a layout choice, and this design spends that choice: the condenser and evaporator are placed adjacent so the run is **4 m**, which is about the practical floor once isolation valves, bend radii and service clearance are allowed for. The charge scales almost linearly with it:

Table 16.2. System charge vs. liquid-line length (suction/discharge held at the adopted 4 / 5 m). The 4 m row is the design; longer rows show what a looser layout would cost.

Liquid-line length	Total charge	vs. the 4 m design
3 m	5.0–8.2 kg	−8 %
4 m (adopted)	**5.5–8.7 kg**	baseline
5 m	5.9–9.1 kg	+8 %
10 m	8.0–11.2 kg	+46 %
15 m	10.1–13.3 kg	+84 %

The message is the same one, now acted on: for an A3 refrigerant, **every metre of liquid line is charge**, and charge is what drives the EN 378 obligations of Part B. A careless 15 m run would have put this plant at 13 kg; the 4 m layout keeps it under 9 kg. The plant-room drawing must therefore **hold the condenser, evaporator and compressor pack tightly together**, and any layout change that lengthens the liquid line has to be treated as a safety regression, not a convenience.

Part B — EN 378 Safety Assessment

16.3. Flammability Assessment

Table 16.3. EN 378 assessment at the upper-bound charge (8.7 kg).

Quantity	Value	Basis
R-290 LFL	38 g/m^3	EN 378-1 Annex E
Practical limit	8 g/m^3	EN 378-1, occupied space
Detection setpoint (20 % LFL)	7.6 g/m^3	EN 378-3
Room volume for full release < practical limit	1084 m^3	$m/0.008$ — impractical
Room volume for full release < 20 % LFL	1141 m^3	$m/(0.2 \cdot \mathrm{LFL})$
Room volume for full release < LFL	228 m^3	m/LFL — no margin
Emergency ventilation airflow	**213 m^3/h** (0.059 m^3/s)	EN 378-3: $\dot{V} = 0.014\, m^{2/3}$

The assessment is decisive and is worth stating plainly. Diluting a full 8.7 kg release below R-290's practical limit by room volume alone would require a **1,084 m^3** plant room; even reaching the bare LFL needs **229 m^3**. A realistically sized machinery room for a 150 kW chiller (≈90 m^3) would reach ≈97 g/m^3 — **2.5× the LFL**. **Room volume is therefore not a viable safety strategy**, and the installation cannot rely on dilution.

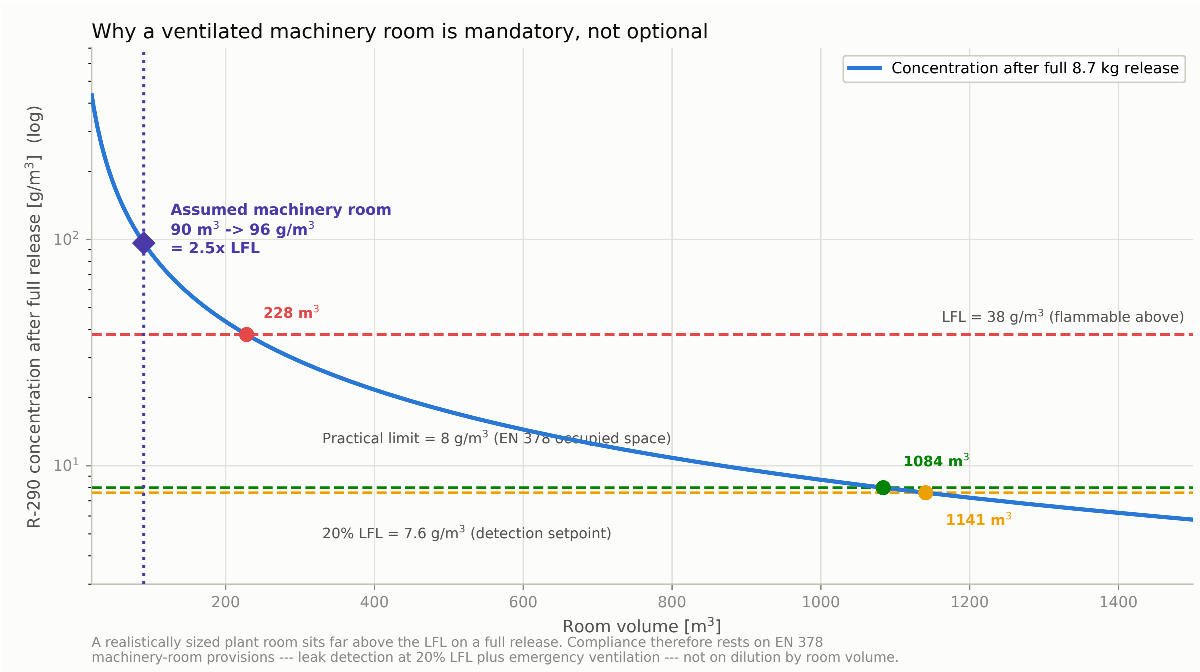


Figure 16.2. Room concentration following a full release of the upper-bound charge, vs. room volume, against the EN 378 thresholds.

This conclusion is robust to every charge decision taken in the report: even the charge-minimized 5.5 kg lower bound would need a 145 m^3 room to reach the bare LFL, and a 688 m^3 room to reach the practical limit. **No plausible plant room dilutes this charge to safety.** The charge-reduction measures — omitting the receiver (Chapter 13) and the 4 m liquid line (§16.2.2) — cut every number in Part B, ventilation from 303 to $\mathbf{213\,m^3/h}$, but they do not, and could not, change the conclusion.

Compliance instead rests on treating the plant room as a **dedicated refrigerating machinery room** under EN 378, whose provisions decouple the permitted charge from room volume:

- **Refrigerant detection** at or below 20 % LFL ($\mathbf{7.6\,g/m^3}$), which alarms and starts emergency ventilation long before a flammable mixture can form.
- **Emergency mechanical ventilation** of $\dot{V} = 0.014\,m^{2/3} = \mathbf{213\,m^3/h}$ ($0.059\,\mathrm{m^3/s}$) per EN 378-3 — a modest fan, and a reminder that *detection plus ventilation*, not volume, is what makes A3 installations safe.
- **Ignition-source control** in the machinery room: no open flames, and electrical equipment either outside the room or rated for the zone.

16.4. Charge-Minimisation and A3 Measures Already in the Design

Several decisions made earlier in this report are, in effect, A3 safety measures. They are consolidated here (subtask 1i) rather than left scattered:

Table 16.4. A3 safety measures already embedded in the design.

Measure	What it does	Where
Double-wall BPHEs	Prevents an R-290 leak reaching the chilled water and hence the occupied data hall — the single most important barrier in the design, and the reason the plant room is the *only* space with a refrigerant inventory	§3, §4
Glycol loop (not direct DX)	The refrigerant never leaves the plant room; the data hall sees only water	Ch. 1
Semi-hermetic compressor	No external shaft seal — removes a leak path an open-drive machine would have	§9.2
Float-valve oil return (not solenoid)	Avoids an electrical ignition source in the refrigerant-wetted oil path	§11.4
Relief discharge routed outdoors	A relieving vessel vents flammable gas to a safe outdoor location, never into the plant room	§11
Pressure vessels rated 28–45 bar	All clear the 19.07 bar 55 °C standstill pressure with large margin	§3, §11
No liquid receiver	Removes 2.97–6.16 kg (a third to a half of the charge) and two more joints — the largest single charge reduction, and it *improves* the COP rather than costing it	Ch. 13
Short (4 m) liquid line	Condenser and evaporator co-located so the only liquid-holding line is 4 m, not a loose 15 m — keeps the system under 9 kg instead of over 13 kg	§16.2.2

16.5. Limitations and Open Items

1. **The inventory is an engineering estimate, not a commissioning charge.** Void fractions in the two-phase exchangers are bounded rather than modelled, and the 6.4 kg spread between low and high bounds reflects that honestly. The real charge is established by weighing at commissioning — which is itself still an outstanding deliverable (subtask 2).
2. **The 15 m line lengths are assumed**, and the charge scales nearly linearly with the liquid line (§16.2.2). This is the dominant uncertainty in the whole chapter and should be closed with a real layout before the receiver is procured.
3. **The 90 m^3 machinery room is assumed**, not specified: the brief gives the data hall footprint (8×12 m) but not the plant room. The EN 378 conclusion is insensitive to it — no plausible plant room dilutes this charge below the LFL — but the ventilation design needs the real geometry.
4. **Ventilation is sized by the EN 378-3 emergency formula only.** A full design must also check normal ventilation, the detector count and placement (R-290 is heavier than air and pools low), and the airflow path.
5. **No formal risk assessment or zone classification** (e.g. ATEX) has been carried out; the ignition-source control above states intent, not a classified drawing.

16.6. References

1. EN 378-1, *Refrigerating systems and heat pumps — Safety and environmental requirements. Part 1: Basic requirements, definitions, classification and selection criteria* — LFL, practical limits, charge limits by occupancy.
2. EN 378-3, *Part 3: Installation site and personal protection* — machinery-room requirements, emergency ventilation ($\dot{V} = 0.014\, m^{2/3}$), detection.
3. IEC 60335-2-40, *Particular requirements for electrical heat pumps, air-conditioners and dehumidifiers* — appliance-level A3 charge limits.
4. ASHRAE 15, *Safety Standard for Refrigeration Systems* — machinery-room and standstill-pressure basis.

Chapter 17

Bill of Materials

17.1. Consolidated Equipment List

Table 17.1 consolidates every major component and accessory selected across the preceding chapters into a single parts list (project subtask 1g). Each line carries the selected vendor model (or, where no vendor selection was made, the specification and its placeholder status), the quantity implied by the dual 75 kW parallel-circuit architecture (Chapter 10), and a cross-reference to the chapter where that selection is developed and justified.

Not yet vendor-selected (carried as placeholders, flagged in their own chapters): the CRAH/in-row units (item 15, Chapter 9) and the expansion valve final model (item 5, a flow-area shortlist rather than a catalog part, Chapter 7). All pressure-vessel and rotating-equipment selections (compressor, both BPHEs, dry cooler, economizer, oil separator, receiver, both pumps) rest on real vendor data. Isolation valves (items 8–9) are specified by function for the P&ID rather than by catalog model at this stage.

Table 17.1. Bill of materials — major equipment and accessories. Model numbers and key ratings are drawn from each component's own chapter; quantities reflect the confirmed architecture — **one shared refrigerant circuit** fed by two parallel 75 kW compressors, hence one evaporator, one condenser and one EEV (Chapter 10, §18.4.1).

Item	Component	Qty	Selected model / spec	Ref.
Refrigerant loop				
1	Working fluid (R-290)	—	Propane, ASHRAE A3, GWP 3; **5.5–8.7 kg** system charge (4 m liquid line, no receiver), critically charged (§16.2)	Ch. 2, 16
2	Compressor	2	Bitzer 4FEP-35Z, semi-hermetic reciprocating, R-290, with VFD for capacity control across the 5:1 turndown (§14.1.1)	Ch. 10, 14
3	Evaporator BPHE	1	Kelvion HP DW 500H, 45 bar double-wall, 0.117 m^2/plate (195×600 mm); ≈**84 plates** of the 120-plate frame ($U = 2{,}400$ DW-derated, pending Kelvion confirmation, §14.7.1)	Ch. 3, 14
4	Condenser BPHE	1	Kelvion HP DW 500H, 45 bar double-wall, 0.117 m^2/plate (195×600 mm); ≈**104 plates** of the 120-plate frame ($U = 2{,}000$ DW-derated, pending Kelvion confirmation, §14.7.1)	Ch. 4, 14
5	Electronic expansion valve	**1**	Sized for the *full* 150 kW flow (single shared circuit): ≈25.7 mm^2 orifice ($d \approx 5.7$ mm); Danfoss ETS / Emerson EX class. 82.6 % open at design, 24.3 % at the 30 kW minimum — inside its controllable range (§18.4)	Ch. 7, 18
6	Oil separator	1	Bitzer OA1954, primary, A3-rated, 28 bar, 18 L oil charge	Ch. 11
7	Liquid receiver	**0**	**Not required by this architecture** — a through-flow receiver would destroy the condenser's 5 K subcooling (AHRI COP 3.53 → 3.35, failing target) and add ≈25 % charge. System is critically charged (Ch. 13)	Ch. 13
8	Discharge check valve	2	One per compressor discharge branch (idle-unit isolation)	Ch. 10
9	Suction shut-off solenoid	2	One per compressor suction branch (idle-unit isolation)	Ch. 10
Heat rejection & free cooling (glycol loop)				
10	Dry cooler	1	**Kelvion ULF-PA106K4V-091F095** (§6.6): 670.36 MBH at 42/38 °C glycol / 35 °C air, 1,949 m^2, 6×0.910 m EC fans @ 950/1050 rpm, 60 dB(A), 8 psi. **Full datasheet still to be pulled** (fan power, dimensions, weight)	Ch. 6, 14
11	Free-cooling economizer	1	Kelvion GB-series, single-wall,	Ch. 5

Chapter 18

Automatic Control of the Compressor–Expansion Device Interaction

18.1. Scope: Why This Is a Discussion, Not a Simulation

The brief asks for the “**consideration of** automatic control of compressor — expansion device interaction” (subtask 3b). The verb is deliberate: section 3 asks for consideration, where section 1 asks for “calculation and/or selection”.

That framing matches the physics. Control interaction is inherently **transient** — it lives in the mass and energy storage of the evaporator, the valve’s actuator dynamics, and the tuning of two feedback loops. Every model in this report is **steady-state**. A credible dynamic study would need a different model entirely, and inventing one would produce numbers with no more authority than the discussion below. It is therefore not attempted, and that boundary is stated rather than blurred.

What the steady-state model *can* settle, it does: the valve’s control authority across the real operating envelope (§18.4), which is a genuine sizing question with a genuine finding.

18.2. Control Architecture

Two loops act on the same evaporator, and the entire subject of this chapter is that they are not independent.

Table 18.1. The two interacting refrigerant-side control loops.

	Compressor (VFD + staging)	**EEV**
Role	**Master** — sets capacity	**Slave** — sets superheat
Controlled variable	Leaving chilled-water temperature (15 °C setpoint)	Evaporator outlet superheat (8 K setpoint)
Manipulated variable	Shaft speed; number of units running	Valve orifice opening
Physical effect	Refrigerant mass flow, hence duty	Split of the evaporator between boiling and superheat
Timescale	Slow (thermal mass of the CHW loop)	Fast (evaporator refrigerant-side dynamics)

The plant-level supervisory logic sits above both: it selects **mechanical** or **free-cooling** mode on outdoor air temperature (Chapter 6, §6.9), and in free cooling the compressors and the EEV are simply off — the refrigerant loop is idle and this chapter's subject does not arise for ≈29 % of the year.

18.3. The Interaction

The loops are coupled because **both act on the evaporator's refrigerant inventory**, from opposite ends. The compressor withdraws vapour from it; the EEV admits liquid into it. Superheat is the observable that reports the balance between the two.

A capacity change propagates as follows:

1. Cooling demand rises; the master loop raises compressor speed.
2. Higher swept volume withdraws vapour faster than the EEV is feeding liquid, so **suction pressure falls** and the evaporator's liquid inventory starts to deplete.
3. The boiling zone shortens, the superheat zone lengthens, and the measured **superheat rises**.
4. The slave loop sees the superheat error and **opens the EEV**, restoring liquid feed until superheat returns to setpoint.

The compressor therefore acts as a **disturbance** to the superheat loop, and the EEV as a disturbance to the evaporating pressure the compressor is working against. Neither loop can be tuned in isolation.

18.3.1 The Two Failure Modes

- **Hunting.** If the EEV loop is tuned too aggressively — or too close in bandwidth to the compressor loop — the two oscillate against each other: the valve overshoots, superheat undershoots, the valve closes, superheat spikes, and the cycle repeats. Symptoms are oscillating suction pressure and capacity, and it is the more common of the two faults.
- **Floodback.** If the EEV loop is too slow, or the superheat setpoint is too low, a fast capacity reduction leaves the evaporator over-fed and **liquid refrigerant reaches the compressor suction**. For a semi-hermetic reciprocating machine this is the destructive failure — liquid is incompressible, and slugging damages valves and bearings.

18.3.2 Why This Design Is More Exposed Than Most

Two decisions taken elsewhere in this report make the interaction harder here, and both are consequences worth owning explicitly:

1. **The VFD makes the disturbance continuous.** A fixed-speed bank disturbs the superheat loop only at discrete stage changes, which are infrequent and predictable. The VFD adopted for efficiency compliance (§14.1.1) modulates continuously, so the superheat loop is *always* tracking a moving target. The efficiency gain is real, but it buys a continuously-coupled control problem.

2. **There is no liquid receiver to absorb an excursion.** The receiver was omitted (Chapter 13) because it would have broken AHRI compliance and added ≈25 % charge — decisions this report stands behind. But a receiver is also a buffer: it damps inventory swings between the condenser and evaporator. Without one the system is **critically charged**, and a superheat excursion has nothing to absorb it. **Floodback protection therefore rests entirely on the control system**, not on hardware forgiveness.

18.4. EEV Control Authority Across the Envelope

This is the part the steady-state model settles. The valve must modulate across the project's full **5:1 turndown** (150 kW design to the 30 kW minimum IT load), and at each load the condenser-entering-air temperature is lower, which lowers t_c and therefore the pressure drop across the valve. Both effects shrink the required opening, and they compound.

Table 18.2. EEV control authority across the operating envelope. Each load point sits at its own reduced condenser-entering-air temperature per AHRI 550/590. The single-EEV column is the selected arrangement; the two-EEV column is retained as the evidence for that choice.

Load	T_{air} [°C]	t_c [°C]	Δp [bar]	1 EEV (selected)	2 EEVs (rejected)
100 % (design)	35.0	45.0	8.98	82.6 %	41.3 %
75 %	26.7	36.7	6.31	67.4 %	33.7 %
50 %	18.3	28.3	3.98	52.1 %	26.0 %
25 %	12.8	22.8	2.63	30.4 %	15.2 %
20 % (30 kW min IT load)	12.8	22.8	2.63	24.3 %	12.2 % †

† below the ≈15 % authority floor, where the valve's characteristic goes non-linear and superheat control degrades.

18.4.1 Result: the Single-Circuit Architecture Is Confirmed

This analysis settled an architecture question the report had previously left inconsistent. Chapter 10 once described "two parallel refrigerant *circuits*, each with its own . . . EEV", while `main.py` had always modelled a **single shared circuit** — one condenser, one evaporator, one expansion valve, fed by two compressors on a common manifold.

The single-circuit architecture is adopted, and the valve authority is one of the reasons why. Table 18.2 evaluates both readings:

- **One EEV on the full flow (adopted)**: **82.6 % open at design, falling to 24.3 %** at the 30 kW minimum load. Comfortable authority across the whole 5:1 turndown, and the E3V65 is well matched to the duty.
- **Two EEVs on split flow (rejected)**: only 41.3 % open at design — the E3V65 would be **oversized** for half the flow — falling to **12.2 %** at minimum load, *below* the ≈15 % floor at which an EEV's characteristic goes non-linear and stepper resolution and stiction begin to dominate. Superheat control would degrade exactly where floodback protection matters

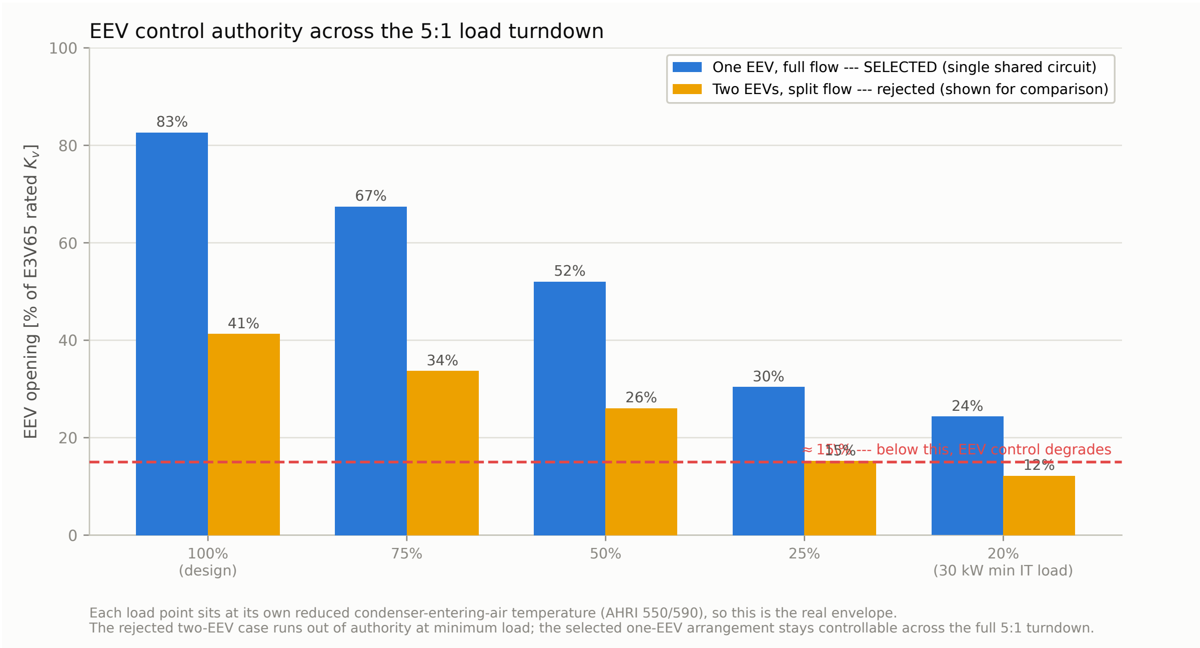


Figure 18.1. EEV opening required across the 5:1 load turndown, for both candidate circuit architectures.

most, and where there is no receiver to forgive it (§18.3.2).

The rejected column is retained deliberately: it shows that the single-circuit choice is not merely a modelling convenience but the arrangement that keeps the expansion valve inside its controllable range at minimum load. Had the two-circuit reading been adopted, the E3V65 would have had to be replaced with a smaller (E3V45-class) valve to move half-flow nearer the middle of its range.

Consequences of the confirmed architecture, now applied throughout the report:

- **One EEV**, not two — the BOM quantity is corrected accordingly.
- **Compressor redundancy only.** Two parallel compressors protect against a compressor failure, but a fault in the single condenser, evaporator or expansion valve stops the plant. The redundancy claim is restated on that narrower basis (§19.3).
- **Fewer components and less charge** than two independent circuits — consistent with the A3 charge-minimisation objective (§16.4).

18.5. The Superheat Setpoint — and a Trap

It is tempting to select the superheat setpoint from the compressor map. That is a mistake, and the mistake is instructive.

Read naively, the map says **more superheat is better**: COP rises from 4.55 at 4 K to 4.88 at

Table 18.3. Design-point COP vs. EEV superheat setpoint, evaluated at FIXED t_o. The rising trend is an artifact of the map's useful-superheat convention, not a design lever — see the discussion.

Superheat setpoint	Design COP	Bank capacity at full speed
4 K	4.553	199.9 kW
6 K	4.612	200.4 kW
8 K **(design)**	4.670	200.8 kW
10 K	4.729	201.2 kW
12 K	4.787	201.6 kW
15 K	4.875	202.3 kW

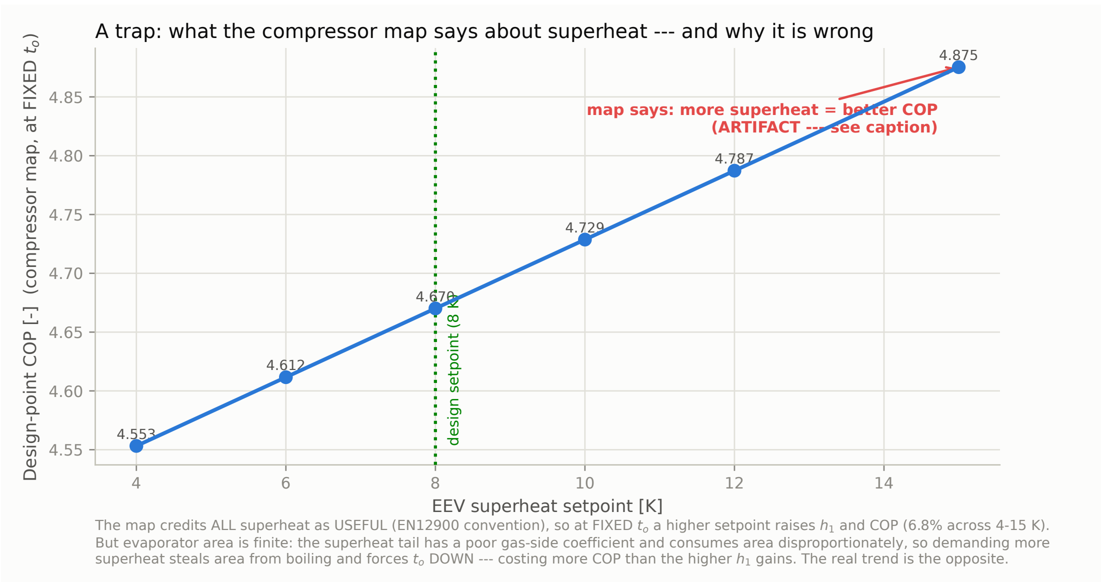


Figure 18.2. What the compressor map says about the superheat setpoint at fixed t_o — and why the trend is an artifact rather than a design lever.

15 K, a 6.8 % span that is far from negligible. **This is an artifact.** The AHRI-540/EN12900 convention credits *all* superheat as *useful* — it assumes the superheat is picked up inside the evaporator, so it adds to the refrigerating effect $h_1 - h_3$. Holding t_o fixed and raising the setpoint therefore raises h_1, and COP follows.

Physically the evaporator's area is finite and the trend reverses. From the evaporator's T–Q profile (Figure 15.3), boiling occupies ≈95 % of the duty and the superheat tail only ≈5 % — but the superheat zone transfers heat to *gas*, with a far poorer film coefficient, so it consumes area disproportionately. Demanding more superheat at fixed area **steals area from boiling**, which forces t_o down, and a lower t_o costs more COP than the higher h_1 gains. Capturing this properly requires an area-constrained evaporator solve, which this report does not perform (§18.7).

The correct conclusion is the opposite of the chart: superheat should be set *as low as is safely controllable*, not high. It is a **compressor-protection setpoint**, not an efficiency knob. The 8 K design value (Chapter 10) is retained: it matches the compressor vendor's own selection basis, and it leaves margin above the floodback boundary for a system with no receiver to forgive an excursion.

18.6. Recommended Control Measures

18.7. Open Items

1. **EEV vendor confirmation.** With the single-circuit architecture confirmed (§18.4.1), the E3V65 sits at 82.6 % open at design and 24.3 % at minimum load — comfortable, but computed from the incompressible-orifice K_v estimate of Chapter 7, which is explicitly a short-listing figure rather than a vendor two-phase capacity rating (§7.5.1). Confirm the opening range against the manufacturer's own capacity table before ordering.
2. **No dynamic model, therefore no tuning constants.** This chapter gives the architecture, the failure modes and the measures, but not PID gains or a stability proof. Those need a transient model and, ultimately, commissioning on the real plant (subtask 2).
3. **The superheat/evaporator-area coupling is argued, not computed** (§18.5). The claim that the real trend opposes the map's rests on the T–Q profile and the poor gas-side coefficient, which is sound but qualitative. An area-constrained evaporator solve would settle it.
4. **Critical-charge sensitivity is unquantified** (§13.7), and it interacts directly with this chapter: with no receiver, a charge error shows up as a superheat control problem.
5. **Free-cooling transitions are not analysed.** Mode switches at the 4 °C threshold start and stop the compressors; the associated refrigerant migration and restart superheat behaviour are not modelled.

Table 18.4. Recommended measures for stable compressor–EEV interaction.

Measure	Purpose	Why here
Timescale separation — EEV loop tuned 5–10× faster than the compressor capacity loop	Lets the slave reject the master's disturbance before it propagates	The standard defence against hunting; the VFD makes it essential (§18.3.2)
Compressor ramp-rate limit	Bounds how fast the disturbance can arrive	Keeps the superheat loop inside its tracking capability
Minimum-superheat clamp	Hard floor below which the EEV closes regardless of the loop	Last-line flood-back protection — load-bearing here, as there is no receiver

Chapter 19

System Assessment: Pros and Cons

19.1. What Was Chosen, and What Was Prescribed

An honest assessment has to separate the decisions this design *made* from the ones the brief *imposed*. The brief prescribes water to the CRAH units, and **heat rejection via a dry cooler with an intermediate water/propylene-glycol loop**. That single prescription determines most of what follows — including the design's central weakness — so it is not fair to score it as a free choice. What was genuinely chosen: the refrigerant, the compressor arrangement and staging, the approach temperatures, the VFD, and the omission of the liquid receiver.

19.2. Pros

19.3. Cons

19.4. The Central Trade, Stated Plainly

The glycol loop is simultaneously the design's greatest weakness and the enabler of its greatest strength, and the two cannot be separated:

- It **costs** ≈7 K of extra condensing approach against a direct air-cooled condenser — worth roughly 20 % of the full-load COP, and the entire reason the efficiency revision (larger dry cooler, more plates, VFDs) was necessary at all.
- It **buys** the waterside economizer, and with it **40.1 % of the year's cooling energy delivered without meaningful compressor work** — 29 % of hours fully free at PUE 1.01 and a further 15.5 % integrated at PUE ≈1.03 (§14.5) — plus the ability to keep an A3 refrigerant entirely out of the data hall.

In Champaign's climate this trade is clearly worth making. The design-day penalty is paid on a handful of hours a year (Fig. 14.5); the free-cooling benefit is collected on thousands. A direct air-cooled DX chiller would post a better AHRI number and a worse annual one — which is a fair summary of why standardized full-load ratings flatter architectures that this application does not want.

The honest caveat: **the compliance metrics are met, but not comfortably.** The 1 % AHRI margin rests on vendor confirmations that have not yet come back (§14.7.1, §6.13). The design is sound; the numbers are not yet safe.

Table 19.1. Strengths of the selected system.

Strength	Evidence	Why it matters
Free cooling dominates the annual picture	2,542 h (29 %) full free cooling, 1,358 h (15.5 %) partial — **44.5 % of the year** (Fig. 14.5). Winter PUE ≈1.01, total-system COP in the high tens	The strongest feature of the design, and the payoff that justifies the glycol loop's efficiency cost. In Champaign's climate zone 5A this is worth far more than the design-day penalty
Very low GWP with a mature ecosystem	R-290, GWP 3 against a <150 limit — a 50× margin (Ch. 2)	Not merely compliant but future-proof against tightening regulation, and every pressure vessel and the compressor were available A3-rated from a single vendor
Warm chilled water exploited	15 °C LCHW (ASHRAE W17) gives design COP 4.67 against 3.53 at the AHRI 6.7 °C rating point (§14.2)	The design is tuned for the condition it actually runs at, not the rating condition. Data-center CHW envelopes reward this
Compressor redundancy	Two 75 kW compressors on a common manifold; losing one leaves 50 % capacity (§9.1). **Compressors only** — the shared circuit's exchangers and EEV are unduplicated	For 24/7/365 IT load, partial failure beats total failure, and the compressor is the likeliest thing to fail. But this is not full N+1: see §20.2
Genuine 5:1 turndown	VFD + staging covers 150 kW to the 30 kW minimum without hot-gas bypass (§14.3)	Resolves the turndown floor the fixed-speed design could not, and lifts IPLV.IP from 4.43 to 5.23
Low refrigerant charge	5.5–8.7 kg, no receiver (Ch. 13)	For an A3 fluid, charge is the safety currency. Omitting

Table 19.2. Weaknesses of the selected system.

Weakness	Evidence	Assessment
The glycol loop costs three stacked approaches	Dry cooler + glycol rise + condenser = 10 K air-to-refrigerant even with all three driven as tight as the hardware allows, against ≈1 approach for a direct air-cooled condenser (Fig. 15.2)	**The design's central weakness — and it is inherited, not chosen.** It is why the COP target (drawn from an *air-cooled chiller* table) is structurally harder to hit here, and why compliance costs a 1,949 m^2 dry cooler
Thin AHRI compliance margin	3.53 against ≥3.5 — ≈1 % (§14.6)	Contingent on the design approaches surviving vendor re-selection, and on assumed U values (§14.7.1). **The most fragile claim in the report**
A3 flammability drives real obligations	No plausible plant room dilutes the charge below LFL; compliance rests on detection at 7.6 $\mathrm{g/m^3}$ + 213 $\mathrm{m^3/h}$ ventilation (§16.3)	A genuine cost of R-290 — the price paid for GWP 3. Manageable and well-precedented, but not free
Critical charge, no buffer	No receiver, so charge must be correct rather than sufficient (§13.5)	Accepted deliberately: the receiver's cost (COP and charge) exceeded its benefit. But it removes hardware forgiveness and pushes floodback protection entirely onto

Chapter 20

Suggestions for Improvement

20.1. Priority 1 — Close the Open Vendor Confirmations

These are not improvements so much as **debts**: the compliance case currently rests on them, and none is expensive to close.

Table 20.1. Outstanding vendor confirmations, in order of how much rests on them.

Item	What is assumed	What rests on it
BPHE double-wall U derate	$U = 2{,}400$ (evap) / 2,000 (cond), a 20 % DW derate estimated from a 10–30 % literature range (§14.7.1)	Plate counts, and hence whether both exchangers stay in the 120-plate frame. Ask Kelvion Select PHE for the DW derate *relative to the equivalent single-wall plate*
Dry cooler fan power	10.6 kW, fan-law scaled — Select RT does not report it (§6.13)	The design-day PUE. **The single largest uncertainty in that figure**
Glycol pump duty point	Ch. 8's system curve predates the revised flow and the re-selected dry cooler's 8 psi (§14.4)	Pump frame selection; 2AD was already near its envelope top, 2.5AC now likely
CRAH coil ΔP	50 kPa placeholder — no unit ever selected (§9.3)	CHW pump selection. The only major component with no vendor basis at all

20.2. Priority 2 — Consider Circuit Redundancy Against the Single Point of Failure

The circuit architecture is now **settled: one shared refrigerant circuit** fed by two parallel compressors (§18.4.1). That choice is right on the merits — it keeps the EEV inside its controllable range at minimum load (24.3 % open, against 12.2 % and unworkable on a split-flow arrangement), and it means fewer components and less charge, which an A3 system wants.

It does, however, leave an honest weakness worth naming: **the shared circuit is a single point of failure**. Two parallel compressors protect against a compressor fault, but the condenser, the evaporator and the expansion valve are each unduplicated — a fault in any of them stops all 150 kW of cooling. For a facility whose IT load runs 24/7/365, that is a real exposure, and the redundancy the design claims (§19.2) is narrower than a casual reading suggests.

Options, none free:

- **Accept it** and manage the risk operationally (spares held, service contract). Reasonable for an edge facility, and the status quo.
- **Two fully independent circuits** — restores graceful degradation to 50 % on any single fault, at the cost of duplicated exchangers, more charge, higher capital, and a smaller EEV to keep authority at part load.
- **Duplicate only the cheapest single points** (e.g. a parallel EEV with isolation valves), which buys partial coverage for modest cost.

Recommendation: **accept the single circuit**, but state the exposure explicitly to the client. The strengths table (§19.2) is worded to that standard — "compressor redundancy", not "N+1" — and any client-facing summary should keep it that way.

20.3. Priority 3 — Protect the Short Liquid Line at Layout Stage

The design already takes the liquid line to **4 m** by co-locating the condenser and evaporator, which is most of why the charge is 5.5–8.7 kg rather than over 13 kg (§16.2.2). **The risk is that this is undone on the drawing board**: a real plant-room layout, routed for pipefitter convenience or to clear other equipment, can silently stretch that run and add charge back. For an A3 refrigerant that is a safety regression, not a detailing choice. The recommendation is therefore to carry the 4 m liquid line as a **fixed constraint into the P&ID and the plant-room general arrangement**, and to re-run the charge and EN 378 numbers if it cannot be held.

20.4. Priority 4 — Float the Economizer Changeover

This is the best return per dollar in the whole report: it is a control-logic change with no hardware attached.

The design follows the project's stated rule — activate the economizer at OAT $\leq 10\,^{\circ}\mathrm{C}$ — and Figure 5.1 shows what that rule costs. At 10 °C the economizer is delivering **75 kW, half the design load**, and then the changeover switches it off. Compressor power jumps from 6.4 to 13.0 kW across a threshold the ambient does not notice. That step is not physics; it is an artefact of a fixed changeover temperature.

The economizer stops being useful only when the glycol it is fed is no longer colder than the chilled water returning from the hall. With glycol at OAT + 5 K and a 21 °C return, that crossover is at **OAT** = 16 **°C**, not 10 °C:

Table 20.2. Economizer duty above the fixed 10 °C changeover, rated with the same ε–NTU model (§5.5.1). This capacity exists and is currently discarded.

OAT [°C]	Glycol supply [°C]	Economizer duty
10	15	75.0 kW (50 % of load) — *switched off here*
11	16	62.5 kW (41.7 %)
12	17	50.0 kW (33.3 %)
13	18	37.5 kW (25.0 %)
14	19	25.0 kW (16.7 %)
15	20	12.5 kW (8.3 %)
16	21	0 kW — true crossover

Champaign spends **1,568 h/yr (17.9 % of the year)** between 10 and 16 °C. Every one of those hours currently runs the chiller at full 150 kW duty while an already-purchased exchanger sits idle with 0–50 kW of free cooling available.

Recommendation: replace the fixed 10 °C rule with a floating changeover — enable the economizer whenever the available glycol supply is below the CHW return by some useful margin (say 2 K), and let the chiller trim whatever is left. This *exceeds* the ASHRAE 90.1 §6.5.1 requirement rather than conflicting with it: 90.1 mandates activation *by* 10 °C, and a floating changeover activates earlier. The economizer, the valves and the piping are all already in the BOM; only the control sequence changes. The one real cost is more hours of low-load chiller operation, where the machine cycles rather than modulates (§5.5.1) — so the sequence should carry a minimum-runtime guard.

20.5. Priority 5 — Efficiency Headroom

The targets are met, but not comfortably — the full-load COP clears by 0.8 %, worth about 0.3 K of condensing temperature (§14.2.1). If more margin is wanted, these are the levers, in descending order of effect:

20.6. Priority 6 — Modelling Fidelity

Ranked by how much the current simplification could mislead:

1. **Glycol-loop hydraulics in the partial band**, where the economizer and the condenser are both served by one loop and one dry cooler. The loop's hot return is closed on an energy balance at the free-cooling flow rather than resolved as two branches; and cycling losses below the drive's minimum speed (679 h/yr) are not modelled (§5.5.1). Both are bounded by the few kW of compressor duty in that band.

2. **Fixed approach temperatures across all ambients.** $t_c = T_{air} + 10$ and the glycol

approaches are held constant rather than re-solving the dry cooler's ε-NTU model at each hour. Cheap and defensible, but it does not capture the fan-speed/approach trade at part load.

3. **Area-constrained evaporator solve**, which would settle the superheat argument of §18.5 rather than leaving it qualitative.

4. **Han correlation applied to R-290.** Regressed on R-410A and R-22; transfer to propane is accepted practice but unvalidated (§2.9), and it underpins every refrigerant-side pressure drop.

5. **Dynamic model** for the control interaction (§18.7).

20.7. Priority 7 — Deliverables Still Outstanding

- **Piping & instrumentation diagram** (subtask 1h) — the last unresolved cross-reference in this report.
- **Commissioning procedure** (subtask 2) — now carrying real weight: with a critically-charged, receiver-less system, **charge weighing and the 5 K subcooling check are acceptance criteria**, not routine steps (§13.5).
- **Specification sheet extracts**, which the brief requires be attached.

Table 20.3. Further efficiency measures, with their costs.

Measure	**Effect**	**Cost**
Reduce the condensing approach further (10 K → 7 K)	AHRI full-load COP ≈3.48 → 3.7+; lifts every IPLV point (§14.2.1)	A larger dry cooler again — already the design's dominant capital item
Float the economizer changeover instead of fixing it at 10 °C (§20.4)	A further 1,568 h/yr (17.9 % of the year) in which the economizer can still carry 50 kW down to 0 kW of the load but is switched off	**Controls only — no hardware.** The cheapest improvement now available
Increase subcooling beyond 5 K	+6 % COP at 10 K, +11 % at 15 K	More condenser area; diminishing returns; interacts with the charge inventory
Economised / vapour-injection cycle	Improves	Rejected: